US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12410434
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Cuero Rengifo; Raul et al.

Genelight cultures and extracts and applications thereof

Abstract

Described herein are genelight cultures, extracts and related methods. In one aspect, the method of making a genelight culture or extract includes the steps of (a) making a DNA construct containing genes for producing a heat shock protein, RuBisCO large subunit 1, tonB, hydrogenase, and a P-type ATPase, (b) introducing the DNA construct into host microbial cells via transformation or transfection, and (c) culturing the microbial cells to produce the genelight cultures and extracts. The compositions of these cultures and extracts can be tailored to have specific properties such as the ability to provide power to a light emitting diode. The cultures and extracts have further uses including enhancing the growth of plants and as supplemental nutrients of cultures of industrially important microorganisms. The cultures and extracts further have UV-protective properties. Also described herein are microbial electric circuits containing the cultures and extracts described herein.

Inventors: **Cuero Rengifo; Raul (Cypress, TX), Varela Toro; Tatiana (Manizales, CO), Giraldo Badillo; Isabella (Aranzazu, CO), Díaz Sánchez; María (Bogotá, CO), Sanchez Giraldo; Johny Mateo (Manizales, CO), Londoño Murillo; Juliana (Manizales, CO)**

Applicant: **Bio Capital Holdings, LLC** (Houston, TX)

Family ID: **74853422**

Assignee: **Bio Capital Holdings, LLC (Houston, TX)**

Appl. No.: **17/653395**

Filed: **March 03, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220183301 A1	Jun. 16, 2022

Related U.S. Application Data

continuation-in-part parent-doc WO PCT/US2020/049425 20200904 PENDING child-doc US 17653395
us-provisional-application US 62896832 20190906

Publication Classification

Int. Cl.: **C12N15/52** (20060101); **A01N63/20** (20200101); **A01N63/32** (20200101); **A01P21/00** (20060101); **A61K8/60** (20060101); **A61K8/99** (20170101); **A61Q17/04** (20060101); **C07K14/245** (20060101); **C07K14/395** (20060101); **C12N1/14** (20060101); **C12N1/20** (20060101); **C12N9/00** (20060101); **C12N9/02** (20060101); **C12N9/88** (20060101); **C12N15/87** (20060101)

U.S. Cl.:

CPC **C12N15/52** (20130101); **A01N63/20** (20200101); **A61K8/606** (20130101); **C12N1/20** (20130101); **C12N9/00** (20130101); **C12N9/0004** (20130101); **C12N9/88** (20130101); **C12N15/87** (20130101); **C12Y401/01039** (20130101); A01P21/00 (20210801); A61K8/99 (20130101); C12N1/14 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006/0130173	12/2005	Lee et al.	N/A	N/A
2007/0042047	12/2006	Hamblin et al.	N/A	N/A
2010/0129408	12/2009	Wang et al.	N/A	N/A
2015/0140592	12/2014	Oard	N/A	N/A
2017/0360695	12/2016	Cuero et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2020257524	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report for PCT/US2020/049425 mailed Feb. 9, 2021. cited by applicant

Primary Examiner: Rao; Manjunath N

Assistant Examiner: Lee; Jae W

Attorney, Agent or Firm: Thomas | Horstemeyer, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of International Patent Application No. PCT/US2020/049425, filed Sep. 4, 2020, which claims the benefit of U.S. Provisional Application No. 62/896,832, filed on Sep. 6, 2019, which is incorporated herein by reference in its entirety.

CROSS REFERENCE TO SEQUENCE LISTING

(1) The genetic components described herein are referred to by a sequence identifier number (SEQ ID NO). The SEQ ID NO corresponds numerically to the sequence identifiers <400>1, <400>2, etc. The Sequence Listing, in written computer readable format (CRF), is incorporated by reference in its entirety.

BACKGROUND

(2) Battery-powered devices are useful in applications ranging from telecommunications to medical devices to providing light during power outages. Current battery technology has several drawbacks, however. Although alkaline batteries can be recycled, few facilities exist for doing so and many end up in landfills each year. Alkaline batteries are also prone to leakage as they age and are used, which can ruin electronic devices. Lead-acid batteries are bulky, contain toxic metals, and may overheat during charging; they may also leak electrolytes (including corrosive acids) under improper storage conditions. Lithium-ion batteries are typically small and rechargeable but charge cycles are limited and transportation restrictions exist due to the possibility of short circuits leading to fires.

(3) It would be advantageous to develop a new, inexpensive, portable power source that is effective and poses no environmental hazards during the disposal process. It would further be advantageous if this power source could make use of water-based electrolyte solutions and did not present a fire hazard during transportation or storage. It would further be advantageous if the raw materials used to generate the electrolyte solutions, or the electrolyte solutions themselves had other applications in agriculture, including plant tissue culture applications; commercial microorganism culture, including growth of organisms that produce industrially important compounds such as ethanol, acetic acid, rennet, insulin, and related compounds; and the like, either as a separate use or as a use for electrolyte solutions that have aged out of useful life. The present invention addresses these needs.

SUMMARY

(4) Described herein are genelight cultures and extracts and methods of making and using thereof. In one aspect, the method of making a genelight culture or extract includes the steps of (a) making a DNA construct containing genes for producing a heat shock protein, RuBisCO large subunit 1, tonB, hydrogenase, and a P-type ATPase, (b) introducing the DNA construct into host microbial cells via transformation or transfection, and (c) culturing the microbial cells to produce the genelight cultures and extracts. The compositions of these cultures and extracts can be tailored to have specific properties such as the ability to provide power to a light emitting diode. The cultures and extracts have further uses including enhancing the growth of plants and as supplemental nutrients of cultures of industrially important microorganisms. The cultures and extracts further have UV-protective properties. Also described herein are microbial electric circuits containing the cultures and extracts described herein.

(5) The advantages of the invention will be set forth in part in the description that follows, and in part will be obvious from the description, or may be learned by practice of the aspects described below. The advantages described below will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the

foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several aspects described below.

(2) FIG. **1** shows a linear schematic of a constructed pBAD plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(3) FIG. **2** shows a circular schematic of a constructed pBAD plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(4) FIG. **3** shows a linear schematic of a constructed pYES2 plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(5) FIG. **4** shows a circular schematic of a constructed pYES2 plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(6) FIG. **5** shows a linear schematic of a constructed pYES2 plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(7) FIG. **6** shows a circular schematic of a constructed pYES2 plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(8) FIG. **7** shows a linear schematic of a constructed pBAD plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(9) FIG. **8** shows a circular schematic of a constructed pBAD plasmid showing the direction, placement, and size of genetic parts used for an exemplary DNA device described herein.

(10) FIG. **9** shows a circuit constructed with the microbial extracts described herein including zinc and copper electrodes (left) and a standard circuit (right).

(11) FIGS. **10**A-**10**B show LED light production from a circuit constructed with the microbial extracts disclosed herein (both panels).

(12) FIGS. **11**A-**11**B shows *Bacillus subtilis* colonies after 30 minutes of UV exposure (7A) and after 2 hours of UV exposure (7B). Samples labeled 1 are *B. subtilis* cultures treated with extracts from the *E. coli* devices disclosed herein. Samples labeled 2 are *B. subtilis* cultures treated with extracts from the *S. cerevisiae* devices disclosed herein. Samples labeled 3 are *B. subtilis* controls (treated with water only).

(13) FIG. **12** shows quantitation of genomic DNA from human fibroblasts with and without treatment using the devices and extracts disclosed herein. Lane 1 shows DNA expression prior to any irradiation or treatment. Lane 2 shows DNA expression in human fibroblasts treated with extracts from *E. coli* devices as disclosed herein and then exposed to UV irradiation. Lane 3 shows DNA expression in human fibroblasts treated with extracts from *S. cerevisiae* devices as disclosed herein and then exposed to UV irradiation. Lane 4 shows DNA expression in human fibroblasts treated with water only and then exposed to UV irradiation. Lane 5 is a 1 kb DNA ladder.

(14) FIG. **13** shows the Raman spectrum of an extract of a genelight *E. coli* device.

DETAILED DESCRIPTION

(15) Before the present compounds, compositions, articles, devices, and/or methods are disclosed and described, it is to be understood that the aspects described below are not limited to specific compounds, synthetic methods, or uses, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

(16) In this specification and in the claims that follow, reference will be made to a number of terms that shall be defined to have the following meanings:

(17) It must be noted that, as used in the specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a restriction enzyme” includes mixtures of two or more such restriction enzymes, and the like.
(18) “Optional” or “optionally” means that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where the event or circumstance occurs and instances where it does not. For example, the phrase “optionally includes a gene for a reporter protein” means that the gene for the reporter protein may or may not be present.
(19) Throughout this specification, unless the context dictates otherwise, the word “comprise,” or variations such as “comprises” or “comprising,” will be understood to imply the inclusion of a stated element, integer, step, or group of elements, integers, or steps, but not the exclusion of any other element, integer, step, or group of elements, integers, or steps.
(20) As used herein, the term “about” is used to provide flexibility to a numerical range endpoint by providing that a given numerical value may be “a little above” or “a little below” the endpoint without affecting the desired result. For purposes of the present disclosure, “about” refers to a range extending from 10% below the numerical value to 10% above the numerical value. For example, if the numerical value is 10, “about 10” means between 9 and 11 inclusive of the endpoints 9 and 11.
(21) “Admixing” or “admixture” refers to a combination of two or more components together wherein there is no chemical reaction or physical interaction. The terms “admixing” and “admixture” can also include the chemical reaction or physical interaction between any of the components described herein upon mixing to produce the composition. The components can be admixed alone, in water, in another solvent, or in a combination of solvents.
(22) “Root tension” as used herein refers to the resistance of a plant or a patch of sod from being uprooted or removed from the ground when pulled. In some aspects, the compositions and extracts disclosed herein, when applied to plants such as, for example, grass, can increase the root tension of the plants. Root tension can be measured using a tension meter and can be given in units of force such as, for example, Newtons (kg.Math.m/s.sup.2).
(23) As used herein, the term “reduce” is defined as the ability to reduce the likelihood of an event (e.g., exposure to UV radiation) from occurring up to about 50%, up to about 60%, up to about 70%, up to about 80%, up to about 90%, up to about 95%, or up to about 99% when compared to not using the methods as described herein. As used herein, the term “reduce” is also defined as the ability to completely eliminate the likelihood of the event from occurring when compared to not using the methods as described herein.
(24) Disclosed are materials and components that can be used for, can be used in conjunction with, can be used in preparation for, or are products of the disclosed compositions and methods. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc., of these materials are disclosed that while specific reference to each various individual and collective combination and permutation of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. For example, if a pharmaceutical composition is disclosed and discussed and a number of pharmaceutically-acceptable carriers are discussed, each and every combination and permutation of pharmaceutical composition and pharmaceutically-acceptable carrier that is possible is specifically contemplated unless specifically indicated to the contrary. For example, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F, and an example of a combination molecule, A-D, is disclosed, then even if each is not individually recited, each is individually and collectively contemplated. Thus, in this example, each of the combinations A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and F; and the example combination A-D. Likewise, any subset or combination of these is also specifically contemplated and disclosed. Thus, for example, the subgroup of A-E, B-F,

and C-E is specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and, F; and the example combination A-D. This concept applies to all aspects of this disclosure including, but not limited to, steps in methods of making and using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed, it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods, and that each such combination is specifically contemplated and should be considered disclosed.

(25) References in the specification and concluding claims to parts by weight, of a particular element or component in a composition or article, denote the weight relationship between the element or component and any other elements or components in the composition or article for which a part by weight is expressed. Thus, in a compound containing 2 parts by weight of component X and 5 parts by weight of component Y, X and Y are present at a weight ratio of 2:5, and are present in such ratio regardless of whether additional components are contained in the compound.

(26) A weight percent of a component, unless specifically stated to the contrary, is based on the total weight of the formulation or composition in which the component is included.

(27) As used herein, a “genelight extract” is an extract produced by the biological devices disclosed herein. In one aspect, the genelight extract contains the biological devices in either a lysed or a whole cell state. In another aspect, the genelight extract contains proteins encoded by the genes in the DNA constructs disclosed herein, or contains the catalytic products of these proteins, or a combination thereof. In one aspect, the genelight extract is capable of providing power to a light source. In a further aspect, the genelight extract can be used to enhance the growth of plants, including plants grown from tissue culture, or can be used to provide supplemental nutrients for cultures of industrially, commercially, and/or scientifically important microorganisms.

(28) Described herein are genelight cultures and extracts and methods of making and using thereof. In one aspect, the method of making a genelight culture or extract includes the steps of (a) making a DNA construct containing genes for producing a heat shock protein, RuBisCO large subunit 1, tonB, hydrogenase, and a P-type ATPase, (b) introducing the DNA construct into host microbial cells via transformation or transfection, and (c) culturing the microbial cells to produce the genelight cultures and extracts. The cultures are grown in standard media for host cells such as, for example, *E. coli* or *S. cerevisiae*. The compositions of these cultures and extracts can be tailored to have specific properties such as, for example, the ability to provide power to a light emitting diode (LED) with a specific voltage. The cultures and extracts have further uses including enhancing the growth of plants, including plants grown from tissue culture, and as supplemental nutrients of cultures of industrially, commercially, and/or scientifically important microorganisms. The cultures and extracts further have UV-protective and/or UV-blocking properties and can be incorporated into or applied on various materials, surfaces, and human or animal subjects for the purpose of protecting those materials, surfaces, and human or animal subjects from the harmful effects of radiation. Also described herein are microbial electric circuits containing the microbial cultures and extracts described herein as well as applications of those circuits.

(29) DNA Constructs and Biological Devices

(30) DNA constructs are provided herein for the production of genelight extracts. It is understood that one way to define the variants and derivatives of the genetic components and DNA constructs described herein is in terms of homology/identity to specific known sequences. Those of skill in the art readily understand how to determine the homology of two nucleic acids. For example, the homology can be calculated after aligning two sequences so that the homology is at its highest level. Another way of calculating homology can be performed according to published algorithms (see Zuker, M., *Science*, 244:48-52, 1989; Jaeger et al, *Proc. Natl. Acad. Sci. USA*, 86:7706-7710, 1989; Jaeger et al, *Methods Enzymol.*, 183:281-306, 1989, which are herein incorporated by reference for at least material related to nucleic acid alignment).

(31) As used herein, "conservative" mutations are mutations that result in an amino acid change in the protein produced from a sequence of DNA. When a conservative mutation occurs, the new amino acid has similar properties as the wild type amino acid and generally does not drastically change the function or folding of the protein (e.g., switching isoleucine for valine is a conservative mutation since both are small, branched, hydrophobic amino acids). "Silent mutations," meanwhile, change the nucleic acid sequence of a gene encoding a protein but do not change the amino acid sequence of the protein.

(32) It is understood that the description of mutations and homology can be combined together in any combination, such as embodiments that have at least about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, or about 99% homology to a particular sequence wherein the variants are conservative or silent mutations. It is understood that any of the sequences described herein can be a variant or derivative having the homology values listed above.

(33) In one aspect, a database such as, for example, GenBank, can be used to determine the sequences of genes and/or regulatory regions of interest, the species from which these elements originate, and related homologous sequences.

(34) In one aspect, provided herein are DNA constructs having at least the following genetic components: a) a gene that expresses a heat shock protein; b) a gene that expresses RuBisCO large subunit 1; c) a gene that expresses tonB; d) a gene that expresses hydrogenase; and e) a gene that expresses P-type ATPase.

(35) Each component of the DNA constructs are described in detail below.

(36) In one aspect, the nucleic acids (e.g., genes that express a heat shock protein, RuBisCO large subunit 1, tonB, hydrogenase, and P-type ATPase) used in the DNA constructs described herein can be amplified using polymerase chain reaction (PCR) prior to being ligated into a plasmid or other vector. Typically, PCR-amplification techniques make use of primers, or short, chemically-synthesized oligonucleotides that are complementary to regions on each respective strand flanking the DNA or nucleotide sequence to be amplified. A person having ordinary skill in the art will be able to design or choose primers based on the desired experimental conditions. In general, primers should be designed to provide for both efficient and faithful replication of the target nucleic acids. Two primers are required for the amplification of each gene, one for the sense strand (that is, the strand containing the gene of interest) and one for the anti sense strand (that is, the strand complementary to the gene of interest). Pairs of primers should have similar melting temperatures that are close to the PCR reaction's annealing temperature. In order to facilitate the PCR reaction, the following features should be avoided in primers: mononucleotide repeats, complementarity with other primers in the mixture, self-complementarity, and internal hairpins and/or loops. Methods of primer design are known in the art; additionally, computer programs exist that can assist the skilled practitioner with primer design. Primers can optionally incorporate restriction enzyme recognition sites at their 5′ ends to assist in later ligation into plasmids or other vectors.

(37) PCR can be carried out using purified DNA, unpurified DNA that is integrated into a vector, or unpurified genomic DNA. The process for amplifying target DNA using PCR consists of introducing an excess of two primers having the characteristics described above to a mixture containing the sequence to be amplified, followed by a series of thermal cycles in the presence of a heat-tolerant or thermophilic DNA polymerase, such as, for example, any of Taq, Pfu, Pwo, Tfl, rTth, Tli, or Tma polymerases. A PCR "cycle" involves denaturation of the DNA through heating, followed by annealing of the primers to the target DNA, followed by extension of the primers using the thermophilic DNA polymerase and a supply of deoxynucleotide triphosphates (i.e., dCTP, dATP, dGTP, and TTP), along with buffers, salts, and other reagents as needed. In one aspect, the DNA segments created by primer extension during the PCR process can serve as templates for additional PCR cycles. Many PCR cycles can be performed to generate a large concentration of target DNA or genes. PCR can optionally be performed in a device or machine with programmable temperature cycles for denaturation, annealing, and extension steps. Further, PCR can be performed

on multiple genes simultaneously in the same reaction vessel or microcentrifuge tube since the primers chosen will be specific to selected genes. PCR products can be purified by techniques known in the art such as, for example, gel electrophoresis followed by extraction from the gel using commercial kits and reagents.

(38) In a further aspect, the plasmid can include an origin of replication, allowing it to use the host cell's replication machinery to create copies of itself.

(39) As used herein, “operably linked” refers to the association of nucleic acid sequences on a single nucleic acid fragment so that the function of one affects the function of another. For example, if sequences for multiple genes are inserted into a single plasmid, their expression may be operably linked. Alternatively, a promoter is said to be operably linked with a coding sequence when it is capable of affecting the expression of that coding sequence.

(40) As used herein, “expression” refers to transcription and/or accumulation of an mRNA derived from a gene or DNA fragment. Expression may also be used to refer to translation of mRNA into a peptide, polypeptide, or protein.

(41) In one aspect, the DNA constructs disclosed herein include a gene for a heat shock protein. Heat shock proteins are a group of proteins that is produced by cells in response to exposure to stressful conditions. In a further aspect, heat shock proteins can be expressed in response to heat shock but also to cold, UV light, wound healing, exposure to toxic chemicals such as, for example heavy metals, including but not limited to arsenic, cadmium, copper, mercury, and the like, as well as during tissue remodeling. In a further aspect, a heat shock protein may function as a chaperone protein by assisting in the refolding process of proteins damaged by cell stress. In a further aspect, the heat shock protein can be HSP60, HSP70, or HSP90, where the number refers to the size in kilodaltons of the protein. In one aspect, the heat shock protein is HSP70 and has a weight of about 70 kDa.

(42) In a further aspect, the gene that expresses a heat shock protein expresses HSP70. In another aspect, the gene that expresses a heat shock protein is isolated from a fungus. In a still further aspect, the fungus is a yeast such as, for example, *Saccharomyces cerevisiae*. Further in this aspect, the *S. cerevisiae* strain can be strain ySR128, Y169, SY14, BY4742, CEN.PK113-7D, S288c, YJM1326, ySR127, S288C, EC1118, Makgeolli, UWOPS03-461.4, DBVPG6765, YJM1447, YJM981, YJM627, YJM1401, YJM1356, YJM978, YJM320, YJM1527, YJM1355, YJM554, HB_S_GIMBLETTROAD_16, WI_S_OAKURA_4, HB_S_BILANCHER_6, HB_S_GIMBLETTROAD_14, T52, T52_5E HB_S_GIMBLETTROAD_22, HB_C_OMARUNUI_7, HB_C_TUKITUKI1_16, HB_C_TUKITUKI2_10, WI_S_JASA_13, WI_S_JASA_5, HB_C_OMARUNUI_14, WA_C_MATES_13, WI_C_MBSP_4, WA_C_CODDINGTON_2, T8, Soil7-1, WA_C_MATES_10, HB_S_GIMBLETTROAD_5, HCNKIsf_G7, HPRMIsf_H7, T16, MARARsf_A10, MTKSKsf_E2, WSETAwf_B1, HB_S_GIMBLETTROAD_9, TNPLST-4-S-2, CDRDR_sf_H, CRIRIwf_A11, T78, NSERVsf_F8, HB_C_TUKITUKI2_4, T.52_3A, YJM1526, YJM969, YJM1478, YJM1338, YJM993, YJM1477, YJM1387, YJM453, YJM1242, YJM683, YJM987, YJM450, UOA_M2, YJM1415, KSD-Yc, YPS128, YJM1573, YJM1190, YJM555, YJM1400, YJM1273, T63, HB_S_BILANCHER_12, HB_C_KOROKIPO_12, Sol7-2, HCNTHsf_F8, HB_C_KOROKIPO_3, HB_C_OMARUNUI_6, WA_C_WAITAKEREROAD_7, HCNTHsf_C5, T.52_5A, WA_C_KINGSMILL_10, NSEBRsf_A9, HPRMAwf_D10, T.52_2H, T.52_3C, YJM1129, WSERCsf_G4, or another *S. cerevisiae* strain. In a further aspect, the gene that expresses a heat shock protein has SEQ ID NO. 1 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(43) Other sequences expressing a heat shock protein or related or homologous genes can be identified in a database such as, for example, GenBank. In one aspect, the gene that expresses a heat shock protein is isolated from *S. cerevisiae* strain ySR128 having GI number CP036483.1 in the GenBank database. In one aspect, sequences useful herein include those with GI numbers listed

in Table 1:
(44) TABLE-US-00001 TABLE 1 Heat Shock Protein Genes Source Organism Sequence Description GI Number *Saccharomyces cerevisiae* chromosome IV sequence CP036483.1 *Saccharomyces cerevisiae* chromosome IV sequence CP033473.1 *Saccharomyces cerevisiae* chromosome I sequence CP029160.1 *Saccharomyces cerevisiae* chromosome IV sequence CP029160.1 *Saccharomyces cerevisiae* chromosome IV sequence CP029160.1 *Saccharomyces cerevisiae* chromosome IV sequence CP029160.1 *Saccharomyces cerevisiae* chromosome IV sequence CP029160.1 *Saccharomyces cerevisiae* chromosome IV sequence CP026298.1 *Saccharomyces cerevisiae* chromosome IV sequence CP022969.1 *Saccharomyces cerevisiae* HSP70 family CP020126.1 *Saccharomyces cerevisiae* chromosome IV sequence CP004710.2 *Saccharomyces cerevisiae* chromosome IV sequence CP011550.1 *Saccharomyces cerevisiae* heat shock cognate gene BK006938.2 *Saccharomyces cerevisiae* genomic DNA NM_001180289.1 *Saccharomyces cerevisiae* chromosome IV sequence FN393064.1 *Saccharomyces cerevisiae* chromosome IV sequence Z74277.1 *Saccharomyces cerevisiae* chromosome I sequence X13713.1 *Saccharomyces cerevisiae* chromosome IV sequence EF058944.1 *Saccharomyces cerevisiae* chromosome IV sequence CP025100.1 *Saccharomyces cerevisiae* chromosome IV sequence CP020228.1 *Saccharomyces cerevisiae* chromosome IV sequence CP020160.1 *Saccharomyces cerevisiae* chromosome IV sequence CP004738.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004688.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004678.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004727.2 *Saccharomyces cerevisiae* chromosome I sequence CP004717.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004687.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004667.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004746.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004716.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004676.2 *Saccharomyces cerevisiae* chromosome IV sequence CP008239.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008324.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008273.1 *Saccharomyces cerevisiae* chromosome I sequence CP008256.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008222.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008409.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008392.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008375.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008358.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008341.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008494.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008443.1 *Saccharomyces cerevisiae* chromosome I sequence CP008579.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008511.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008647.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008630.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008596.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008188.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008171.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008154.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008681.1 *Saccharomyces cerevisiae* chromosome I sequence CP008137.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008120.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008086.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008052.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008035.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008001.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007984.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007950.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007899.1 *Saccharomyces cerevisiae* chromosome I sequence CP007882.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007831.1 *Saccharomyces cerevisiae* chromosome IV sequence CP004745.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004684.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004743.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004713.2 *Saccharomyces cerevisiae*

chromosome IV sequence CP004692.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004742.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004722.2 *Saccharomyces cerevisiae* chromosome I sequence CP004672.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004701.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004681.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004690.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004670.2 *Saccharomyces cerevisiae* chromosome IV sequence CP011082.1 *Saccharomyces cerevisiae* chromosome IV sequence CP004729.1 *Saccharomyces cerevisiae* chromosome IV sequence M25395.1 *Saccharomyces cerevisiae* chromosome IV sequence CP023998.1 *Saccharomyces cerevisiae* chromosome I sequence CP020211.1 *Saccharomyces cerevisiae* chromosome IV sequence CP004748.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004697.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004677.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004726.2 *Saccharomyces cerevisiae* chromosome IV sequence CP004706.2 *Saccharomyces cerevisiae* chromosome IV sequence CP008307.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008290.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008477.1 *Saccharomyces cerevisiae* chromosome I sequence CP008460.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008426.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008562.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008664.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008613.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008103.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008069.1 *Saccharomyces cerevisiae* chromosome IV sequence CP008018.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007967.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007933.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007916.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007865.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007848.1 *Saccharomyces cerevisiae* chromosome IV sequence CP007814.1 *Saccharomyces cerevisiae* chromosome IV sequence CP004695.2 *Saccharomyces cerevisiae* chromosome IV sequence CP008205.1

(45) In one aspect, the DNA constructs disclosed herein include a gene that expresses ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO) large subunit 1. RuBisCO is an enzyme involved in carbon fixation and can be isolated from plants, algae, cyanobacteria, and phototrophic and/or chemoautotrophic proteobacteria. In a further aspect, the RuBisCO large subunit is typically encoded by chloroplast DNA. In a still further aspect, RuBisCO typically catalyzes the formation of two molecules of glycerate-3-phosphate from ribulose-1,5-bisphosphate and carbon dioxide. In an alternative aspect, RuBisCO is capable of catalyzing the formation of phosphoglycolate and 3-phosphoglycerate from ribulose-1,5-bisphosphate and molecular oxygen.

(46) In one aspect, the gene that expresses RuBisCO large subunit 1 is isolated from an alga. In another aspect, the alga can be selected from *Guillardia theta*, *Storeatula* sp. CCMP1668, Hanusia phi, a *Rhodomonas* species, *Teleaulax amphioxeia, a Cryptomonas* species, *Pyrenomonas helgolandii*, a *Botrydium* species, a *Xanthonema* species, *Chrysoparadoxa australica, Ophiocytium majus, Chroomonas* sp. SAG 980-1, *Porphyridium aerugineum*, a *Bumilleria* species, a *Hemiselmis* species, a *Bumilleriopsis* species, *Ophiocytium capitatum, Erythrotrichia carnea, Tribonema* sp. UTEX 639, *Heterothrix debilis, Mischococcus sphaerocephalus*, a *Tribonema* species, *Ophiocytium parvulum, Porphyridium sordidum, Nitzschia* sp. SZCZP 1124, or *Exanthemachrysis gayraliae*. In a further aspect, the gene that expresses RuBisCO large subunit 1 is isolated from a plastid or chloroplast present in these or other algae. In a still further aspect, the gene that expresses RuBisCO large subunit 1 has SEQ ID NO. 2 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(47) Other sequences expressing RuBisCO large subunit 1 or related or homologous genes can be identified in a database such as, for example, GenBank. In one aspect, the gene that expresses

RuBisCO large subunit 1 is isolated from *Guillardia theta* and has GI number AF041468.1 in the GenBank database. In another aspect, sequences useful herein include those with the GI numbers listed in Table 2:

(48) TABLE-US-00002 TABLE 2 RuBisCO Large Subunit 1 Genes Sequence Source Organism Description GI Number *Guillardia theta* plastid gene AF041468.1 *Guillardia theta* RuBisCO large subunit MK818447.1 *Guillardia theta* chloroplast gene KT428890.1 *Storeatula* sp. CCMP 1668 plastid gene KY856940.1 Hanusia phi RuBisCO large subunit KX777654.1 *Rhodomonas salina* chloroplast gene EF508371.1 *Teleaulax amphioxeia* plastid gene KP899713.1 *Cyrptomonas curvata* plastid gene KY856939.1 *Chroomonas placoidea* plastid gene KY856941.1 *Chroomonas mesostigmatica* chloroplast gene KY860574.1 *Cryptomonas pyrenoidifera* RuBisCO large subunit AM051217.1 *Cryptomonas marssonii* RuBisCO large subunit AM051209.1 *Cryptomonas tetrapyrenoidosa* RuBisCO large subunit AM051219.1 *Cryptomonas* sp. CCAP 979/46 RuBisCO large subunit AM051223.1 *Cryptomonas pyrenoidifera* RuBisCO large subunit AM051216.1 *Cryptomonas* sp. CCAC 0109 RuBisCO large subunit AM051222.1 *Cryptomonas curvata* RuBisCO large subunit AM051204.1 *Cyrptomonas tetrapyrenoidosa* RuBisCO large subunit AM051220.1 *Cryptomonas curvata* RuBisCO large subunit AM051205.1 *Cryptomonas* sp. CCAC 0031 RuBisCO large subunit AM051221.1 *Cryptomonas ovata* RuBisCO large subunit AM051210.1 *Pyrenomonas helgolandii* RuBisCO large subunit AY119782.1 *Botrydium stoloniferum* RuBisCO large subunit AF465707.1 *Botrydium stoloniferum* RuBisCO large subunit EF455981.1 *Xanthonema muciculum* RuBisCO large subunit EF455957.1 Hanusia phi RuBisCO large subunit MK818448.1 *Cryptomonas ovata* RuBisCO large subunit AM051211.1 *Chrysoparadoxa australica* RuBisCO large subunit MK189080.1 *Botrydium granulatum* RuBisCO large subunit EF455980.1 *Cryptomonas* sp. M0741 RuBisCO large subunit AM051224.1 *Rhodomonas* sp. isolate CCMP 757 RuBisCO large subunit KX777650.1 *Xanthonema* sp. SAG 60.94 RuBisCO large subunit EF455977.1 *Porphyridium aerugineum* RuBisCO large subunit X17597.1 *Xanthonema* sp. SAG 2192 RuBisCO large subunit EF426794.1 *Xanthonema* cf. RuBisCO large subunit EF455953.1 *hormidioides* strain SAG 2194 *Ophiocytium majus* RuBisCO large subunit EF455971.1 *Chroomonas* sp. SAG 980-1 RuBisCO large subunit AY119781.1 *Xanthonema* sp. CCAP 836/5 RuBisCO large subunit EF455940.1 *Xanthonema exile* RuBisCO large subunit EF455929.1 *Xanthonema bristolianum* RuBisCO large subunit EF455955.1 *Xanthonema* sp. SAG 2189 RuBisCO large subunit EF455954.1 *Xanthonema* sp. SAG 2179 RuBisCO large subunit EF426796.1 *Xanthonema debile* RuBisCO large subunit EF455975.1 *Xanthonema hormidioides* RuBisCO large subunit EF455939.1 *Bumilleria exilis* RuBisCO large subunit EF455937.1 *Xanthonema hormidioides* RuBisCO large subunit EF455922.1 *Cryptomonas gyropyrenoidosa* RuBisCO large subunit AM051206.1 *Hemiselmis* sp. isolate SUR21-C3 RuBisCO large subunit KX777660.1 *Hemiselmis pacifica* RuBisCO large subunit KX777646.1 *Hemiselmis rufescens* RuBisCO large subunit KX777647.1 *Bumilleriopsis* sp. SAG 33.93 RuBisCO large subunit EF431849.1 *Xanthonema solidum* RuBisCO large subunit EF455973.1 *Xanthonema sessile* RuBisCO large subunit EF455935.1 *Bumilleriopsis* sp. SAG 57.94 RuBisCO large subunit EF431850.1 *Bumilleriopsis peterseniana* RuBisCO large subunit EF455942.1 *Xanthonema mucicolum* RuBisCO large subunit AJ874332.1 *Botrydium cystosum* RuBisCO large subunit AF465708.1 *Botrydium stoloniferum* RuBisCO large subunit AF064743.1 *Rhodomonas salina* RuBisCO large subunit MK818469.1 *Botrydium becharianum* RuBisCO large subunit EF455979.1 *Ophiocytium capitatum* RuBisCO large subunit EF455959.1 *Xanthonema* sp. SAG 2184 RuBisCO large subunit EF455931.1 *Xanthonema* sp. 907 RuBisCO large subunit AJ874714.1 *Botrydium becharianum* RuBisCO large subunit AF465706.1 *Erythrotrichia carnea* plastid gene NC_031176.2 *Erythrotrichia carnea* plastid gene KX284721.2 *Tribonema* sp. UTEX 639 RuBisCO large subunit EF455921.1 *Bumilleria* sp. CCAP 806/3 RuBisCO large subunit EF455938.1 *Heterothrix debilis* RuBisCO large subunit EF455920.1 *Ophiocytium majus* RuBisCO large subunit AJ874699.1 *Rhodomonas lens* RuBisCO large subunit MK818450.1 *Hemiselmis andersenii*

RuBisCO large subunit KX777655.1 *Xanthonema* sp. SAG 2183 RuBisCO large subunit EF455930.1 *Mischococcus sphaerocephalus* RuBisCO large subunit EF455972.1 *Xanthonema bristolianum* RuBisCO large subunit AJ874331.1 *Xanthonema* cf. *debile* RuBisCO large subunit EF455932.1 *Bumilleria* sp. SAG2160 RuBisCO large subunit EF460494.1 *Bumilleriopsis* sp. SAG 58.94 RuBisCO large subunit EF455963.1 *Ophiocytium parvulum* RuBisCO large subunit EF455961.1 *Xanthonema debile* RuBisCO large subunit EF455933.1 *Porphyridium sordidum* plastid gene KX284720.1 *Xanthonema debile* RuBisCO large subunit AF084612.1 *Hemiselmis rufescens* RuBisCO large subunit MK818472.1 *Tribonema viride* RuBisCO large subunit EF460493.1 *Bumilleria klebsiana* RuBisCO large subunit EF460492.1 *Bumilleria* sp. SAG 2157 RuBisCO large subunit EF426792.1 *Tribonema regulare* RuBisCO large subunit EF455928.1 *Xanthonema* sp. 773 RuBisCO large subunit AJ874713.1 *Bumilleriopsis* sp. SAG 33.93 RuBisCO large subunit AJ874706.1 *Botrydium granulatum* RuBisCO large subunit AJ874698.1 *Xanthonema sessile* RuBisCO large subunit AJ874329.1 *Tribonema intermixum* RuBisCO large subunit AF465709.1 *Nitzschia* sp. SZCZP1124 RuBisCO large subunit LC385875.1 *Cryptomonas pyrenoidifera* RuBisCO large subunit MK818480.1 *Exanthemachrysis gayraliae* RuBisCO large subunit AB043701.1 *Tribonema intermixum* RuBisCO large subunit EF460491.1 *Tribonema* sp. SAG 2176 RuBisCO large subunit EF455950.1 *Tribonema viride* RuBisCO large subunit EF455966.1 *Cyrptomonas rostratiformis* RuBisCO large subunit MK818486.1 *Rhodomonas abbreviata* RuBisCO large subunit MK818476.1

(49) In one aspect, the DNA constructs disclosed herein incorporate a gene that expresses tonB. TonB is a beta barrel protein found in the outer membrane of gram-negative bacteria. In a still further aspect, tonB proteins are involved with the uptake and/or transport of large substrates including iron siderophore complexes, heme, and other chelated forms of iron.

(50) In one aspect, the gene that expresses tonB is isolated from a bacterium such as, for example, *Enterococcus faecalis, Stenotrophomonas rhizophila*, or a *Pseudomonas* species. In another aspect, the *Pseudomonas* species is selected from *P. entomophila, P*. sp. CCOS191, *P. plecoglossicida, P. mosselii, P. putida, P*. sp. PONIH3, *P. soli, P*. sp. LTGT-11-2Z, *P. monteilii, P*. sp. DRA525, *P. syringae, P. avellanae, P. chloroaphis, P*. sp. GR 6-02, *P. frederiksbergensis, P*. sp. ATCC 43928, *P. chloroaphis, P. parafulva, P. fulva, P*. sp. BJP69, *P*. sp. Leaf58, *P*. sp. SW144, *P*. sp. VLB120, *P*. sp. SW16, *P*. sp. 02C 26, *P*. sp. URMO17WK12:I11, *P*. sp. SW17, *P*. sp. SGair0191, *P. cremoricolorata, P*. sp. SNU WTI, *P*. aklylphenolica, *P*. sp. DG56-2, *P*. sp. HLS-6, *P. rhizosphaerae, P. viridiflava, P. cichorii, P*. sp. MRSN12121, or *P. fluorescens*. In a further aspect, the gene that expresses tonB has SEQ ID NO. 3 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(51) Other sequences expressing tonB or related or homologous genes can be identified in a database such as, for example, GenBank. In one aspect, the gene that expresses tonB is isolated from *Pseudomonas entomophila* and can be identified by the GI number CT573326.1 in the GenBank database. In another aspect, sequences useful herein include those with GI numbers listed in Table 3:

(52) TABLE-US-00003 TABLE 3 TonB Genes Sequence Source Organism Description GI Number *Pseudomonas entomophila* chromosomal DNA CT573326.1 *Pseudomonas* sp. CCOS191 chromosomal DNA LN847264.1 *Pseudomonas entomophila* genomic DNA CP034337.1 *Pseudomonas entomophila* genomic DNA CP034338.1 *Pseudomonas plecoglossicida* genomic DNA CP031146.1 *Pseudomonas mosselii* genomic DNA CP024159.1 *Pseudomonas mosselii* genomic DNA CP023299.1 *Pseudomonas putida* genomic DNA CP014343.1 *Pseudomonas* sp. PONIH3 genomic DNA CP026386.1 *Pseudomonas soli* genomic DNA CP009365.1 *Pseudomonas putida* genomic DNA CP018846.1 *Pseudomonas putida* genomic DNA CP010979.1 *Pseudomonas* sp. LTGT-11-2Z genomic DNA CP033104.1 *Pseudomonas monteilii* genomic DNA CP022562.1 *Pseudomonas* sp. DRA525 genomic DNA CP018743.1 *Pseudomonas putida* genomic DNA

CP003738.1 *Pseudomonas monteilii* genomic DNA CP014062.1 *Pseudomonas putida* genomic DNA AP013070.1 *Pseudomonas* sp. FGI182 genomic DNA CP007012.1 *Pseudomonas putida* genomic DNA CP002870.1 *Pseudomonas putida* genomic DNA CP000926.1 *Pseudomonas* sp. XWY-1 genomic DNA CP026332.1 *Pseudomonas* sp. JY-Q genomic DNA CP011525.1 *Pseudomonas putida* genomic DNA CP003734.1 *Pseudomonas putida* TonB AF315582.1 *Pseudomonas putida* genomic DNA CP017073.1 *Pseudomonas* sp. KBS0802 genomic DNA CP042180.1 *Pseudomonas putida* genomic DNA CP022561.1 *Pseudomonas putida* chromosomal DNA LR134299.1 *Pseudomonas* sp. SWI36 genomic DNA CP026675.1 *Pseudomonas putida* chromosomal DNA LT799039.1 *Pseudomonas putida* genomic DNA AE015451.2 *Pseudomonas plecoglossicida* genomic DNA CP010359.1 *Pseudomonas putida* genomic DNA CP007620.1 *Pseudomonas monteilii* genomic DNA CP006979.1 *Pseudomonas monteilii* genomic DNA CP006978.1 *Pseudomonas putida* chromosomal DNA LT707061.1 *Pseudomonas putida* genomic DNA CP016212.1 *Pseudomonas putida* genomic DNA CP015876.1 *Pseudomonas putida* genomic DNA CP015202.1 *Pseudomonas putida* genomic DNA CP009974.1 *Pseudomonas putida* genomic DNA CP003588.1 *Pseudomonas putida* genomic DNA CP002290.1 *Pseudomonas putida* genomic DNA CP000712.1 *Stenotrophomonas rhizophila* genomic DNA CP031729.1 *Pseudomonas putida* genomic DNA AP015029.1 *Pseudomonas putida* genomic DNA CP005976.1 *Pseudomonas* sp. BJP69 genomic DNA CP041933.1 *Pseudomonas putida* genomic DNA CP030750.1 *Pseudomonas putida* genomic DNA CP011789.1 *Pseudomonas* sp. Leaf58 genomic DNA CP032677.1 *Pseudomonas putida* genomic DNA CP024086.1 *Pseudomonas putida* genomic DNA CP024085.1 *Pseudomonas putida* TonB X70139.1 *Pseudomonas putida* genomic DNA CP016634.1 *Pseudomonas* sp. SWI44 genomic DNA CP026674.1 *Pseudomonas* sp. VLB120 genomic DNA CP003961.1 *Pseudomonas* sp. SWI6 genomic DNA CP026676.1 *Pseudomonas putida* genomic DNA CP039371.1 *Pseudomonas parafulva* genomic DNA CP031641.1 *Pseudomonas parafulva* genomic DNA CP009747.1 *Pseudomonas putida* genomic DNA CP000949.1 *Pseudomonas* sp. 02C 26 chromosomal DNA CP025262.1 *Pseudomonas fulva* genomic DNA CP014025.1 *Pseudomonas* sp. UPMO17WK12:I11 chromosomal DNA LN865164.1 *Pseudomonas fulva* genomic DNA CP023048.1 *Pseudomonas* sp. SWI7 genomic DNA CP040930.1 *Pseudomonas* sp. SGair0191 genomic DNA CP025035.2 *Enterococcus faecalis* genomic DNA CP022312.1 *Pseudomonas parafulva* genomic DNA CP019952.1 *Pseudomonas cremoricolorata* genomic DNA CP009455.1 *Pseudomonas monteilii* genomic DNA CP013997.1 *Pseudomonas* sp. SNU WT1 genomic DNA CP035952.1 *Pseudomonas alkylphenolica* genomic DNA CP009048.1 *Pseudomonas* sp. DG56-2 genomic DNA CP032311.1 *Pseudomonas* sp. HLS-6 genomic DNA CP024478.1 *Pseudomonas rhizosphaerae* genomic DNA CP009533.1 *Pseudomonas viridiflava* chromosomal DNA LT855380.1 *Pseudomonas cichorii* genomic DNA CP007039.1 *Pseudomonas chloroaphis* genomic DNA CP027707.1 *Pseudomonas chloroaphis* chromosomal DNA LT629761.1 *Pseudomonas* sp. MRSN12121 genomic DNA CP010892.1 *Pseudomonas fluorescens* genomic DNA CP012830.1 *Pseudomonas frederiksbergensis* genomic DNA CP017886.1 *Pseudomonas chloroaphis* genomic DNA CP011110.1 *Pseudomonas chloroaphis* genomic DNA CP027744.1 *Pseudomonas* sp. ATCC 43928 genomic DNA CP041753.1 *Pseudomonas frederiksbergensis* genomic DNA CP023466.1 *Pseudomonas* sp. GR 6-02 genomic DNA CP011567.1 *Pseudomonas chloroaphis* genomic DNA CP027718.1 *Pseudomonas chloroaphis* genomic DNA CP027717.1 *Pseudomonas chloroaphis* genomic DNA CP027746.1 *Pseudomonas chloroaphis* genomic DNA CP027743.1 *Pseudomonas syringae* genomic DNA CP032631.1 *Pseudomonas syringae* genomic DNA CP032871.1 *Pseudomonas avellanae* genomic DNA CP026562.1 *Pseudomonas syringae* chromosomal DNA LT963408.1 *Pseudomonas syringae* genomic DNA CP024712.1 *Pseudomonas syringae* genomic DNA CP019732.1 *Pseudomonas syringae* genomic DNA CP019730.1

(53) In one aspect, the DNA constructs disclosed herein incorporate a gene that expresses hydrogenase. Hydrogenase is a protein that catalyzes the reversible oxidation of molecular

hydrogen. Numerous hydrogenases are recognized including [NiFe] hydrogenases, [NiFeSe] hydrogenases, [FeFe] hydrogenases, and [Fe]-only hydrogenases, where the chemical symbols in brackets indicate the metal ions at the catalytic centers of the protein.

(54) In one aspect, the gene that expresses hydrogenase is isolated from a bacterium. In another aspect, the bacterium can be selected from *Burkholderia pseudomallei, Burkholderia thailandensis, Burkholderia oklahomensis*, or another *Burkholderia* species, *Acidithiobacillus ferridurans, Acidithiobacillus ferrivorans, Acidithiobacillus ferrooxidans, Acidithiobacillus caldus*, a *Paraburkholderia* species, a *Leptospirillum* species, a *Thiomonas* species, *Rhodoferax antarcticus, Azoarcus aromaticum, Sulfuricella denitrificans, Variovorax* sp. HS608, *Candidatus symbiobacter mobilis, Azoarcus* sp. DN11, *Acidiferrobacter* sp. SPIII_3, *Rugosibacter aromaticivorans, Cupriavidus metallidurans, Sideroxydans lithotrophicus, Ralstonia syzygii, Rhodoferax ferrireducens, Sulfuritalea hydrogenivorans, Azoarcus* sp. KH32C, *Polaromonas* sp. JS666, *Cupriavidus oxalaticus, Ralstonia eutropha, Ralstonia pickettii, Cupriavidus necator, Variovorax paradoxus, Gallionella capsiferriformans, Burkholderiales bacterium* GJ-E10, or an uncultured bacterium. In a further aspect, the gene that expresses hydrogenase has SEQ ID NO. 4 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(55) Other sequences expressing hydrogenase or related or homologous genes can be identified in a database such as, for example, GenBank. In one aspect, the gene that expresses hydrogenase is isolated from *Acidithiobacillus ferridurans* and can be identified by the GI number AP018795.1 in the GenBank database. In another aspect, sequences useful herein include those with the GI numbers listed in Table 4:

(56) TABLE-US-00004 TABLE 4 Hydrogenase Genes Sequence Source Organism Description GI Number *Acidithiobacillus ferridurans* genomic DNA AP018795.1 *Acidithiobacillus ferrivorans* genomic DNA LT841305.1 *Acidithiobacillus ferrivorans* genomic DNA CP002985.1 *Acidithiobacillus ferrooxidans* genomic DNA CP001219.1 *Acidithiobacillus ferrooxidans* genomic DNA CP001132.1 *Acidithiobacillus caldus* genomic DNA CP005986.1 *Acidithiobacillus caldus* genomic DNA CP002573.1 *Acidithiobacillus caldus* genomic DNA CP026328.1 *Cupriavidus necator* genomic DNA CP002879.1 *Variovorax paradoxus* genomic DNA CP003912.1 *Gallionella capsiferriformans* genomic DNA CP002159.1 Uncultured bacterium CRISPR-Cas KU516114.1 system sequence *Burkholderia* sp. HB1 genomic DNA CP012193.1 *Burkholderiales bacterium* GJ-E10 genomic DNA AP014683.1 Uncultured bacterium genomic DNA KX576128.1 *Cupriavidus necator* genomic DNA CP039287.1 *Paraburkholderia xenovorans* genomic DNA CP008761.1 *Ralstonia pickettii* genomic DNA CP006667.1 *Ralstonia eutropha* genomic DNA AM260479.1 *Paraburkholderia xenovorans* genomic DNA CP000272.1 *Cupriavidus oxalaticus* genomic DNA CP038636.1 *Paraburkholderia terricola* genomic DNA CP024941.1 *Polaromonas* sp. JS666 genomic DNA CP000316.1 *Paraburkholderia* genomic DNA CP022991.1 *aromaticivorans Paraburkholderia* sp. 7MH5 plasmid DNA CP038152.1 *Azoarcus* sp. KH32C genomic DNA AP012304.1 *Sulfuritalea hydrogenivorans* genomic DNA AP012547.1 *Paraburkholderia xenovorans* genomic DNA CP008762.1 *Paraburkholderia xenovorans* genomic DNA CP000271.1 *Rhodoferax ferrireducens* genomic DNA CP000267.1 *Ralstonia syzygii* genomic DNA FR854086.1 *Sideroxydans lithotrophicus* genomic DNA CP001965.1 *Cupriavidus metallidurans* megaplasmid DNA CP000353.2 *Paraburkholderia* sp. DCR13 chromosomal DNA CP029641.1 *Rugosibacter aromaticivorans* genomic DNA CP010554.1 *Paraburkholderia hospital* chromosomal DNA CP024939.1 *Acidiferrobacter* sp. SPIII 3 chromosomal DNA CP027663.1 *Paraburkholderia hospital* chromosomal DNA CP026107.1 *Paraburkholderia caribensis* chromosomal DNA CP012747.1 *Azoarcus* sp. DN11 genomic DNA CP021731.1 *Burkholderia thailandensis* chromosomal DNA CP023498.1 *Candidatus symbiobacter mobilis* genomic DNA CP004885.1 *Burkholderia thailandensis* chromosomal DNA CP004096.1 *Paraburkholderia phymatum* plasmid

DNA CP001045.1 *Variovorax* sp. HS608 chromosomal DNA LT607803.1 *Sulfuricella denitrificans* genomic DNA AP013066.1 *Burkholderia* sp. Bp5365 chromosomal DNA CP013382.1 *Burkholderia* sp. 2002721687 chromosomal DNA CP009548.1 *Thiomonas* sp. Str. 3As chromosomal DNA FP475956.1 *Paraburkholderia caribensis* chromosomal DNA CP026103.1 *Paraburkholderia caribensis* chromosomal DNA CP013349.1 *Paraburkholderia caribensis* plasmid DNA CP013104.1 *Thiomonas intermedia* genomic DNA CP002021.1 *Burkholderia oklahomensis* chromosomal DNA CP013359.1 *Burkholderia oklahomensis* chromosomal DNA CP009556.1 *Thiomonas* sp. CB2 genome assembly LK931600.1 scaffold DNA *Azoarcus aromaticum* genomic DNA CR555306.1 *Burkholderia* sp. DHOD12 chromosomal DNA CP040077.1 *Burkholderia thailandensis* chromosomal DNA CP020391.1 *Burkholderia thailandensis* chromosomal DNA CP013410.1 *Burkholderia thailandensis* chromosomal DNA CP004090.1 *Leptospirillum ferriphilum* genomic DNA CP002919.1 *Rhodoferax antarcticus* genomic DNA CP019240.1 *Burkholderia thailandensis* chromosomal DNA CP013361.1 *Burkholderia thailandensis* chromosomal DNA CP008915.2 *Burkholderia thailandensis* chromosomal DNA CP022216.1 *Burkholderia thailandensis* chromosomal DNA CP022215.1 *Burkholderia thailandensis* chromosomal DNA CP020389.1 *Burkholderia thailandensis* chromosomal DNA CP013413.1 *Burkholderia thailandensis* chromosomal DNA CP010018.1 *Burkholderia thailandensis* chromosomal DNA CP009602.1 *Burkholderia thailandensis* chromosomal DNA CP004382.1 *Burkholderia thailandensis* chromosomal DNA CP008786.1 *Burkholderia thailandensis* chromosomal DNA CP004384.1 *Burkholderia thailandensis* chromosomal DNA CP004118.1 *Burkholderia thailandensis* chromosomal DNA CP004098.1 *Burkholderia thailandensis* chromosomal DNA CP000085.1 *Burkholderia* sp. BDU8 chromosomal DNA CP013388.1 *Leptospirillum* sp. Group II CF-1 genomic DNA CP012147.1 *Leptospirillum ferriphilum* genomic DNA CP007243.1 *Burkholderia thailandensis* chromosomal DNA CP013408.1 *Burkholderia thailandensis* chromosomal DNA CP004386.1 *Burkholderia pseudomallei* chromosomal DNA CP041221.1 *Burkholderia pseudomallei* chromosomal DNA CP041219.1 *Burkholderia pseudomallei* chromosomal DNA CP040551.1 *Burkholderia pseudomallei* chromosomal DNA CP040532.1 *Burkholderia pseudomallei* chromosomal DNA CP038806.1 *Burkholderia pseudomallei* chromosomal DNA CP038194.1 *Burkholderia pseudomallei* chromosomal DNA CP037969.1 *Burkholderia pseudomallei* chromosomal DNA CP037971.1 *Burkholderia pseudomallei* chromosomal DNA CP037975.1 *Burkholderia pseudomallei* chromosomal DNA CP037973.1 *Burkholderia pseudomallei* chromosomal DNA CP037757.1 *Burkholderia pseudomallei* chromosomal DNA CP037759.1 *Burkholderia pseudomallei* chromosomal DNA CP036451.1 *Burkholderia pseudomallei* chromosomal DNA CP036453.1 *Burkholderia pseudomallei* chromosomal DNA CP033704.1 *Burkholderia pseudomallei* chromosomal DNA CP033706.1 *Burkholderia pseudomallei* chromosomal DNA CP033701.1 *Burkholderia pseudomallei* chromosomal DNA CP023776.1

(57) In one aspect, the DNA constructs disclosed herein include a gene that expresses a P-type ATPase. P-type ATPases are typically found in bacteria, archaea, and eukaryotes, and function as ion pumps and/or lipid pumps. P-type ATPases are also known as E.sub.1-E.sub.2 ATPases due to their interconversion between two forms (i.e., E.sub.1 and E.sub.2). P-type ATPases have a primarily α-helical structure and harness energy from ATP hydrolysis to transport a ligand across a cell membrane. Numerous P-type ATPases are recognized, typically classified into families based on affinity for particular ions (i.e., potassium, heavy metals, calcium, sodium/potassium, proton/potassium, sodium, proton, magnesium, and/or phospholipids).

(58) In one aspect, the gene that expresses P-type ATPase is isolated from a bacterium such as, for example, *Escherichia coli*. Further in this aspect, the *E. coli* strain is selected from ECOL-18-VL-LA-PA-Ryan-0026, S17-1, KR2009, EK2009, BE104, EC-129, RRL B-1109, PigCaeca_2, PF9285, CJ236, BH212, BH214, BW25113, WCHE025970, NCTC12655, WPB121, WPB102, 4/2-1, ER1709, M217, ECCHD184, W5-6, K12 substr. MG1655, E706, 2452, MT102, 3426, T06, BR32-

DEC, L53, WCHEC03503S1G0, L37, NCTC9107, NCTC0192, C600, SCEC020023, 99-3165, RTdelA_B_UU3, WCHEC005237, WCHEC005784, 51008369SK1, DA33133, VH1, H9, AR435, WCHEC050613, ME8067, J53, 26561, 675SK2, CFSAN064036, APEC 01, DTU-1, 2A, 1A, 2F_0, 4FA, 4A, 5A, 1FA, 3FA, 8A, 5FA, 7A, 7FA, 9FA, 9A, 6FA, 6A, 2_0, 8FA, 2FA, 3A, ECONIH6, LIM, CAR, CIT, SCEC020007, CRE1493, DH5alpha, AR_0015, AR_0011, CREC-532, CCREC-629, WG5, O6:H16 strain M9682-C1, O6:H16 strain F6699, F5656C1, O15:H11 strain 90-9272, O6:H16 strain 2014EL-1346-6, FDAARGOS_434, 1223, NIVEDI_C53, 360/16, O6:H16 strain 2011EL-1370-2, NCTC122, DS1, AT01, or another *E. coli* strain. In a further aspect, the gene that expresses P-type ATPase has SEQ ID NO. 5 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(59) Other sequences expressing P-type ATPase or related or homologous sequences can be identified in a database such as, for example, GenBank. In one aspect, the gene that expresses P-type ATPase is isolated from *E. coli* strain ECOL-18-VL-LA-PA-Ryan-0026 and can be identified in the GenBank database by the GI number CP041392.1. In one aspect, sequences useful herein include those with the GI numbers listed in Table 5:

(60) TABLE-US-00005 TABLE 5 P-Type ATPase Genes Sequence Source Organism Description GI Number *Escherichia coli* genomic DNA CP041392.1 *Escherichia coli* genomic DNA CP040667.1 *Escherichia coli* genomic DNA CP040664.1 *Escherichia coli* genomic DNA CP040663.1 *Escherichia coli* genomic DNA CP040643.1 *Escherichia coli* genomic DNA CP038453.1 *Escherichia coli* genomic DNA CP039753.1 *Escherichia coli* genomic DNA CP038857.1 *Escherichia coli* genomic DNA CP038791.1 *Escherichia coli* genomic DNA CP029238.1 *Escherichia coli* genomic DNA CP029239.1 *Escherichia coli* genomic DNA CP029240.1 *Escherichia coli* genomic DNA CP037857.1 *Escherichia coli* genomic DNA CP036177.1 *Escherichia coli* genomic DNA LR134083.1 *Escherichia coli* genomic DNA CP034426.1 *Escherichia coli* genomic DNA CP034428.1 *Escherichia coli* genomic DNA CP023834.1 *Escherichia coli* genomic DNA CP030240.1 *Escherichia coli* genomic DNA AP019189.1 *Escherichia coli* genomic DNA CP033250.1 *Escherichia coli* genomic DNA CP032992.1 *Escherichia coli* genomic DNA CP032667.1 *Escherichia coli* genomic DNA CP029687.1 *Escherichia coli* genomic DNA CP031833.1 *Escherichia coli* genomic DNA CP034953.2 *Escherichia coli* genomic DNA LS992185.1 *Escherichia coli* genomic DNA LS992166.1 *Escherichia coli* genomic DNA CP035350.1 *Escherichia coli* genomic DNA CP034734.1 *Escherichia coli* genomic DNA CP034595.1 *Escherichia coli* genomic DNA CP034589.1 *Escherichia coli* genomic DNA LR134240.1 *Escherichia coli* genomic DNA LR134227.1 *Escherichia coli* genomic DNA CP031214.1 *Escherichia coli* genomic DNA CP025950.3 *Escherichia coli* genomic DNA CP029981.1 *Escherichia coli* genomic DNA CP023749.1 *Escherichia coli* genomic DNA CP026580.2 *Escherichia coli* genomic DNA CP028578.2 *Escherichia coli* genomic DNA CP029973.1 *Escherichia coli* genomic DNA CP029574.1 *Escherichia coli* genomic DNA CP028704.1 *Escherichia coli* genomic DNA CP029180.1 *Escherichia coli* genomic DNA CP029115.1 *Escherichia coli* genomic DNA CP019213.2 *Escherichia coli* genomic DNA CP028703.1 *Escherichia coli* genomic DNA CP028702.1 *Escherichia coli* genomic DNA CP027118.1 *Escherichia coli* genomic DNA CP027701.1 *Escherichia coli* genomic DNA CP028166.1 *Escherichia coli* genomic DNA CP028310.1 *Escherichia coli* genomic DNA CP027060.1 *Escherichia coli* genomic DNA CP026612.1 *Escherichia coli* genomic DNA CP026358.1 *Escherichia coli* genomic DNA CP026361.1 *Escherichia coli* genomic DNA CP026357.1 *Escherichia coli* genomic DNA CP026352.1 *Escherichia coli* genomic DNA CP026353.1 *Escherichia coli* genomic DNA CP026351.1 *Escherichia coli* genomic DNA CP026360.1 *Escherichia coli* genomic DNA CP026354.1 *Escherichia coli* genomic DNA CP026345.1 *Escherichia coli* genomic DNA CP026350.1 *Escherichia coli* genomic DNA CP026347.1 *Escherichia coli* genomic DNA

CP026346.1 *Escherichia coli* genomic DNA CP026342.1 *Escherichia coli* genomic DNA CP026343.1 *Escherichia coli* genomic DNA CP026348.1 *Escherichia coli* genomic DNA CP026349.1 *Escherichia coli* genomic DNA CP026359.1 *Escherichia coli* genomic DNA CP026344.1 *Escherichia coli* genomic DNA CP026356.1 *Escherichia coli* genomic DNA CP026355.1 *Escherichia coli* genomic DNA CP026199.1 *Escherichia coli* genomic DNA CP026027.1 *Escherichia coli* genomic DNA CP026026.1 *Escherichia coli* genomic DNA CP026028.1 *Escherichia coli* genomic DNA CP025627.1 *Escherichia coli* genomic DNA CP019071.1 *Escherichia coli* genomic DNA CP025520.1 *Escherichia coli* genomic DNA CP025268.1 *Escherichia coli* genomic DNA CP024859.1 *Escherichia coli* genomic DNA CP024855.1 *Escherichia coli* genomic DNA CP024830.1 *Escherichia coli* genomic DNA CP024815.1 *Escherichia coli* genomic DNA CP024090.1 *Escherichia coli* genomic DNA CP024275.1 *Escherichia coli* genomic DNA CP024266.1 *Escherichia coli* genomic DNA CP024260.1 *Escherichia coli* genomic DNA CP024239.1 *Escherichia coli* genomic DNA CP024232.1 *Escherichia coli* genomic DNA CP023870.1 *Escherichia coli* genomic DNA CP023383.1 *Escherichia coli* genomic DNA CP017061.1 *Escherichia coli* genomic DNA CP023201.1 *Escherichia coli* genomic DNA CP022912.1 *Escherichia coli* genomic DNA LT906474.1 *Escherichia coli* genomic DNA CP022466.1 *Escherichia coli* genomic DNA CP022414.1

(61) In one aspect, the DNA construct has the following genetic components: a) a gene that expresses a heat shock protein, b) a gene that expresses RuBisCO large subunit 1, c) a gene that expresses tonB, d) a gene that expresses hydrogenase, and e) a gene that expresses P-type ATPase.

(62) In one aspect, the DNA constructs disclosed herein optionally include a gene that expresses dehydrogenase. The gene that expresses hydrogenase can be positioned before or after any of the genetic components used to produce the constructs described herein. In another aspect, a dehydrogenase catalyzes the removal of hydrogen atoms from a particular molecule, including, but not limited to, molecules involved in the electron transport chain reactions of cellular respiration or in anaerobic and other non-standard energy-generation metabolisms, particularly in conjunction with coenzymes such as, for example, nicotinamide adenine dinucleotide (NAD), flavin adenine dinucleotide (FAD), or flavin mononucleotide (FMN). In a further aspect, dehydrogenases are classified as oxidoreductases. In some aspects, the dehydrogenases disclosed herein are quinoproteins that use pyrrolo-quinoline quinone (also known as methoxatin) as a redox cofactor. In a further aspect, the dehydrogenases disclosed herein can stimulate bacterial growth. In a still further aspect, a dehydrogenase oxidizes its substrate by transferring a hydrogen to an electron acceptor.

(63) In one aspect, the gene that expresses dehydrogenase is isolated from a bacterium such as, for example, *Acidithiobacillus ferrooxidans* or another *Acidithiobacillus* species. In another aspect, the bacterium can be a *Chloroflexi* species, a *Dehalococcoidia* species, a *Microbispora* species, a *Spirochaetes* species, an *Acidobacteria* species, a *Deltaproteobacteria* species, an *Ardenticatena* species, an *Anaerolina* species, an *Arthospira* species, a *Methanothermobacter* species, a *Xanthomonas* species, or another bacterium. In another aspect, the gene that expresses dehydrogenase is isolated from an archaeon. In a further aspect, the archaeon is a *Halorubrum* species, a *Natronorubrum* species, a *Natronobacterium* species, a *Natronolimnobius* species, a *Halopiger* species, a *Thermoplasmata* species, a *Bathyarchaeota* species, a *Halorhabdus* species, a *Halonotius* species, a *Methanolinea* species, a *Thermococcus* species, a *Hadesarchaea* species, a *Halostella* species, a *Salinarchaeum* species, or another archaeon. In one aspect, the bacterium or archaeon is an extremophile such as an acidophile, *halophile*, or thermophile, or a combination thereof. In another aspect, the bacterium or archaeon is autotrophic (such as, for example, a cyanobacterium) or has a non-standard metabolism such as, for example, a methanogenic, organohalide respiration, a sulfate- and/or sulfur-reducing metabolism, or the like. In one aspect, the gene that expresses dehydrogenase is isolated from a primate such as, for example, a human or

a golden snub-nosed monkey, a plant such as, for example, soybean, or a virus such as, for example, Rubella virus or hepatitis C virus. In a further aspect, the gene that expresses dehydrogenase has SEQ ID NO. 9 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(64) Other sequences expressing dehydrogenase or related or homologous sequences can be identified in a database such as, for example, GenBank. In one aspect, the gene that expresses dehydrogenase is isolated from *A. ferrooxidans* and the dehydrogenase protein can be identified in the GenBank database by the GI number CP040511.1. In one aspect, sequences useful herein include those with the GI numbers listed in Table 6:

(65) TABLE-US-00006 TABLE 6 Dehydrogenase Genes Sequence Source Organism Description GI Number *Acdithiobacillus* sp. “AMD genomic DNA CP044411.1 consortium” *Acidithiobacillus ferridurans* genomic DNA AP018795.1 *Acidithiobacillus ferrooxidans* genomic DNA CP040511.1 *Acidithiobacillus ferrooxidans* genomic DNA CP001219.1 *Acidithiobacillus ferrooxidans* genomic DNA CP001132.1 *Sphingomonas* sp. Cra20 genomic DNA CP024924.1 *Pseudomonas rhodesiae* genomic DNA CP054205.1 *Rhinopithecus roxellana* dispatched XM_010353518.2 RND transporter family member 3 *Scophthalmus maximus* genomic DNA CP026246.1 *Rhinopithecus bieti* genomic DNA XM_017866453.1 Rubella virus E1 protein gene KU884920.1 Rubella virus E1 protein gene KU884918.1 *Streptomyces* sp. Jing01 genomic DNA CP053189.1 *Neptunomonas concharum* genomic DNA CP043869.1 *Pigmentiphaga* sp. H8 genomic DNA CP033966.1 *Zobellia galactanivorans* genomic DNA FP476056.1 *Homo sapiens* genomic DNA AC108734.10 *Crassostrea gigas* integrator XM_034450993.1 complex subunit 13 *Crassostrea gigas* integrator XM_034450992.1 complex subunit 13 *Crassostrea gigas* integrator XM_034450991.1 complex subunit 13 *Lutra lutra* genomic DNA LR738418.1 *Aeromonas schubertii* genomic DNA CP039611.1 *Xanthomonas hortorum* genomic DNA CP016878.1 *Xanthomonas hortorum* genomic DNA CP018731.1 *Xanthomonas hortorum* genomic DNA CP018728.1 *Aeromonas schubertii* genomic DNA CP013067.1 *Acomys russatus* genomic DNA LR877214.1 *Xanthomonas hortorum* genomic DNA LR828264.1 *Xanthomonas hortorum* genomic DNA LR828257.1 *Xanthomonas gardneri* genomic DNA LR828253.1 *Xanthomonas cynarae* genomic DNA LR828251.1 Hepatitis C virus polyprotein gene HQ318852.1 Hepatitis C virus polyprotein gene HQ318842.1 Hepatitis C virus polyprotein gene HQ318841.1 *Glycine max* genomic DNA AC235463.1 *Glycine max* genomic DNA AC235121.1 Hepatitis C virus polyprotein AM271579.1 gene, E1-E2 region Hepatitis C virus polyprotein AM271511.1 gene, E1-E2 region Synthetic construct envelope EF043090.1 glycoprotein gene

(66) In another aspect, said construct further includes a) a promoter, b) a terminator or stop sequence, c) a gene that confers resistance to an antibiotic (a “selective marker”), d) a reporter protein, or any combination thereof.

(67) In one aspect, the construct includes a regulatory sequence. In a further aspect, the regulatory sequence is already incorporated into a vector such as, for example, a plasmid, prior to genetic manipulation of the vector. In another aspect, the regulatory sequence can be incorporated into the vector through the use of restriction enzymes or any other technique known in the art.

(68) In one aspect, the regulatory sequence is a promoter. The term “promoter” refers to a DNA sequence capable of controlling the expression of a coding sequence. In another aspect, the coding sequence to be controlled is located 3′ to the promoter. In still another aspect, the promoter is derived from a native gene. In an alternative aspect, the promoter is composed of multiple elements derived from different genes and/or promoters. A promoter can be assembled from elements found in nature, from artificial and/or synthetic elements, or from a combination thereof. It is understood by those skilled in the art that different promoters can direct the expression of a gene in different tissues or cell types, at different stages of development, in response to different environmental or physiological conditions, and/or in different species. In one aspect, the promoter functions as a

switch to activate the expression of a gene.

(69) In one aspect, the promoter is “constitutive.” A constitutive promoter is a promoter that causes a gene to be expressed in most cell types at most times. In another aspect, the promoter is “regulated.” A regulated promoter is a promoter that becomes active in response to a specific stimulus. A promoter may be regulated chemically, such as, for example, in response to the presence or absence of a particular metabolite (e.g., lactose or tryptophan), a metal ion, a molecule secreted by a pathogen, or the like. A promoter also may be regulated physically, such as, for example, in response to heat, cold, water stress, salt stress, oxygen concentration, illumination, wounding, or the like.

(70) Promoters that are useful to drive expression of the nucleotide sequences described herein are numerous and familiar to those skilled in the art. Suitable promoters include, but are not limited to, the following: T3 promoter, T7 promoter, an iron promoter, araBAD promoter, and GAL1 promoter. In a further aspect, the promoter is a native part of the vector used herein. Variants of these promoters are also contemplated. The skilled artisan will be able to use site-directed mutagenesis and/or other mutagenesis techniques to modify the promoters to promote more efficient function. The promoter may be positioned, for example, from 10-100 nucleotides from a ribosomal binding site.

(71) In one aspect, the promoter is a GAL1 promoter. In another aspect, the GAL1 promoter is native to the plasmid used to create the vector. In another aspect, a GAL1 promoter is positioned before the gene that expresses hydrogenase, the gene that expresses P-type ATPase, the gene that expresses tonB, the gene that expresses a heat shock protein, the gene that expresses RuBisCO large subunit 1, or any combination thereof. In another aspect, the promoter is a GAL1 promoter obtained from or native to the pYES2 plasmid.

(72) In another aspect, the promoter is an araBAD promoter. In a further aspect, the araBAD promoter is native to the plasmid used to create the vector. In still another aspect, an araBAD promoter is positioned before the gene that expresses a heat shock protein, the gene that expresses RuBisCO large subunit 1, the gene that expresses tonB, the gene that expresses hydrogenase, the gene that expresses P-type ATPase, or any combination thereof. In one aspect, the araBAD promoter is positioned before the gene that expresses hydrogenase. In another aspect, the promoter is an araBAD promoter obtained from or native to the pBAD plasmid.

(73) In another aspect, the regulatory sequence is a terminator or stop sequence. As used herein, a terminator is a sequence of DNA that marks the end of a gene or operon to be transcribed. In a further aspect, the terminator is an intrinsic terminator or a Rho-dependent transcription terminator. As used herein, an intrinsic terminator is a sequence wherein a hairpin structure can form in the nascent transcript that disrupts the mRNA/DNA/RNA polymerase complex. As used herein, a Rho-dependent transcription terminator requires a Rho factor protein complex to disrupt the mRNA/DNA/RNA polymerase complex. In one aspect, the terminator is an rrnB terminator obtained from or native to the pBAD plasmid. In an alternative aspect, the terminator is a CYC1 terminator obtained from or native to the pYES2 plasmid.

(74) In a further aspect, the regulatory sequence includes both a promoter and a terminator or stop sequence. In a still further aspect, the regulatory sequence can include multiple promoters or terminators. Other regulatory elements, such as enhancers, are also contemplated. Enhancers may be located from about 1 to about 2000 nucleotides in the 5′ direction from the start codon of the DNA to be transcribed, or may be located 3′ to the DNA to be transcribed. Enhancers may be “cis-acting,” that is, located on the same molecule of DNA as the gene whose expression they affect.

(75) In another aspect, the vector contains one or more ribosomal binding sites. As used herein, a “ribosomal binding site” is a sequence of nucleotides located 5′ to the start codon of an mRNA that recruits a ribosome to initiate protein translation. In one aspect, the ribosomal binding site can be positioned before one or more or all genes in the DNA construct, or a before a subset of genes in a DNA construct.

(76) In one aspect, when the vector is a plasmid, the plasmid can also contain a multiple cloning site or polylinker. In a further aspect, the polylinker contains recognition sites for multiple restriction enzymes. The polylinker can contain up to 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more than 20 recognition sites for restriction enzymes. Further, restriction sites may be added, disabled, or removed as required, using techniques known in the art. In one aspect, the plasmid contains restriction sites for any known restriction enzyme such as, for example, HindIII, KpnI, SacI, BamHI, BstXI, EcoRI, BasBI, NotI, XhoI, XphI, XbaI, ApaI, SalI, ClaI, EcoRV, PstI, SmaI, XmaI, SpeI, EagI, SaclI, or any combination thereof. In a further aspect, the plasmid contains more than one recognition site for the same restriction enzyme.
(77) In one aspect, the restriction enzyme can cleave DNA at a palindromic or an asymmetrical restriction site. In a further aspect, the restriction enzyme cleaves DNA to leave blunt ends; in an alternative aspect, the restriction enzyme cleaves DNA to leave “sticky” or overhanging ends. In another aspect, the enzyme can cleave DNA at a distance of from 20 bases to over 1000 bases away from the restriction site. A variety of restriction enzymes are commercially available and their recognition sequences, as well as instructions for use (e.g., amount of DNA needed, precise volumes of reagents, purification techniques, as well as information about salt concentration, pH, optimum temperature, incubation time, and the like) are provided by enzyme manufacturers.
(78) In one aspect, a plasmid with a polylinker containing one or more restriction sites can be digested with one restriction enzyme and a nucleotide sequence of interest can be ligated into the plasmid using a commercially-available DNA ligase enzyme. Several such enzymes are available, often as kits containing all reagents and instructions required for use. In another aspect, a plasmid with a polylinker containing two or more restriction sites can be simultaneously digested with two restriction enzymes and a nucleotide sequence of interest can be ligated into the plasmid using a DNA ligase enzyme. Using two restriction enzymes provides an asymmetric cut in the DNA, allowing for insertion of a nucleotide sequence of interest in a particular direction and/or on a particular strand of the double-stranded plasmid. Since RNA synthesis from a DNA template proceeds from 5′ to 3′, usually starting just after a promoter, the order and direction of elements inserted into a plasmid can be especially important. If a plasmid is to be simultaneously digested with multiple restriction enzymes, these enzymes must be compatible in terms of buffer, salt concentration, and other incubation parameters.
(79) In some aspects, prior to ligation using a ligase enzyme, a plasmid that has been digested with a restriction enzyme is treated with an alkaline phosphatase enzyme to remove 5′ terminal phosphate groups. This prevents self-ligation of the plasmid and thus facilitates ligation of heterologous nucleotide fragments into the plasmid.
(80) In one aspect, different genes can be ligated into a plasmid in one pot. In this aspect, the genes will first be digested with restriction enzymes. In certain aspects, the digestion of genes with restriction enzymes provides multiple pairs of matching 5′ and 3′ overhangs that will spontaneously assemble the genes in the desired order. In another aspect, the genes and components to be incorporated into a plasmid can be assembled into a single insert sequence prior insertion into the plasmid. In a further aspect, a DNA ligase enzyme can be used to assist in the ligation process.
(81) In another aspect, the ligation mix may be incubated in an electromagnetic chamber. In one aspect, the incubation lasts for about 1 minute, about 2 minutes, about 5 minutes, about 10 minutes, about 15 minutes, about 20 minutes, about 30 minutes, or about 1 hour.
(82) The DNA construct described herein can be part of a vector. In general, plasmid vectors containing replicon and control sequences that are derived from species compatible with the host cell are used in connection with the hosts. The vector ordinarily carries a replication site as well as marking sequences that are capable of performing phenotypic selection in transformed cells. Plasmid vectors are well known and commercially available. Such vectors include, but are not limited to, pWLneo, pSV2cat, pOG44, pXT1, pSG, pSVK3, pBSK, pYES, pYES2, pBSKII, pUC, pUC19, pBAD, and pETDuet-1 vectors.

(83) Plasmids are double-stranded, autonomously-replicating, genetic elements that are not integrated into host cell chromosomes. Further, these genetic elements are usually not part of the host cell's central metabolism. In bacteria, plasmids may range from 1 kilobase (kb) to over 200 kb. Plasmids can be engineered to encode a number of useful traits including the production of secondary metabolites, antibiotic resistance, the production of useful proteins, degradation of complex molecules and/or environmental toxins, and others. Plasmids have been the subject of much research in the field of genetic engineering, as plasmids are convenient expression vectors for foreign DNA in, for example, microorganisms. Plasmids generally contain regulatory elements such as promoters and terminators and also usually have independent replication origins. Ideally, plasmids will be present in multiple copier per host cell and will contain selectable markers (such as genes for antibiotic resistance) to show the skilled artisan to select host eels that have been successfully transfected with the plasmids (for example, by growing the host cells in a medium containing the antibiotic).
(84) In one aspect, the vector encodes a selection marker. In a further aspect, the selection marker is a gene that confers resistance to an antibiotic. In certain aspects, during fermentation of host cells transformed with the vector, the cells are contacted with the antibiotic. For example, the antibiotic may be included in the culture medium. Cells that have not been successfully transformed cannot survive in the presence of the antibiotic; only cells containing the vector, which confers antibiotic resistance, can survive. Optimally, only cells containing the vector to be expressed will be cultured, as this will result in the highest production efficiency of the desired gene products (e.g., peptides). Cells that do not contain the vector would otherwise compete with transformed cells for resources. In one aspect, the antibiotic is tetracycline, neomycin, kanamycin, ampicillin, hygromycin, chloramphenicol, amphotericin B, bacitracin, carbapenam, cephalosporin, ethambutol, fluoroquinolones, isoniazid, methicillin, oxacillin, vancomycin, streptomycin, quinolines, rifampin, rifampicin, sulfonamides, cephalothin, erythromycin, streptomycin, gentamycin, penicillin, other commonly-used antibiotics, or a combination thereof.
(85) In certain aspects, the DNA construct can include a gene that expresses a reporter protein. The selection of the reporter protein can vary. For example, the reporter protein can be a yellow fluorescent protein, a red fluorescent protein, a green fluorescent protein, or a cyan fluorescent protein. In one aspect, the reporter protein is a yellow fluorescent protein and the gene that expresses the reporter protein has SEQ ID NO. 6 or at least 70% homology thereto. The amount of fluorescence that is produced can be correlated to the amount of DNA incorporated into the transfected cells. The fluorescence produced can be detected and quantified using techniques known in the art. For example, spectrofluorometers are typically used to measure fluorescence.
(86) The DNA construct described herein can be part of a vector. In one aspect, the vector is a plasmid, a phagemid, a cosmid, a yeast artificial chromosome, a bacterial artificial chromosome, a virus, a phage, or a transposon.
(87) Exemplary methods for producing the DNA constructs described herein are provided in the Examples. Restriction enzymes and purification techniques known in the art can be used to assemble the DNA constructs. Backbone plasmids and synthetic inserts can be mixed together for ligation purposes at different ratios ranging from 1:1, 1:2, 1:3, 1:4, and up to 1:5. In one aspect, the ratio of backbone plasmid to synthetic insert is 1:4. After the vector comprising the DNA construct has been produced, the resulting vector can be incorporated into the host cells using the methods described below.
(88) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses RuBisCO large subunit 1, (3) a gene that expresses tonB, (4) a gene that expresses hydrogenase, and (5) a gene that expresses P-type ATPase.
(89) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70%

homology thereto, a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto.

(90) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses RuBisCO large subunit 1, (3) a gene that expresses tonB, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase, and (7) a gene that expresses P-type ATPase.

(91) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 90% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 90% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 90% homology thereto, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 90% homology thereto, and (7) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 90% homology thereto.

(92) In still another aspect, the construct is a pBAD plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (7) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto.

(93) In still another aspect, the construct is a pBAD plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (7) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, and (8) a gene that expresses a reporter protein having SEQ ID NO. 6 or at least 70% homology thereto.

(94) In another aspect, the DNA construct has SEQ ID NO. 7 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(95) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto.

(96) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses a dehydrogenase, (3) a gene that expresses RuBisCO large subunit 1, (4) a gene that expresses tonB, (5) an rrnB terminator, (6) an araBAD promoter, (7) a gene that expresses hydrogenase, and (8) a gene that expresses P-type ATPase.

(97) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 90% homology thereto, (2) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 90% homology thereto, (3) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 90% homology thereto, (4) a gene that expresses tonB having SEQ ID NO. 3 or at least

90% homology thereto, (5) an rrnB terminator, (6) an araBAD promoter, (7) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 90% homology thereto, and (8) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 90% homology thereto.

(98) In still another aspect, the construct is a pBAD plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (3) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (4) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (5) an rrnB terminator, (6) an araBAD promoter, (7) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (8) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto.

(99) In still another aspect, the construct is a pBAD plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (3) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (4) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (5) an rrnB terminator, (6) an araBAD promoter, (7) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (8) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, and (9) a gene that expresses a reporter protein having SEQ ID NO. 6 or at least 70% homology thereto.

(100) In another aspect, the DNA construct has SEQ ID NO. 11 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(101) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a gene that expresses P-type ATPase, (3) a gene that expresses tonB, (4) a gene that expresses a heat shock protein, and (5) a gene that expresses RuBisCO large subunit 1.

(102) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, and a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto.

(103) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses RuBisCO large subunit 1, and (14) a CYC1 terminator.

(104) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 90% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 90% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 90% homology thereto, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 90% homology thereto, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 90% homology thereto, and (14) a CYC1 terminator.

(105) In still another aspect, the construct is a pYES2 plasmid having from 5′ to 3′ the following

genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, and (14) a CYC1 terminator.

(106) In still another aspect, the construct is a pYES2 plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (14) a CYC1 terminator, (15) a GAL1 promoter, and (16) a gene that expresses a reporter protein having SEQ ID NO. 6 or at least 70% homology thereto.

(107) In another aspect, the DNA construct has SEQ ID NO. 8 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(108) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a gene that expresses P-type ATPase, (3) a gene that expresses tonB, (4) a gene that expresses a heat shock protein, (5) a gene that expresses dehydrogenase, and (6) a gene that expresses RuBisCO large subunit 1.

(109) In one aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, and a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto.

(110) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses dehydrogenase, (14) a CYC1 terminator, (15) a GAL1 promoter, and (16) a gene that expresses a gene that expresses RuBisCO large subunit 1.

(111) In another aspect, the construct includes from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 90% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 90% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 90% homology thereto, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 90% homology thereto, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 90% homology thereto, (14) a CYC1 terminator, (15) a GAL1 promoter, and (16) a gene that expresses

RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 90% homology thereto.

(112) In still another aspect, the construct is a pYES2 plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (14) a CYC1 terminator, (15) a GAL1 promoter, and (16) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto.

(113) In still another aspect, the construct is a pYES2 plasmid having from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses P-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (8) a CYC1 terminator, (9) a GAL1 promoter, (10) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (11) a CYC1 terminator, (12) a GAL1 promoter, (13) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (14) a CYC1 terminator, (15) a GAL1 promoter, (16) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (17) a CYC1 terminator, (18) a GAL1 promoter, (19) a gene that expresses a reporter protein having SEQ ID NO. 6 or at least 70% homology thereto.

(114) In another aspect, the DNA construct has SEQ ID NO. 10 or at least 70% homology thereto, at least 75% homology thereto, at least 80% homology thereto, at least 85% homology thereto, at least 90% homology thereto, at least 95% homology thereto, or at least 99% homology thereto.

(115) Biological Devices

(116) In one aspect, a “biological device” is formed when a microbial cell is transfected with the DNA construct described herein. The biological devices are generally composed of microbial host cells, where the host cells are transformed with a DNA construct described herein.

(117) In one aspect, the DNA construct is carried by the expression vector into the cell and is separate from the host cell's genome. In another aspect, the DNA construct is incorporated into the host cell's genome. In still another aspect, incorporation of the DNA construct into the host cell enables the host cell to produce a genelight solution such as, for example, those disclosed herein. “Heterologous” genes and proteins are genes and proteins that have been experimentally inserted into a cell that are not normally expressed by the cell. A heterologous gene may be cloned or derived from a different cell type or species than the recipient cell or organism. Heterologous genes may be introduced into cells by transduction or transformation.

(118) An “isolated” nucleic acid is one that has been separated from other nucleic acid molecules and/or cellular material (peptides, proteins, lipids, saccharides, and the like) normally present in the natural source of the nucleic acid. An “isolated” nucleic acid may optionally be free of the flanking sequences found on either side of the nucleic acid as it naturally occurs. An isolated nucleic acid can be naturally occurring, can be chemically synthesized, or can be a cDNA molecule (i.e., is synthesized from an mRNA template using reverse transcriptase and DNA polymerase enzymes).

(119) “Transformation” or “transfection” as used herein refers to a process for introducing heterologous DNA into a host cell. Transformation can occur under natural conditions or may be induced using various methods known in the art. Many methods for transformation are known in the art and the skilled practitioner will know how to choose the best transformation method based on the type of cells being transformed. Methods for transformation include, for example, viral infection, electroporation, lipofection, chemical transformation, and particle bombardment. Cells

may be stably transformed (i.e., the heterologous DNA is capable of replicating as an autonomous plasmid or as part of the host chromosome) or may be transiently transformed (i.e., the heterologous DNA is expressed only for a limited period of time).
(120) “Competent cells” refers to microbial cells capable of taking up heterologous DNA. Competent cells can be purchased from a commercial source, or cells can be made competent using procedures known in the art. Exemplary procedures for producing competent cells are provided in the Examples.
(121) The host cells as referred to herein include their progeny, which are any and all subsequent generations formed by cell division. It is understood that not all progeny may be identical due to deliberate or inadvertent mutations. A host cell may be “transfected” or “transformed,” which refers to a process by which an exogenous nucleic acid is transferred or introduced into the host cell.
(122) A transformed cell includes the primary subject cell and its progeny. The host cells can be naturally-occurring cells or “recombinant” cells. Recombinant cells are distinguishable from naturally-occurring cells in that naturally-occurring cells do not contain heterologous DNA introduced through molecular cloning procedures. In one aspect, the host cell is a prokaryotic cell such as, for example, *Escherichia coli*. In other aspects, the host cell is a eukaryotic cell such as, for example, the yeast *Saccharomyces cerevisiae*. Host cells transformed with the DNA construct described herein are referred to as “biological devices.”
(123) The DNA construct is first delivered into the host cell. In one aspect, the host cells are naturally competent (i.e., able to take up exogenous DNA from the surrounding environment). In another aspect, cells must be treated to induce artificial competence. This delivery may be accomplished in vitro, using well-developed laboratory procedures for transforming cell lines. Transformation of bacterial cell lines can be achieved using a variety of techniques. One method involves calcium chloride. The exposure to the calcium ions renders the cells able to take up the DNA construct. Another method is electroporation. In this technique, a high-voltage electric field is applied briefly to cells, producing transient holes in the membranes of the cells through which the vector containing the DNA construct enters. Another method involves exposing intact yeast cells to alkali cations such as, for example, lithium. In one aspect, this method includes exposing yeast to lithium acetate, polyethylene glycol, and single-stranded DNA such as, for example, salmon sperm DNA. Without wishing to be bound by theory, the single-stranded DNA is thought to bind to the cell wall of the yeast, thereby blocking plasmids from binding. The plasmids are then free to enter the yeast cell. Enzymatic and/or electromagnetic techniques can also be used alone, or in combination with other methods, to transform microbial cells. Exemplary procedures for transforming yeast and bacteria with specific DNA constructs are provided in the Examples. In certain aspects, two or more types of DNA can be incorporated into the host cells. Thus, different metabolites can be produced from the same host cells at enhanced rates.
(124) Preparation of Genelight Extracts
(125) A satisfactory microbiological culture contains available sources of hydrogen donors and acceptors, carbon, nitrogen, sulfur, phosphorus, inorganic salts and, in certain cases, vitamins or other growth-promoting substances. For example, the addition of peptone provides a readily-available source of nitrogen and carbon. Furthermore, the use of different types of media results in different growth rates and different stationary phase densities. A rich media results in a short doubling time and higher cell density at stationary phase. Minimal media results in slow growth and low final cell densities. Efficient agitation and aeration increase final cell densities.
(126) Culturing or fermenting of host cells can be accomplished by any technique known in the art. In one aspect, batch fermentation can be conducted. In batch fermentation, the composition of the culture medium is set at the beginning and the system is closed to future alterations. In some aspects, a limited form of batch fermentation may be carried out, wherein factors such as oxygen concentration and pH are manipulated, but additional carbon is not added. Continuous fermentation methods are also contemplated. In continuous fermentation, equal amounts of a defined medium

are continuously added to and removed from a bioreactor. In other aspects, microbial cells are immobilized on a substrate. Fermentation may be carried out on any scale and may include methods in which literal “fermentation” is carried out as well as other culture methods that are non-fermentative.

(127) In one aspect, the microorganisms can be cultured for a period of from 2 days to 2 weeks, or for about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, about 12, about 13, or about 14 days, where any value can be the lower or upper endpoint of a range (e.g., about 3 days to about 13 days, about 8 days to about 12 days, etc.). In one aspect, the microorganisms are cultured for about 10 days.

(128) In another aspect, the microorganisms can be cultured at any temperature appropriate for the microorganisms, with the understanding that the temperature may vary according to the microorganism (for example, a thermophilic microorganism may require a higher culture temperature than a mesophile). In one aspect, the microorganisms are cultured at a temperature of from about 20 to about 37° C., or are cultured at about 20° C., about 21° C., about 22° C., about 23° C., about 24° C., about 25° C., about 26° C., about 27° C., about 28° C., about 29° C., about 30° C., about 31° C., about 32° C., about 33° C., about 34° C., about 35° C., about 36° C., or about 37° C., where any value can be the lower or upper endpoint of a range, where any value can be the lower or upper endpoint of a range (e.g., about 21° C. to about 36° C., about 25° C. to about 30° C., etc.).

(129) In certain aspects, after culturing the microorganisms for a sufficient time, the microbial cells can be lysed with one or more enzymes. For example, when the microbial cells are fungal, the fungal cells can be lysed with lyticase. In one aspect, the lyticase concentration can be about 500 μL, about 600 μL, about 700 μL, about 800 μL, about 900 μL, or about 1,000 μL per liter of culture, where any value can be the lower or upper endpoint of a range, where any value can be the lower or upper endpoint of a range (e.g., about 500 μL to about 900 μL, about 600 μL to about 800 μL, etc.).

(130) In addition to or in place of enzymes, other components can be used to facilitate lysis of the microbial cells. In one aspect, chitosan can be used in combination with an enzyme to lyse the microbial cells. Chitosan is generally composed of glucosamine units and N-acetylglucosamine units and can be chemically or enzymatically extracted from chitin, which is a component of arthropod exoskeletons and fungal and microbial cell walls. In certain aspects, the chitosan can be acetylated to a specific degree of acetylation. In one aspect, the chitosan is from about 60% to about 100% acetylated, or about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, or about 100% acetylated, where any value can be the lower or upper endpoint of a range, where any value can be the lower or upper endpoint of a range (e.g., about 60% to about 90%, about 70% to about 80%, etc.).

(131) The molecular weight of the chitosan can vary, as well. For example, the chitosan can comprise about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 glucosamine units and/or N-acetylglucosamine units, where any value can be the lower or upper endpoint of a range, where any value can be the lower or upper endpoint of a range (e.g., 2 to 19, 3 to 10, 5 to 7, etc.). In one aspect, chitosan can be added until a concentration of about 0.0015%, about 0.0025%, about 0.005%, about 0.0075%, about 0.01%, about 0.015%, about 0.02%, about 0.03%, about 0.04%, or about 0.05%, where any value can be an upper or lower endpoint of a range (e.g., 0.002% to 0.04%, 0.05% to 0.015%, etc.).

(132) In one aspect, the microbial cultures are used without further processing in the applications that follow (i.e., with whole, growing/reproducing cells). In an alternative aspect, the cells are filtered out using a filter membrane such as, for example, filter with a 1-2 μm size cutoff. Further in this aspect, the filtrate is preserved and used as the microbial extract in the applications that follow. In still another aspect, the filtrate can be partially or fully dried to produce a solid or powder (such as, for example, by lyophilization) prior to use. Further in this aspect, the filtrate can be

resuspended in water until it dissolves, with stirring and heating if necessary, or can be partially resuspended into a suspension or a slurry. In still another aspect, cells can be lysed as described previously and the composition including lysed cells can be used as the microbial extract in the applications that follow, either as-is or dried. In some aspects, the dried composition containing lysed cells can be reconstituted as a solution or slurry as described above for the filtrate. In still other aspects, the composition containing lysed cells can be further purified by filtration or another method prior to use as an extract, and can be used as-is or dried and reconstituted as described above.
(133) In one aspect, a dried filtrate, lysed cell culture, or whole cell culture can be reconstituted in water as a solution, suspension, or slurry and used in further applications. In another aspect, the filtrate, lysed cell culture, or whole cell culture can be used as-is, without further processing. In still another aspect, any of the solutions, suspensions, or slurries described above can be mixed together, with or without water or another solvent, and used in the applications that follow.
(134) Genelight Extracts as Power Sources
(135) In one aspect, described herein are circuits containing the genelight extracts and compositions described herein. In another aspect, the circuits incorporate the extracts and compositions as electrolyte solutions, i.e., as in a battery. In a further aspect, the circuits include metal electrodes immersed in the extracts and compositions and wired to one another to provide for transfer of electric charge. In one aspect, the electrodes are made from zinc, cadmium, copper, silver, graphite, rhodium, or lead. In one aspect, the electrodes are copper and zinc. In a further aspect, the circuits are arranged as series or as parallel circuits.
(136) In another aspect, described herein is a battery containing the extracts and compositions described herein. In one aspect, the battery uses the extracts and compositions as electrolytes and incorporates a circuit of the type previously described. In another aspect, the battery can be used to power a device or part of a device such as those described herein. In one aspect, the battery is sufficient power for the device and no external power supply is needed. In an alternative aspect, the battery acts as a supplement to another power source.
(137) In a further aspect, the battery and/or circuit described herein use only environmentally safe, natural materials from the extracts and compositions as electrolytes and can be disposed of and/or recycled by the consumer. In a still further aspect, the battery or circuit does not contain toxic chemicals or heavy metals. In another aspect, the battery or circuit does not present a fire hazard. In yet another aspect, at the end of its useful life, the battery or circuit can be disassembled and the electrolyte removed and used for another application described herein (e.g., enhancing the growth of plants or microorganisms).
(138) In one aspect, the equations in Table 7 below can be useful in modeling, describing, and/or predicting the behavior of microbial circuits and electronic devices using those circuits constructed using the extracts and compositions described herein:
(139) TABLE-US-00007 TABLE 7 Equations Useful for Circuits Incorporating Microbial Extracts Property Formula Variable Descriptions Units Biological BQ = OIDC × {CR × CR: culture redox mv × mg Charge pH × [C](IC.sub.e − pH: culture pH mL IC.sub.c)EZ} [C]: cell concentration (cells/mL) EZ: extract size IC: iron concentration OIDC: organic-inorganic dilution coefficient Electric P = V × I V: electric potential watts (W) Power I: electric current Genelight W.sub.BL = OIDC × V: electric potential watts (W) Electric V × I I: electric current Power.sup.a OIDC: organic-inorganic dilution coefficient Microbial I = OIDC × P/V V: electric potential amperes (A) Electric P: microbial electric power Current OIDC: organic-inorganic dilution coefficient Photon PED = (P × P: microbial electric power W × mv × mg Emission BQ × λ) BQ: biological charge mL Duration.sup.a λ: color wavelength .sup.aEquations specific to circuits including an LED or other light-producing element.
Applications of Genelight Extracts
(140) In one aspect, the compositions and extracts described herein are useful in a variety of

applications such as, for example, powering lighting sources, powering warming and cooling devices, and powering battery operated or other electrical devices in a variety of environments and settings.

(141) Lighting Applications

(142) In one aspect, for any device or in any situation where lighting is required and where use of an LED is suitable, the compositions and extracts disclosed herein can be incorporated into a circuit as described previously, wherein the circuit can provide lighting for one or more of: cell phones, medical instruments, holiday lighting displays, handheld flashlights, indicator lights on radar or sonar equipment, digital displays of alphanumeric information, billboards, automobiles (both interior and exterior), airport runways, train tracks, subways, highways, street signs, traffic lights, auxiliary lighting for building safety, exit signs and other lighted signs, flares, night-vision equipment, telescopes, binoculars, rifle scopes, distance finders, and the like.

(143) Warming and Cooling Applications

(144) In another aspect, wherever a temperature differential is required, the compositions and extracts disclosed herein can be incorporated into a circuit as described previously, wherein the circuit can provide warming and/or cooling for one or more of: clothing including, but not limited to, gloves, hats, coats, shoes, socks, blankets, or baby clothing; climate control in buildings including, but not limited to, floor heating, roof heating, building heating, water heaters, fans, and the like; food preparation and storage including but not limited to, refrigerators, freezers, beverages coolers and warmers, food warmers, ice chests, cooking apparatuses including stoves and grills, and the like. In still another aspect, the compositions and extracts disclosed herein can be used as add-ins or backups to other power sources such as, for example, gas, coal, nuclear, solar, or wind energy, or can be used during power outages instead of or in addition to generators.

(145) Battery-Powered and Electrical Devices

(146) In one aspect, the compositions and methods described herein can be incorporated into a circuit as described previously, wherein the circuit can be incorporated into an electronic device such as a battery of any size including a rechargeable battery, a key fob or remote control, an electronic reading device, a laptop or portable computer, or similar. In another aspect, the circuit can be incorporated into a kitchen appliance such as, for example, a microwave, toaster, coffee maker, oven, electric knife, food and beverage slicers, grinders, mixers, and dispensers, blenders, and the like. In still another aspect, the circuit can be incorporated into a television, portable video device, radio, speaker, microphone, video game console or controller, headphones, sound canceling equipment, or similar. In still another aspect, the circuit can be incorporated into a toy, a bicycle or tricycle, a musical instrument, a prosthetic limb, pacemaker, a CPAP device, a device for stimulating muscle growth and regeneration, or an insulin pump or other powered medical implant, a thermometer, a hearing aid, eyeglasses, another medical device, a scientific instrument such as, for example, a microscope, a power tool such as an air compressor, a gas detection device, a smoke detector, or an electric toothbrush or other small personal hygiene or grooming device. In one aspect, the circuit can be used in the transportation industry, for example, in automobiles, trucks, trains, buses and other means of public transportation, subways, watercraft of various sizes, aircraft, drones, motorcycles, golf carts, and the like. In another aspect, the circuit can be used in any type of pump (i.e., water, sewage, oil field, marine, swimming pool or hot tub, artificial heart). In yet another aspect, the circuit can be incorporated into electric fencing egress and control, a clock or other timekeeping device, irrigation systems and landscaping equipment, a cash register, voting equipment, buzzers, security systems, pest control devices, and equipment for industrial or home cleaning and sterilizing.

(147) In one aspect, the devices can be wireless devices. In an alternative aspect, the devices may optionally require wires for operation.

(148) Environmental Suitability

(149) Devices and applications of the compositions and extracts disclosed herein are suitable for

use in a variety of environments. In one aspect, the devices can be used in applications under water, at high altitudes, at temperature extremes both high and low, in situations where traditional batteries and/or power sources would present a flammability hazard or explosion risk, in space, in laboratories and industrial facilities, in remote locations, and in areas with weather extremes including wind, rain, sandstorms, and the like.

(150) Methods for Enhancing the Growth of Microorganisms

(151) In one aspect, the compositions and extracts disclosed herein can be used as a culture medium for commercially important microorganisms, or can be used as a supplement to an existing medium. Further in this aspect, culturing microorganisms in the presence of the compositions and extracts disclosed herein can lead to enhanced growth rates and/or enhanced production of desirable metabolites.

(152) In a further aspect, the compositions and extracts disclosed herein are useful in culturing the following types or organisms: (1) *Saccharomyces cerevisiae* for use in yeast doughs, brewing beer and wine, genetic research, and production of desirable secondary metabolites including the enzymes invertase and raffinase; *Kluyveromyces* species for the commercial production of lactase; and *Candida* species for the commercial production of lipase; (2) *Lactobacillus* species for use in making fermented foods such as yogurt, kefir, cheese, sauerkraut, pickles, hard cider, wine, and beer as well as for the commercial production of lactic acid and engineered *Lactobacillus* species engineered to produce protein drugs such as, for example, insulin; (3) *Pyrococcus furiosus, Thermus aquaticus, Bacillus stearothermophilus, Thermus filiformis, Thermus* thermophiles, and other thermophiles for production of heat stable polymerases for use in the polymerase chain reaction (PCR); (4) *Xanthomonas* species for production of xanthan gum, used in a variety of food and cosmetic products; (5) *Aspergillus niger*, used in the production of citric acid and fermentation of sake and other alcoholic beverages and this and other *Aspergillus* species for commercial production of α-amylase, aminoacylase, glucoamylase, catalase, glucose oxidase, lactase, pectinase, pectin lyase, and protease; *Trichoderma* species for the commercial production of cellulose; *Mucor miehei* for the commercial production of rennet; *Rhizopus* species for the commercial production of lipase; and *Mortierella* species for the commercial production of raffinase; (6) *Clostridium* species for production of botulinum toxin for cosmetic and medical purposes as well as the production of butanol (i.e., from *Clostridium acetobutylicum*); (7) *Streptomyces* species for production of antibiotics, antiparasitic, antineoplastic, and antifungal compounds including, but not limited to, chloramphenicol, daptomycin, fosfomycin, lincomycin, neomycin, nourseothricin, puromycin, streptomycin, tetracycline, oleandomycin, tunicamycin, mycangimycin, boromycin, bambermycin, clavulanic acid, guadinomine, ivermectin, migrastatin, bleomycin, erythromycin, geldanamycin, and the like; (8) *Penicillium* species for the production of penicillin and other beta-lactam antibiotics and precursors to semi-synthetic beta-lactam antibiotics; (9) *Acetobacter aceti* for production of acetic acid; (10) *Bacillus* species for commercial production of α-amylase, β-amylase, glucose isomerase, penicillin amidase, and protease; *E. coli* for commercial production of asparaginase; and *Klebsiella* species for commercial production of pullulanase; (11) *Trichoderma polysporum* for the production of cyclosporine A, (12) yeasts such as *Monascus purpureus* for the production of statin drugs for lowering blood cholesterol; (13) *Bacillus thuringiensis* for the production of insecticides, and other commercially important bacteria, fungi, algae, and cyanobateria.

(153) In an alternative aspect, the compositions and extracts disclosed herein can be used as an animal food or animal supplement. Further in this aspect, the compositions and extracts disclosed herein may contain vitamins, minerals, proteins, peptides, carbohydrates, and/or other nutrients essential for animal growth and development and/or maintenance of animal health.

(154) Methods for Enhancing the Physiological Properties of Plants

(155) The compositions and extracts described herein can enhance or improve the physiological properties of a plant. The term “physiological property” as defined herein includes any physical,

chemical, or biological feature that is improved using the compositions and extracts described herein. In one aspect, the compositions and extracts can enhance the growth rate of the plant. In another aspect, the physiological property includes increased root tension, root length, hormone production, drought tolerance, disease resistance, photosynthesis, or any combination thereof.
(156) Herein, “plant” is used in a broad sense to include, for example, any species of woody, ornamental, crop, cereal, fruit, or vegetable plant, as well as photosynthetic green algae. “Plant” also refers to a plurality of plant cells that are differentiated into a structure that is present at any stage of the plant's development. Such structures include, but are not limited to, fruits, shoots, stems, leaves, flower petals, roots, tubers, corms, bulbs, seeds, gametes, cotyledons, hypocotyls, radicles, embryos, gametophytes, tumors, and the like. “Plant cell,” “plant cells,” or “plant tissue” as used herein refer to differentiated and undifferentiated tissues of plants including those present in any of the tissues described above, as well as to cells in culture such as, for example, single cells, protoplasts, embryos, calluses, etc.
(157) The selection of the plant used in the methods described herein can vary depending on the application. For example, a specific plant can be selected that produces certain desirable metabolites. Current techniques for producing most plant metabolites are expensive. For example, large amounts of fresh plant biomass must be cultivated and harvested and expensive and time-consuming extraction methods must be used. The compositions and extracts described herein enhance the production of metabolites from plants that naturally produce these metabolites.
(158) In one aspect, plant cells when contacted with the compositions and extracts described herein exhibit enhanced production of various desirable metabolites. Recipient cell targets include, but are not limited to, meristem cells, Type I, Type II, and Type III callus, immature embryos and gametic cells such as microspores, pollen, sperm, and egg cells. It is contemplated that any cell from which a fertile plant may be regenerated is useful as a recipient cell. Type I, Type II, and Type III callus may be initiated from tissue sources including, but not limited to, immature embryos, immature inflorescences, seedling apical meristems, microspores, and the like. Those cells that are capable of proliferating as callus are also useful herein. Methods for growing plant cells are known in the art. In one aspect, plant calluses grown from 2 to 4 weeks can be used herein. The plant cells can also be derived from plants varying in age. For example, plants that are 80 days to 120 days old after pollination can be used to produce calluses useful herein.
(159) The plant cells can be contacted with the compositions and extracts described herein in a number of different ways. In one aspect, the compositions and extracts described herein can be added to media containing the plant cells, or can be the media containing the plant cells. In another aspect, the compositions and extracts can be injected into the plant cells via syringe. The amount of extract and the duration of exposure to the extract can vary as well.
(160) Once the plant cells have been in contact with the compositions and extracts for a sufficient time to produce a desired metabolite, the metabolite is isolated. In one aspect, the metabolite is extracted from the media containing the plant cells. The selection of extraction solvent can vary depending on the solubility of the metabolite.
(161) In other aspects, the compositions and extracts described herein can increase the growth rate of a plant. In particular, the compositions and extracts described herein are effective in accelerating plant development in the early stages of tissue culturing. By accelerating plant development in the early stages, it is possible to harvest more metabolites from the plant. Additionally, traditional methods for tissue culture involve the use of synthetic growth factors such as 2,4-dichlorophenoxyacetic acid (2,4-D), which can pose environmental concerns. The compounds and extracts described herein avoid the need for such compounds.
(162) In certain aspects, any of the compositions and extracts described above can be used in combination with a polysaccharide to enhance one or more physiological properties of the plant. In one aspect, the plant (e.g., cells, seeds, callus, mature plant) is first contacted with the compositions and extracts described herein, then subsequently contacted with the polysaccharide. In another

aspect, the plant is first contacted with the polysaccharide, then subsequently contacted with the compositions or extracts described herein. In a still further aspect, the plant cells are contacted simultaneously with the polysaccharide and the compositions and extracts described herein.

(163) Inn one aspect, the polysaccharide includes a polyactive carbohydrate as described and produced in WO 2019/055456, which is incorporated by reference in its entirety with respect to polyactive carbohydrates. In another aspect, the polysaccharide is a polyactive carbohydrate produced by a biological device transformed with a DNA construct as depicted in FIGS. **1**A-**1** and **2**A-**2**B in WO 2019/055456. In another aspect, the polysaccharide is a polyactive carbohydrate produced by a biological device transformed with a DNA construct having SEQ ID NO. 5 or at least 90% homology thereof or SEQ ID NO. 7 or at least 90% homology thereof as provided in WO 2019/055456.

(164) In one aspect, the polysaccharide includes chitosan, glucosamine (GlcN), N-acetylglucosamine (NAG), or any combination thereof. Chitosan is generally composed of GlcN and NAG units and can be chemically or enzymatically extracted from chitin, which is a component of arthropod exoskeletons and fungal and microbial cell walls. In certain aspects, the chitosan can be acetylated to a specific degree of acetylation in order to enhance tissue growth during culturing as well as metabolite production. In one aspect, chitosan isolated from shells of crab, shrimp, lobster, and/or krill is useful herein.

(165) In one aspect, the polysaccharide is in a solution of water and acetic acid at less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, or less than 0.1% by weight. In another aspect, the amount of chitosan that is applied to the plant cells is from 0.01 wt % to 0.1 wt % by weight, or 0.01 wt %, 0.02 wt %, 0.03 wt %, 0.04 wt %, 0.05 wt %, 0.06 wt %, 0.07 wt %, 0.08 wt %, 0.09 wt %, or 0.1 wt %, where any value can be an upper or lower endpoint of a range (e.g., 0.01 wt % to 0.09 wt %, 0.02 wt % to 0.08 wt %, etc.). The polysaccharides used herein are generally natural polymers and thus present no environmental concerns. Additionally, the polysaccharides can be used in acceptably low concentrations. In certain aspects, however, the polysaccharides can be used in combination with one or more plant growth regulators.

(166) In one aspect, the polysaccharide is in a solution of water and acetic acid at less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, or less than 0.1% by weight. In another aspect, the amount of chitosan that is applied to the plant cells is from 0.01 wt % to 0.1 wt % by weight, or 0.01 wt %, 0.02 wt %, 0.03 wt %, 0.04 wt %, 0.05 wt %, 0.06 wt %, 0.07 wt %, 0.08 wt %, 0.09 wt %, or 0.1 wt %, where any value can be an upper or lower endpoint of a range (e.g., 0.01 wt % to 0.09 wt %, 0.02 wt % to 0.08 wt %, etc.). The polysaccharides used herein are generally natural polymers and thus present no environmental concerns. Additionally, the polysaccharides can be used in acceptably low concentrations. In certain aspects, however, the polysaccharides can be used in combination with one or more plant growth regulators.

(167) In one aspect, disclosed herein are compositions composed of the genelight extracts described herein with one or more polysaccharides. In one aspect, the composition includes water. In one aspect, the composition includes the genelight extract, chitosan, polyactive carbohydrate, and water. In one aspect, the genelight extract is less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, less than 0.1% by weight, less than 0.5% by weight, or less than 0.3% by weight of the composition. In one aspect, the polysaccharide is chitosan and is less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, less than 0.1% by weight, less than 0.5% by weight, or less than 0.3% by weight of the composition. In one aspect, the polysaccharide is a polyactive carbohydrate and is less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, less than 0.1% by weight, less than 0.5% by weight, or less than 0.3% by weight of the composition.

(168) In one aspect, the plant growth regulator is an auxin, a cytokinin, a gibberellin, abscisic acid, or a polyamine. In a further aspect, the auxin is a natural or synthetic auxin. In a still further aspect, the auxin is indole-3-acetic acid (IAA), 4-chloroindole-3-acetic acid (4-Cl-IAA), 2-phenylacetic acid (PAA), indole-3-butyric acid (IBA), 2,3-dichlorophenoxyacetic acid (2,4-D), α-naphthalene acetic acid (α-NAA), 2-methoxy-3,6-dichlorobenzoic acid (dicamba), 4-amino-3,5,6-trichloropicolinic acid (torden or picloram), 2,4,5-trichloropicolinic acid (2,4,5-T), or a combination thereof. In another aspect, the cytokinin is zeatin, kinetin, 6-benzylaminopurine, diphenylurea, thidizuron (TDZ), 6-(γ,γ-dimethylallylamino)purine, or a combination thereof. In another aspect, the gibberellin is gibberellin A1 (GA1), gibberellic acid (GA3), ent-gibberellane, ent-kaurene, or a combination thereof. In yet another aspect, the polyamine is putrescine, spermidine, or a combination thereof.
(169) In one aspect, the plant cell or callus is first contacted with a polysaccharide and subsequently contacted with a plant growth regulator. In another aspect, the plant cell or callus is first contacted with a plant growth regulator and subsequently contacted with a polysaccharide. In an alternative aspect, the plant cell or callus is simultaneously contacted with a polysaccharide and a plant growth regulator. In a further aspect, the plant cell or callus is only contacted with a polysaccharide and is not contacted with a plant growth regulator.
(170) The plant cells can be contacted with the polysaccharide using a number of techniques. In one aspect, the plant cells or reproductive organs (e.g., a plant embryo) can be cultured in agar and medium with a solution of the polysaccharide. In other aspects, the polysaccharide can be applied to a plant callus by techniques such as, for example, coating the callus or injecting the polysaccharide into the callus. In this aspect, the age of callus can vary depending on the type of plant. The amount of polysaccharide can vary depending upon, among other things, the selection and number of plant cells. The use of the polysaccharide in the methods described herein permits rapid tissue culturing at room temperature. Due to the ability of the polysaccharide to prevent microbial contamination, the tissue can grow for extended periods of time ranging from days to several weeks. Moreover, tissue culturing with the polysaccharide can occur in the dark and/or light. As discussed above, the plant cells are also contacted with any of the compositions or extracts described above. Thus, the use of the polysaccharides and compositions and extracts described herein is a versatile way to culture and grow plant cells—and, ultimately, plants of interest—with enhanced physiological properties.
(171) In other aspects, the plant cells can be cultured in a liquid medium on a larger scale in a bioreactor. For example, plant cells can be cultured in agar and medium, then subsequently contacted with the compositions and extracts described herein. After a sufficient culturing time (e.g., two to four weeks), the plant cells are introduced into a container with the same medium used above and, additionally, the polysaccharide. In certain aspects, the polysaccharide can be introduced with anionic polysaccharides including, but not limited to, alginates (e.g., sodium alginate, calcium alginate, potassium alginate, etc.). After the introduction of the polysaccharide, if using, the solution is mixed for a sufficient time to produce a desired result (e.g., production of a desired metabolite). Alternatively, the initial liquid medium in the bioreactor can include any of the compositions and/or extracts described herein.
(172) In one aspect, provided herein is a plant grown by the process that involves contacting plant gamete cells or a plant reproductive organ with the extracts disclosed herein. In a further aspect, the plant is produced by the following method: (a) contacting a plant callus with the extracts; (b) culturing the plant callus; and (c) growing the plant from the plant callus.
(173) In a further aspect, the same method can be applied to other plant parts including fruits, stems, roots, tubers, corms, bulbs, flowers, buds, seeds, and the like. In a still further aspect, the same method can be applied to an entire plant.
(174) In one aspect, the plant callus is immersed in a solution of polysaccharide (e.g., chitosan), then inoculated with the compositions and/or extracts described herein. In another aspect, the plant

callus can be from 2 days up to 20 days old prior to inoculation with the compositions and/or extracts described herein. The plant callus is then allowed to grow until it is of sufficient weight and size. In one aspect, the plant callus is allowed to grow (i.e., culture) for 1 to 10 weeks after inoculation. Following growth or culture of the callus for a sufficient period of time, desired metabolites can be collected according to methods known in the art; said methods are specific to the desired metabolites and make use of properties ranging from molecular size to charge to hydrophobicity or hydrophilicity to other properties useful for collection and purification of the metabolites.

(175) Turf Applications

(176) In one aspect, the compositions and extracts disclosed herein can be used to improve the health and appearance of grass and/or turf such as, for example, grass or turf on sports playing fields, lawns, golf courses, and the like. In one aspect, application of the compositions and extracts enhances root growth.

(177) In a further aspect, plant hormones can promote or stimulate plant growth and development. In one aspect, application of the compounds or extracts disclosed herein stimulates endogenous production of plant hormones, in turn enhancing root growth. In a further aspect, application of the compounds or extracts increases the production of cytokinins by the plants. Cytokinins are responsible for the growth of roots and anchoring of a plant in soil. Still further in this aspect, root strength is especially important on golf courses due to foot traffic as well as contact with golf clubs. In another aspect, application of the compounds or extracts increases the production of salicylic acid by the plants. Salicylic acid promotes and/or enhances photosynthesis, transpiration, and mineral uptake. In one aspect, increased production of salicylic acid maintains an aesthetically-pleasing green appearance in, for example, turf grasses, as well as protecting the plant against pathogenic microorganisms. In still another aspect, application of the compounds or extracts increases the production of jasmonic acid by plants. Jasmonic acid protects against environmental damage, microbial pathogens, and insects, as well as protecting against early senescence of plants. Jasmonic acid is particularly important for field grasses (e.g., on golf courses) since the grasses are kept green for most of the year.

(178) In one aspect, application of the compounds or extracts disclosed herein increases root tension (i.e., how difficult it is to pull up a plant or section of sod to which the compounds or extracts have been applied) and/or root length. In another aspect, applications of the compounds or extracts disclosed herein leads to a greener appearance of plants such as, for example, field grasses or turf grasses. In one aspect, the compositions and extracts described herein can increase the root force of sod by about 50% to about 200%, or about 50%, about 75%, about 100%, about 125%, about 150%, about 175%, or about 200% compared to sod that has not been treated with the composition or extract, where any value can be a lower and upper endpoint of a range (e.g., about 50% to about 175%, about 75% to about 125%, etc.).

(179) In one aspect, the compositions and extracts described herein can increase the root force of a plant by about 50% to about 200%, or about 50%, about 75%, about 100%, about 125%, about 150%, about 175%, or about 200% compared to a plant that has not been treated with the composition or extract, where any value can be a lower and upper endpoint of a range (e.g., about 50% to about 175%, about 75% to about 125%, etc.).

(180) In one aspect, the compounds and extracts disclosed herein can be applied at a volume of about 50 to about 200 mL per square meter of, for example, turf, or can be applied at about 50 mL, about 75 mL, about 100 mL, about 125 mL, about 150 mL, about 175 mL, or about 200 mL per square meter of turf, or a combination of any of the foregoing values, or a range encompassing any of the foregoing values. In one aspect, the compounds and extracts are applied to turf at a volume of about 75 mL per square meter. In an alternative aspect, the compounds and extracts are applied to turf at a concentration of about 150 mL per square meter.

(181) In one aspect, the compositions and extracts disclosed herein are applied multiple times to the

plants of interest. Further in this aspect, the compositions and extracts can be applied once per week, twice per week, or three times per week. In one aspect, the compositions and/or extracts are applied to the plants of interest twice per week.
(182) In another aspect, the compositions and extracts disclosed herein can be applied alone or in combination with another product intended to enhance plant growth. In one aspect, the compositions and extracts disclosed herein are applied in combination with a fertilizer such as, for example, activated sewage sludge including, but not limited to, Hou-actinite, or another commercially-available fertilizer. In one aspect, the compositions and extracts disclosed herein are applied after fertilizer application, or can be applied before fertilizer application, or can be applied at the same time as fertilizer is applied to the plants of interest. In still another aspect, another compound such as, for example, chitosan, can be applied to the plants in addition to the compositions and extracts disclosed herein. In one aspect, the chitosan can be applied as a solution with a concentration of from about 0.01% to about 0.05% (w/v), or can be applied as a solution with a concentration of about 0.01, 0.02, 0.03, 0.04, or about 0.05%. In one aspect, the chitosan is applied as a solution with a concentration of about 0.02%. In another aspect, In one aspect, the chitosan can be applied at a concentration of about 50 to about 200 mL per square meter of, for example, turf, or can be applied at about 50, 75, 100, 125, 150, 175, or about 200 mL per square meter of turf, or a combination of any of the foregoing values, or a range encompassing any of the foregoing values. In one aspect, the chitosan is applied to turf at a concentration of 75 mL per square meter. In an alternative aspect, the chitosan is applied to turf at a concentration of 150 mL per square meter. In another aspect, the chitosan can be applied to the turf once per week, twice per week, or three times per week. In one aspect, the chitosan is applied twice per week. In still another aspect, the chitosan can be applied before the compounds and extracts disclosed herein, or after the compounds and extracts disclosed herein, or simultaneously with the compounds and extracts disclosed herein.
(183) In any of the above aspects, application of the compounds and extracts disclosed herein leads to an increase in plant hormones and, in turn, improvement in properties such as earlier rooting, root anchoring or tension, root strength, root length, green coloration, nutrient availability, reduction of irrigation needs, prevention of erosion, and other desirable properties of plants. In another aspect, enhancing plant growth including magnesium uptake and production of chlorophyll a and/or chlorophyll b can be used to trap carbon dioxide from the atmosphere.
(184) Anti-UV Applications
(185) The genelight extracts produced herein may be applied to any material that may benefit from a reduction in exposure to UV radiation. The exact formulation of the extract plus any carriers can be adjusted based on the desired use. In one aspect, the extract is formulated with only non-toxic components if it is to be used on a human or animal or with another microorganism, such as in a fermentation process or on an agricultural product. In another aspect, the extract can be mixed with other substances to provide UV-protective properties to the overall composition. In still another aspect, if coated on the material to be protected, the extract itself can be covered with a further protective coating to project, for example, against mechanical wear and damage.
(186) Methods of Applying Extracts to Surfaces
(187) In the case when the extract is applied to the surface of an article, it can be applied using techniques known in the art such spraying or coating. In other aspects, the extract can be intimately mixed with a substance or material that ultimately produces the article. For example, the extract can be mixed with molten glass so that the extract is dispersed throughout the final glass product.
(188) In one aspect, the extract is formulated or applied in such a manner as to block approximately about 50%, about 60%, about 70%, about 80%, about 90%, about 95%, or about 99% of the UV radiation that encounters the extract, where any value can be a lower and upper end-point of a range (e.g., about 60% to about 95%, about 70% to about 80%). In a further aspect, the extract can also be formulated to block these percentages of particular UV wavelengths, or, more generally, to

block these percentages of UVA, UVB, or UVC radiation.

(189) Uses of the Extracts

(190) Extracts according to the present disclosure can be used for a variety of purposes. These purposes include, but are not limited to, the following: 1. blocking UV radiation or other types of radiation; 2. protecting human skin against damage and/or skin cancer induced by UV radiation or other types of radiation; 3. protecting against side effects of radiation used in cancer treatments; 4. protecting animals from deleterious effects of UV radiation or other radiation; 5. protecting plastic, fiberglass, glass, rubber, or other solid surfaces from UV radiation or other radiation; 6. providing a UV radiation screen or screen for other types of radiation; 7. protecting astronauts and/or other persons or organisms as well as equipment during space trips; 8. enhancement of industrial fermentation processes or other processes requiring energy by allowing the use of UV radiation in connection with the process to supply additional energy and thus to increase the ultimate energy-requiring output of the cells without substantially killing the fermenting organism; 9. protection of experimentation, fermentation, biochemical, and/or biological processes under the presence of UV radiation, for example in extraterrestrial conditions such as on the moon or Mars; and 10. protection of agricultural plants, particularly agricultural plants in which the revenue-producing part of the plant is above ground, such as fruits, vine vegetables, beans and peas, and leaf vegetables.

Agricultural Applications

(191) In one aspect, the genelight can be applied to an agricultural plant. In one aspect, the plant can be one that produces fruit or vegetable, such as, for example, a watermelon or a tomato. Further in this aspect, the extract can be applied during at least a part of the plant's growth to increase the amounts of one or more nutrients of the fruit or vegetable, such as a vitamin, mineral, or other recommended dietary component. In one specific aspect, the amount of lycopene can be increased (which may be accompanied by a decrease in carotene or other less-valuable nutrients formed by competing pathways). In another aspect, the amount of a flavor-enhancing component, such as glucose, can be increased. Further in this aspect, an increase in glucose can help protect against water loss.

(192) In one aspect, the genelight extract can be applied for about 25%, about 50%, about 75%, about 90%, about 95%, or about 99% of the fruit's or vegetable's on-plant life, where any value can be a lower and upper endpoint of a range (e.g., about 25% to about 95%, about 50% to about 75%, etc.), and where the on-plant life includes the time span from the formation of a separate body that will constitute the fruit or vegetable (in some aspects, excepting flowers) until the fruit or vegetable is harvested. In one aspect, the extract can be first applied when the fruit or vegetable is sufficiently large to no longer be substantially protected from UV radiation by leaves. In another aspect, the extract can first be applied five days, one week, or two weeks prior to harvest. Further in this aspect, application at this later stage can be particularly useful with fruits or vegetables in which an increase in a nutrient or flavor-enhancing component can be obtained by protecting the fruit or vegetable from UV radiation later in its on-plant life.

(193) In one aspect, the genelight extract can be applied once or multiple times to each fruit or vegetable. In another aspect, it can be applied weekly, or it can be reapplied after the fruit or vegetable is exposed to rain or after a turning process. In another aspect, the agricultural plant can be another food crop that grows above ground and is exposed to natural UV radiation, wherein the agricultural product produced can be a fruit, leaf, seed, flower, grain, nut, stem, vegetable, or mushroom.

(194) In another aspect, it is desirable for agricultural plants that do not produce parts typically consumed by humans to be protected from UV irradiation. In a further aspect, these other agricultural plants can includes sources of fibers such as, for example, cotton and linen (flax), of cork, of wood or lumber, of feedstocks for producing ethanol or biodiesel (including, but not limited to, sugar beet, sugarcane, cassava, sorghum, corn, wheat, oil palm, coconut, rapeseed,

peanut, sunflower, soybean, and the like), of animal feedstocks or fodder, or of decorative or horticultural plants.

(195) In one aspect, any part of the plant can be coated with the genelight extract, including, but not limited to, the part of the plant that is collected during harvest. In an alternative aspect, the harvested part of the plant is not coated, but another part can be coated with the extracts disclosed herein. In addition to the aspects already described, in one aspect, coating a plant with the extracts described herein can prolong the life of the plant, increase production capacity of a desired product, can increase the growth rate of the plant relative to an untreated plant of the same type, can increase production of a desired metabolite that might otherwise decrease due to UV-induced stress, can increase yield of a crop of such plants, and the like.

(196) In a further aspect, application can be accomplished with a commercial sprayer. In another aspect, application can be only on the upper portions of the fruit or vegetable, which are exposed to substantially greater amounts of UV radiation than the lower portions of the fruit or vegetable.

(197) Cosmetics and Pharmaceutical Compositions Containing the Extracts

(198) In another aspect, provided herein is a pharmaceutical composition containing the genelight extracts described herein. In one aspect, the pharmaceutical composition can be applied to a subject, wherein the subject is exposed to radiation. In one aspect, the radiation is applied as a strategy to treat cancer. In another aspect, the pharmaceutical composition is used to prevent radiation-induced cellular and DNA damage. In another aspect, dosage ranges of the extract in the pharmaceutical composition can vary from about 0.01 g extract/mL of pharmaceutical composition to about 1 g extract/mL of pharmaceutical composition, or can be about 0.01 g extract/mL, about 0.02 g extract/mL, about 0.025 g extract/mL, about 0.05 g extract/mL, about 0.075 g extract/mL, or about 1 g extract/mL of pharmaceutical composition, where any value can be a lower and upper endpoint of a range (e.g., about 0.01 g extract/mL to about 0.075 g extract/mL, about 0.025 g extract/mL to about 0.05 g extract/mL, etc.). In an alternative aspect, provided herein is a cosmetic composition containing the genelight extracts produced herein. Further in this aspect, the cosmetic composition can be a cleanser, lotion, cream, shampoo, hair treatment, makeup, lip treatment, nail treatment, or related composition. In still a further aspect, the compositions containing the extracts can have both pharmaceutical and cosmetic applications. In yet another aspect, the compositions containing the extracts can be used in veterinary medicine.

(199) The cosmetic compositions can be formulated in any physiologically acceptable medium typically used to formulate topical compositions. The cosmetic compositions can be in any galenic form conventionally used for a topical application such as, for example, in the form of dispersions of aqueous gel or lotion type, emulsions of liquid or semi-liquid consistency of the milk type, obtained by dispersing a fatty phase in an aqueous phase (O/W) or vice versa (W/O), or suspensions or emulsions of soft, semi-solid or solid consistency of the cream or gel type, or alternatively multiple emulsions (W/O/VV or O/W/O), microemulsions, vesicular dispersions of ionic and/or non-ionic type, or wax/aqueous phase dispersions. These compositions are prepared according to the usual methods.

(200) The cosmetic compositions can also contain one or more additives commonly used in the cosmetics field, such as emulsifiers, preservatives, sequestering agents, fragrances, thickeners, oils, waxes or film-forming polymers. In one aspect, in any of the above scenarios, the pharmaceutical, cosmetic, or veterinary composition also includes additional UV-protective compounds or UV-blocking agents such as, for example, zinc oxide, titanium dioxide, carotenoids, oxybenzone, octinoxate, homosalate, octisalate, octocrylene, avobenzone, or a combination thereof.

(201) In one aspect, the composition is a sunscreen. A sunscreen can be formulated with any of the extracts produced herein. In addition to the extract, the sunscreen in certain aspects can be formulated with one or more UV-protective compounds or UV-blocking agents listed above. The sunscreen can be formulated as a paste, lotion, cream, aerosol, or other suitable formulations for topical use. In certain aspects, the sunscreen can be formulated as a transparent composition.

(202) In one aspect, the cosmetic composition can be a film composed of the genelight extracts produced herein that can be directly applied to the skin. For example, the film can be composed of a biocompatible material such as a protein or oligonucleotide, where the extract is coated on one or more surfaces of the film or, in the alternative dispersed throughout the film. For example, the film can be composed of DNA. In this application, the films can be used as a wound covering and provide protection from UV photodamage. The films can also be prepared so that they are optically transparent. Here, it is possible to view the wound without removing the covering and exposing the wound. The films can also include other components useful in cosmetic applications such as, for example, compounds to prevent or reduce wrinkles.
(203) In one aspect, the pharmaceutical, cosmetic, or veterinary compositions described herein are applied to subjects. In one aspect, the subject is a human, another mammal, or a bird. In a further aspect, the mammal is a pet such as a dog or cat or is livestock such as horses, goats, cattle, sheep, and the like. In an alternative aspect, the bird is a pet bird or is poultry such as, for example, a chicken or turkey. In any of these aspects, the compositions can be applied to skin, fur, feathers, wool, hooves, horns, or hair as appropriate and applicable.
(204) In a related aspect, the compositions and extracts disclosed herein and/or the pharmaceutical, cosmetic, or veterinary compositions disclosed herein are applied to isolated human or animal cells. Further in this aspect, the compositions and extracts and/or the pharmaceutical, cosmetic, or veterinary compositions disclosed herein can be applied to the isolated human or animal cells in any desired volume ratio such as, for example, from about 1:7 to about 7:1, or about 1:7, 2:5, 5:4, 5:2, or about 7:1. In one aspect, the compositions or extracts disclosed herein are applied to human fibroblast cells in culture at a volume ratio of about 5:4.
(205) Further in this aspect, the compositions and extracts disclosed herein protect the isolated human or animal cells from UV-induced cell death. Still further in this aspect, this protection can be assessed by known means such as, for example, staining with trypan blue and counting living and/or dead cells under a microscope, or by assessing total amount of genomic DNA present in the sample by a means such as, for example, gel electrophoresis followed by quantification of stain intensity.
(206) Paints, Inks, Dyes, and Stains
(207) In another aspect, provided herein is a paint, dye, stain, or ink containing the genelight extracts disclosed herein. In one aspect, there are several benefits to having a paint that is resistant to UV irradiation. In a further aspect, imparting UV resistance to a paint slows or stops photodegradation, bleaching, or color fading. In another aspect, a paint with UV resistance prevents chemical modification of exposed paint surfaces. Further in this aspect, chemical modification of exposed paint surfaces includes change in finish, structural changes in binders, flaking, chipping, and the like. In one aspect, the paint provided herein resists these changes.
(208) Articles Incorporating the Genelight Extracts
(209) In still another aspect, provided herein is an article coated with the genelight extracts disclosed herein. In one aspect, the article is made of glass, plastic, metal, wood, fabric, or any combination thereof. In one aspect, the article is a construction material such as, for example, steel, concrete or cement, brick, wood, window or door glass, fiberglass, siding, wallboard, a flooring material, masonry, mortar, grout, stone, artificial stone, stucco, shingles, roofing materials, and the like. In an alternative aspect, the material is an aeronautical or aerospace material such as, for example, the metal or metal alloy body of an aircraft or spacecraft, paint on the body of an aircraft or spacecraft, glass windows on an aircraft or spacecraft, carbon fiber composite, titanium or aluminum, a ceramic heat absorbing tile, and the like. In still another aspect, the article is a fabric article such as, for example, clothing, drapes, outdoor upholstery, a tent or outdoor pavilion, a flag or banner, or the like. In another aspect, the extract can be applied to the article to fine artwork, solid pieces (e.g., vases), and historical documents in order to preserve them. In another aspect, the extract can be applied to outdoor signs such as highway billboards and advertising.

(210) In other aspects, the genelight extract can be incorporated within or throughout the article. In one aspect, the extract can be mixed with molten glass to produce glass article that are UV resistant such as, for example, sunglasses, car windshields, window glass, and eyeglasses. In another aspect, the glass article can be a bottle for storing a beverage or food container in order to increase the shelf-life of the beverage or food. It is contemplated that the extract can be applied externally to the glass articles as well.
(211) In another aspect, the genelight extract can be mixed with fiberglass or plastics in order to reduce negative effects to aircraft, watercraft, boats, jet skis, decking, house siding, motor homes, sunroofs, and moon roofs that are constantly exposed to UV radiation. It is contemplated that the extract can be applied externally to the fiberglass or plastic articles as well.
(212) In another aspect, the genelight extract can be mixed with rubber, silicon, or latex used to make a variety of articles such as water hoses, tires, and the like. It is contemplated that the extract can be applied externally to the rubber, silicone, or latex articles as well.
(213) In another aspect, the genelight extract can be mixed with foams used to make a variety of articles such as automotive dashboard padding, seat cushions, and the like. It is contemplated that the extract can be applied externally to the foam articles as well.
(214) In another aspect, the genelight extracts described herein can be incorporated into an optical film. In one aspect, the extract is applied to at least one surface of the film. In another aspect, the extract can be dispersed throughout the film. The film can be transparent, translucent or opaque. The film can be composed of, but not limited to, polyolefin resin, such as polyethylene (PE) or polypropylene (PP); polyester resin, such as polyethylene terephthalate (PET); polyacrylate resin, such as polymethyl (meth)acrylate (PMMA); polycarbonate resin; polyurethane resin or a mixture thereof. The optical film can be applied to any substrate where it is desirable to reduce or prevent UV exposure or damage. For example, the optical film can be applied to windows to reduce or prevent UV radiation from entering a structure (e.g., building, vehicle, etc.).
(215) Methods for Reducing or Preventing Exposure to UV Radiation
(216) In another aspect, provided herein is a method of reducing or preventing the exposure of an item to UV radiation by applying the genelight extracts described herein to the item or incorporating the extract within/throughout the article. Further in this aspect, “reducing” is defined relative to an untreated control. That is, if two like items are exposed to equal amounts of UV radiation for an equal amount of time, but one has been treated with the UV-resistant extracts and the other has not, and some objective response is measured (e.g., color fading, structural degradation, plant size or yield, etc.), the treated item will appear to have been exposed to a lower amount of UV (for example, the color of the treated item will have faded less and will remain closer to the original, or a treated plant will appear larger and more vigorous and will have a greater yield, etc.). In some aspects, treatment with the extracts disclosed herein will prevent UV exposure from occurring. As used herein, “prevent” indicates that a treated item will not be affected, changed, or altered by UV exposure.
(217) In one aspect, the genelight extract blocks from about 50% to about 100% of UV radiation from contacting the item. Further in this aspect, the extract blocks about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, about 99%, or about 100% of UV radiation from contacting the item, where any value can be a lower and upper endpoint of a range (e.g., about 50% to about 90%, about 60% to about 80%, etc.). In another aspect, the extract blocks from about 50% to about 100% of longwave UV radiation from contacting the item. Further in this aspect, the extract blocks about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, about 99%, or about 100% of longwave UV radiation from contacting the item, where any value can be a lower and upper endpoint of a range (e.g., about 50% to about 90%, about 60% to about 80%, etc.). In one aspect, the extract blocks from about 50% to 100% of shortwave UV radiation from contacting the item. Further in this aspect, the extract blocks about 50%, about 55%, about 60%, about 65%,

about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, about 99%, or about 100% of shortwave UV radiation from contacting the item, where any value can be a lower and upper endpoint of a range (e.g., about 50% to about 90%, about 60% to about 80%, etc.).
(218) Depending upon the application, the genelight extract can prevent or reduce damage cause by UV radiation from limited to extended periods of time. By varying the amount of extract that is applied as well as the number of times the extract is applied, the degree of UV protection can be varied. In certain aspects, it may be desirable for the article to be protected from UV damage for a short period of time then subsequently biodegrade.
(219) Preventing or Reducing the Growth of Barnacles
(220) In another aspect, the genelight extracts produced herein can be used to reduce or prevent the growth of barnacles on boats and other water vehicles. In one aspect, the extract can be admixed with a paint that is typically applied to water vehicles, where the paint also includes chitosan. In one aspect, the chitosan can be acetylated to a specific degree of acetylation in order to enhance tissue growth during culturing as well as metabolite production. In one aspect, the chitosan is from about 60% to about 100%, or about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, about 90%, about 95%, about 99%, or about 100%, where any value can be a lower and upper endpoint of a range (e.g., about 60% to about 90%, about 70% to about 80%, etc.). In one aspect, chitosan isolated from the shells of crab, shrimp, lobster, and/or krill is useful herein. The molecular weight of the chitosan can vary, as well. For example, the chitosan comprises about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 glucosamine units and/or N-acetylglucosamine units, where any value can be a lower and upper endpoint of a range (e.g., 3 to about 19, 5 to 7, etc.).
(221) Protection of Microbial Cultures from the Effects of UV Radiation
(222) In one aspect, the compositions and extracts disclosed herein can be applied to cultures of commercially-important microorganisms. Further in this aspect, the compositions and extracts disclosed herein offer protection from UV radiation to the microorganisms. In one aspect, the microorganisms are used for the production of industrially-important enzymes or chemical compounds such as, for example, *Acetobacter aceti* (source of acetic acid), *Aspergillus niger* (source of citrus acid), any microorganism that produces enzymes such as, for example, cellulases, amylase, palatase, lipozyme, lipase, lipopan F, xylose isomerase, resinase, penicillin amidase, or amidase. In another aspect, the microorganisms are important for fermentation in food production, including, but not limited to, *Saccharomyces cerevisiae* for the production of bread and for alcoholic fermentation, bacteria such as *Streptococcus* thermophiles, *Lactobacillus delbrueckii*, and other lactobacilli and bifidobacteria important in the production of yogurt, skyr, kefir, and other fermented dairy products, *Lactococcus* species and *Propionibacter shermani* used in the production of some cheeses, *Clostridium butyricum* useful in the process of retting of jute, hemp, and/or flax, microorganisms used for water treatment and/or sewage treatment, experimental microorganisms useful in molecular biology, genetic engineering, and other laboratory uses including, but not limited to, *E. coli*, *B. subtilis*, and the like, microoragnisms useful in the production of B vitamins and other vitamins, microorganisms such as *Bacillus thuringiensis* useful for agricultural pest control, and the like.
(223) In one aspect, the cultures and extracts disclosed herein can be mixed in any proportion with cultures of commercially important microorganisms for the purpose of providing protection from UV-irradiation to the commercially important microorganisms. In one aspect, the cultures and extracts can be applied in a volume ratio of from 1:7 to 7:1 to cultures of the commercially important microorganisms, or can be applied in a ratio of about 1:7, 2:5, 3:4, 4:3, 5:2, 7:1, or a combination of any of the foregoing values, or a range encompassing any of the foregoing values. In a further aspect, the cultures and extracts are applied in a ratio of 5:2 to cultures of commercially important microorganisms.
(224) In another aspect, treatment with the cultures and extracts disclosed herein protects the

commercially important microorganisms from negative effects of UV radiation for a period of from 10 minutes to 2 hours, or for about 10 minutes, 15 minutes, 20 minutes, 30 minutes, 1 hour, 90 minutes, or about 2 hours, or a combination of any of the foregoing values or a range encompassing any of the foregoing values. In one aspect, successful protection from negative effects of UV radiation can be assessed by culturing cells of the commercially important microorganism in a Petri dish and counting colonies, or by any other common means for assessing cell survival, with higher numbers of surviving cells being indicative of successful UV protection. In any of the above aspects, the cultures and extracts disclosed herein are biocompatible and do not harm the commercially important microorganisms.

ASPECTS

(225) The present disclosure can be described in accordance with the following numbered Aspects, which should not be confused with the claims. Aspect 1. A DNA construct comprising the following genetic components: (a) a gene that expresses a heat shock protein, (b) a gene that expresses RuBisCO large subunit 1, (c) a gene that expresses tonB, (d) a gene that expresses hydrogenase, and (e) a gene that expresses p-type ATPase. Aspect 2. The DNA construct of aspect 1, wherein the gene that expresses a heat shock protein expresses HSP70. Aspect 3. The DNA construct of aspect 2, wherein the gene that expresses HSP70 has SEQ ID NO. 1 or at least 70% homology thereto. Aspect 4. The DNA construct of any preceding aspect, wherein the gene that expresses RuBisCO large subunit 1 has SEQ ID NO. 2 or at least 70% homology thereto. Aspect 5. The DNA construct of any preceding aspect, wherein the gene that expresses tonB has SEQ ID NO. 3 or at least 70% homology thereto. Aspect 6. The DNA construct of any preceding aspect, wherein the gene that expresses hydrogenase has SEQ ID NO. 4 or at least 70% homology thereto. Aspect 7. The DNA construct of any preceding aspect, wherein the gene that expresses p-type ATPase has SEQ ID NO. 5 or at least 70% homology thereto. Aspect 8. The DNA construct in any preceding aspect, wherein the DNA construct further comprises a promoter. Aspect 9. The DNA construct of aspect 8, wherein the promoter comprises a GAL1 promoter, an araBAD promoter, or both. Aspect 10. The DNA construct in any preceding aspect, wherein the DNA construct further comprises a terminator. Aspect 11. The DNA construct of aspect 10, wherein the terminator comprises a CYC1 terminator, an rrnB terminator, or a combination thereof. Aspect 12. The DNA construct of aspect 9, wherein a GAL1 promoter is positioned before (a) the gene that expresses a heat shock protein, (b) the gene that expresses RuBisCO large subunit 1, (c) the gene that expresses tonB, (d) the gene that expresses hydrogenase, (e) the gene that expresses p-type ATPase, or any combination thereof. Aspect 13. The DNA construct of aspect 11, wherein a CYC1 terminator is positioned after (a) the gene that expresses a heat shock protein, (b) the gene that expresses RuBisCO large subunit 1, (c) the gene that expresses tonB, (d) the gene that expresses hydrogenase, (e) the gene that expresses p-type ATPase, or any combination thereof. Aspect 14. The DNA construct of aspect 9, wherein an araBAD promoter is positioned before (a) the gene that expresses a heat shock protein, (b) the gene that expresses RuBisCO large subunit 1, (c) the gene that expresses tonB, (d) the gene that expresses hydrogenase, (e) the gene that expresses p-type ATPase, or any combination thereof. Aspect 15. The DNA construct of aspect 11, wherein an rrnB terminator is positioned after (a) the gene that expresses a heat shock protein, (b) the gene that expresses RuBisCO large subunit 1, (c) the gene that expresses tonB, (d) the gene that expresses hydrogenase, (e) the gene that expresses p-type ATPase, or any combination thereof. Aspect 16. The DNA construct of any preceding aspect, wherein the DNA construct further comprises a gene that confers resistance to an antibiotic. Aspect 17. The DNA construct of aspect 16, wherein the antibiotic comprises tetracycline, neomycin, kanamycin, ampicillin, hygromycin, chloramphenicol, amphotericin B, bacitracin, carbapenam, cephalosporin, ethambutol, fluoroquinolones, isoniazid, methicillin, oxacillin, vancomycin, streptomycin, quinolones, rifampin, rifampicin, sulfonamides, cephalothin, erythromycin, gentamicin, penicillin, or a combination thereof. Aspect 18. The DNA construct in any preceding aspect, wherein the construct further comprises a gene that expresses a reporter protein. Aspect 19.

The DNA construct of aspect 18, wherein the reporter protein is a fluorescent protein. Aspect 20. The DNA construct of aspect 19, wherein the reporter protein comprises a red fluorescent protein, a cyan fluorescent protein, a green fluorescent protein, a yellow fluorescent protein, or a combination thereof. Aspect 21. The DNA construct of aspect 20, wherein the reporter protein is a green fluorescent protein. Aspect 22. The DNA construct of aspect 21, wherein the green fluorescent protein is EGFP. Aspect 23. The DNA construct of aspect 18, wherein the gene that expresses a reporter protein has SEQ ID NO. 6 or at least 70% homology thereto. Aspect 24. The DNA construct of aspect 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses RuBisCO large subunit 1, (3) a gene that expresses tonB, (4) a gene that expresses hydrogenase, and (5) a gene that expresses p-type ATPase. Aspect 25. The DNA construct of aspect 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (5) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 26. The DNA construct of aspect 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (7) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 27. The DNA construct of aspect 1, wherein the DNA construct has SEQ ID NO. 7 or at least 70% homology thereto. Aspect 28. The DNA construct of aspect 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a gene that expresses p-type ATPase, (3) a gene that expresses tonB, (4) a gene that expresses a heat shock protein, and (5) a gene that expresses RuBisCO large subunit 1. Aspect 29. The DNA construct of aspect 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, and (5) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto. Aspect 30. The DNA construct of aspect 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (8) a CYC1 terminator, (8) a GAL1 promoter, (9) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (10) a CYC1 terminator, (11) a GAL1 promoter, (12) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, and (13) a CYC1 terminator. Aspect 31. The DNA construct of aspect 1, wherein the DNA construct has SEQ ID NO. 8 or at least 70% homology thereto. Aspect 32. The DNA construct in any one of aspects 1-31, wherein the construct further comprises a gene that expresses a dehydrogenase. Aspect 33. The DNA construct of aspect 32, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses dehydrogenase, (3) a gene that expresses RuBisCO large subunit 1, (4) a gene that expresses tonB, (5) a gene that expresses hydrogenase, and (6) a gene that expresses p-type ATPase. Aspect 34. The DNA construct of aspect 32, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (3) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (4) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (5) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (6) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 35. The DNA construct of aspect 32, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (3) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (4) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (5) an rrnB terminator, (6) an araBAD promoter, (7) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (8) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 36. The DNA construct of aspect 32, wherein the DNA construct has SEQ ID NO. 11 or at least 70% homology thereto. Aspect 37. The DNA construct of aspect 32, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a gene that expresses p-type ATPase, (3) a gene that expresses tonB, (4) a gene that expresses a heat shock protein, (5) a gene that expresses dehydrogenase, and (6) a gene that expresses RuBisCO large subunit 1. Aspect 38. The DNA construct of aspect 32, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (5) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, and (6) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto. Aspect 39. The DNA construct of aspect 32, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a CYC1 terminator, (3) a GAL1 promoter, (4) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (5) a CYC1 terminator, (6) a GAL1 promoter, (7) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (8) a CYC1 terminator, (8) a GAL1 promoter, (9) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (10) a CYC1 terminator, (11) a GAL1 promoter, (12) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (13) a CYC1 terminator, (14) a GAL1 promoter, and (15) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, and (13) a CYC1 terminator. Aspect 40. The DNA construct of aspect 1, wherein the DNA construct has SEQ ID NO. 10 or at least 70% homology thereto. Aspect 41. A vector comprising the DNA construct in any preceding aspect. Aspect 42. The vector of aspect 41, wherein the vector is a plasmid. Aspect 43. The vector of aspect 42, wherein the plasmid is pWLNEO, pSV2CAT, pOG44, pXTI, pSG, pSVK3, pBSK, pBR322, pYES, pYES2, pBSKII, pUC, or pBAD. Aspect 44. The vector of aspect 43, wherein the plasmid is pBAD. Aspect 45. The vector of aspect 43, wherein the plasmid is pYES2. Aspect 46. A biological device comprising host cells transformed with the DNA construct or vector of any preceding aspect. Aspect 47. The biological device of aspect 46, wherein the host cells comprise yeast or bacteria. Aspect 48. The biological device of aspect 47, wherein the bacteria comprise *Escherichia coli.* Aspect 49. The biological device of aspect 47, wherein the yeast comprise *Saccharomyces cerevisiae.* Aspect 50. An extract produced by culturing the biological device of any of aspects 46-49 in a culture medium. Aspect 51. The extract of aspect 50, wherein the extract is formulated as a liquid, a slurry, a powder, or a mixture thereof. Aspect 52. A battery comprising an electrolyte, wherein the electrolyte comprises the extract of any aspect 50 or 51. Aspect 53. An article or device comprising the extract of aspect 50 or 51 or the battery of aspect 44. Aspect 54. The article or device of aspect 53, wherein the article or device comprises a standalone light, an indicator light on a larger device, a lighted panel on a larger device, a computer, a tablet, a smartphone, a medical device, a portable heating device, a portable cooling device, a grooming or personal care device, a lighted sign, a toy, a scientific instrument, a pump, a transportation device or vehicle, a kitchen device, or a household appliance. Aspect 55. The article or device of aspect 53 or 54, wherein the device includes a light emitting diode. Aspect 56. A method of culturing cells comprising growing the cells in a medium comprising the extract of aspect 50 or 51. Aspect 57. The method of aspect 56, wherein the cells comprise bacteria or fungi. Aspect 58. The method of aspect 57, wherein the bacteria or fungi comprise *Saccharomyces cerevisiae*, a *Kluyveromyces* species, a *Candida* species, a *Lactobacillus* species, *Pyrococcus furiosus, Thermus aquaticus, Bacillus stearothermophilus, Thermus filiformis, Thermus* thermophiles, a *Xanthomonas* species, *Aspergillus niger* or another *Aspergillus* species, a *Trichoderma* species, *Mucor miehei*, a *Rhizopus* species, a *Mortierella* species, *Clostridium acetobutylicum* or another *Clostridium* species, a *Streptomyces* species, a *Penicillium* species, *Acetobacter acetii, Bacillus thuringiensis* or another *Bacillus* species, *E. coli*, a *Klebsiella* species, or *Trichoderma polysporum.* Aspect 59. A plant grown by the process comprising contacting plant gamete cells, a plant reproductive organ, or a plant callus with the extract of aspect 50 or 51. Aspect 60. The plant of aspect 59, wherein the plant is produced by a method comprising the steps of: a. contacting the plant callus with the extract; b. culturing the plant callus; and c. growing the plant from the plant callus. Aspect 61. A pharmaceutical composition comprising the extract of aspect 50 or 51. Aspect 62. The pharmaceutical composition of aspect 61, further comprising a UV-blocking agent. Aspect 63. The pharmaceutical composition of aspect 62, wherein the UV-blocking agent is zinc oxide, titanium dioxide, a carotenoid, oxybenzone, octinoxate, homosalate, octisalate, octocrylene, avobenzone, or any combination thereof. Aspect 64. A sunscreen comprising the extract of aspect 50 or 51. Aspect 65. A paint, ink, dye, or stain comprising the extract of aspect 50 or 51. Aspect 66. A plant coated with the extract of aspect 50 or 51. Aspect 67. An agricultural product coated with the extract of aspect 50 or 51. Aspect 68. The agricultural product of aspect 67, wherein the agricultural product comprises fruits, leaves, seeds, flowers, grains, nuts, stems, vegetables, or mushrooms. Aspect 69. A method for improving a physiological property of a plant, the method comprising applying the extract of aspect 50 or 51 to the plant, wherein the property is improved compared to the property of the same plant that has not been applied the extract. Aspect 70. The method of aspect 69, wherein the property comprises increased root tension, root length, hormone production, drought tolerance, disease resistance, photosynthesis, or any combination thereof. Aspect 71. The method of aspect 69 or 70, wherein the plant is grass, trees, bushes, shrubs, flower, vines, coffee, soybean, or cotton. Aspect 72. The method of aspect 71, wherein the grass is growing on a golf course, a lawn, or an athletic playing field. Aspect 73. The method of any of aspects 60-72, wherein the extract is applied in an amount of from about 50 to about 200 mL per square meter of grass. Aspect 74. The method of aspect 73, wherein about 75 mL of extract are applied per square meter of grass. Aspect 75. The method of aspect 73, wherein about 150 mL of extract are applied per square meter of grass. Aspect 76. The method of any of aspects 69-75, wherein the extract is applied twice per week. Aspect 77. The method of any of aspects 69-75, wherein the extract is applied in combination with a second compound that promotes plant growth. Aspect 78. The method of aspect 77, wherein the second compound is applied before the extract. Aspect 79. The method of aspect

77, wherein the second compound is applied after the extract. Aspect 80. The method of aspect 77, wherein the second compound is applied simultaneously with the extract. Aspect 81. The method of aspect 77, wherein the second compound is a polysaccharide, a fertilizer, or a combination thereof. Aspect 82. The method of any one of aspects 77-81, wherein the second compound is chitosan, a polyactive carbohydrate, or a combination thereof. Aspect 83. The method of aspect 82, wherein the second compound is in an aqueous solution having a concentration of from about 0.01% to about 0.05%. Aspect 84. The method of aspect 83, wherein the second compound has a concentration of about 0.02%. Aspect 85. The method of aspect 81, wherein the fertilizer is activated sewage sludge. Aspect 86. An article comprising the extract of aspect 50 or 51, wherein the article is coated with the extract, the extract is dispersed throughout the article, or a combination thereof. Aspect 87. The article of aspect 86, wherein the article is made of glass, fiberglass, plastic, metal, wood, fabric, foam, rubber, latex, silicone, or any combination thereof. Aspect 88. A method of reducing or preventing exposure of an item to UV radiation comprising applying to the item the extract in aspect 50 or 51. Aspect 89. The method of aspect 88, wherein the extract blocks at least 50% of UV radiation from contacting the item. Aspect 90. The method of aspect 88, wherein the extract blocks at least 50% of longwave UV radiation from contacting the item. Aspect 91. The method of aspect 88, wherein the extract blocks at least 50% of shortwave UV radiation from contacting the item. Aspect 92. The method of aspect 88, wherein the item comprises the skin of a subject. Aspect 93. The method of aspect 88, wherein the item comprises an agricultural product. Aspect 94. The method of aspect 88, wherein the item comprises a construction material, an aeronautical material, or an aerospace material. Aspect 95. The method of aspect 88, wherein the item comprises a culture of a microorganism. Aspect 96. The method of aspect 88, wherein the item comprises a culture of isolated human or animal cells. Aspect 97. A method for reducing or preventing the growth of barnacles on a surface, the method comprising applying a paint comprising the extract in aspect 50 or 51 and chitosan to the surface. Aspect 98. The method of aspect 97, wherein the chitosan is from 60% to 100% acetylated and has from 3 to 20 glucosamine units, N-acetylglucosamine units, or a combination thereof. Aspect 99. A cosmetic composition comprising a physiologically acceptable medium and the extract of aspect 50 or 51. Aspect 100. An optical film comprising the extract of aspect 50 or 51. Aspect 101. A method of reducing or preventing exposure of an item to UV radiation comprising applying to the item the extract of aspect 50 or 51. Aspect 102. The method of aspect 101, wherein the extract blocks at least 50% of UV radiation from contacting the item. Aspect 103. The method of aspect 101, wherein the extract blocks at least approximately 50% of longwave UV radiation from contacting the item. Aspect 104. The method of aspect 101, wherein the extract blocks at least approximately 50% of shortwave UV radiation from contacting the item. Aspect 105. The method of aspect 101, wherein the item comprises the skin of a subject. Aspect 106. The method of aspect 101, wherein the item comprises an agricultural product. Aspect 107. The method of aspect 101, wherein the item comprises a construction material, and aeronautical, or an aerospace material. Aspect 108. An article comprising the extract of aspect 50 or 51, wherein the article is coated with the extract, the extract is dispersed throughout the article, or a combination thereof. Aspect 109. The article of aspect 108, wherein the article is made of glass, fiberglass, plastic, metal, wood, fabric, foam, rubber, latex, silicone, or any combination thereof. Aspect 110. A composition comprising water, genelight extract of aspects 50-51, and one or more one or more polysaccharides. Aspect 111. The composition of aspect 110, wherein the polysaccharide comprises chitosan, a polyactive carbohydrate, or a combination thereof. Aspect 112. The composition of aspects 110-111, wherein the genelight extract is less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, less than 0.1% by weight, less than 0.5% by weight, or less than 0.3% by weight of the composition. Aspect 113. The composition of aspects 110-112, wherein chitosan is less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, less than 0.1% by weight, less than 0.5% by weight, or less than 0.3% by weight of the composition. Aspect 114. The composition of aspects 110-113, wherein the polyactive carbohydrate is less than 1% by weight, less than 0.75% by weight, less than 0.5% by weight, less than 0.25% by weight, less than 0.1% by weight, less than 0.5% by weight, or less than 0.3% by weight of the composition. Aspect 115. A method for improving a physiological property of a plant, the method comprising applying to the plant a composition comprising water, RuBisCO, and a heat shock protein, wherein the property is improved compared to the property of the same plant that has not been applied the composition. Aspect 116. The method of Aspect 115, wherein the property comprises increased root tension, root length, hormone production, drought tolerance, disease resistance, photosynthesis, or any combination thereof. Aspect 117. The method of Aspect 115, wherein the plant is grass, trees, bushes, shrubs, flower, vines, coffee, soybean, or cotton. Aspect 118. The method of Aspect 117, wherein the grass is growing on a golf course, a lawn, or an athletic playing field. Aspect 119. The method of Aspect 115, wherein the heat shock protein comprises HSP70. Aspect 120. The method of Aspect 115, wherein the composition is produced by culturing a biological device in a culture medium, wherein the biological device comprises host cells transformed with a DNA construct comprising the following genetic components: (a) a gene that expresses a heat shock protein, (b) a gene that expresses RuBisCO large subunit 1, (c) a gene that expresses tonB, (d) a gene that expresses hydrogenase, and (e) a gene that expresses p-type ATPase. Aspect 121. The method of Aspect 120, wherein the gene that expresses a heat shock protein expresses HSP70. Aspect 122. The method of Aspect 120, wherein the construct further comprises a gene that expresses a reporter protein. Aspect 123. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses RuBisCO large subunit 1, (3) a gene that expresses tonB, (4) a gene that expresses hydrogenase, and (5) a gene that expresses p-type ATPase. Aspect 124. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (5) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 125. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (7) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 126. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a gene that expresses p-type ATPase, (3) a gene that expresses tonB, (4) a gene that expresses a heat shock protein, and (5) a gene that expresses RuBisCO large subunit 1. Aspect 127. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, and (5) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto. Aspect 128. The method of Aspect 120, wherein the construct further comprises a gene that expresses a dehydrogenase. Aspect 129. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein, (2) a gene that expresses dehydrogenase, (3) a gene that expresses RuBisCO large subunit 1, (4) a gene that expresses tonB, (5) a gene that expresses hydrogenase, and (6) a gene that expresses p-type ATPase. Aspect 130. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (2) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, (3) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto, (4) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (5) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, and (6) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto. Aspect 131. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase, (2) a gene that expresses p-type ATPase, (3) a gene that expresses tonB, (4) a gene that expresses a heat shock protein, (5) a gene that expresses dehydrogenase, and (6) a gene that expresses RuBisCO large subunit 1. Aspect 132. The method of Aspect 120, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having SEQ ID NO. 4 or at least 70% homology thereto, (2) a gene that expresses p-type ATPase having SEQ ID NO. 5 or at least 70% homology thereto, (3) a gene that expresses tonB having SEQ ID NO. 3 or at least 70% homology thereto, (4) a gene that expresses a heat shock protein having SEQ ID NO. 1 or at least 70% homology thereto, (5) a gene that expresses dehydrogenase having SEQ ID NO. 9 or at least 70% homology thereto, and (6) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2 or at least 70% homology thereto. Aspect 133. The method of Aspect 120, wherein the DNA construct has SEQ ID NOS. 7, 8, 10, 11, or at least 70% homology thereto. Aspect 134. The method of Aspect 120, wherein the DNA construct is incorporated in a vector. Aspect 135. The method of Aspect 134, wherein the vector is a plasmid selected from the group consisting of pWLNEO, pSV2CAT, pOG44, pXTI, pSG, pSVK3, pBSK, pBR322, pYES, pYES2, pBSKII, pUC, or pBAD. Aspect 136. The method of Aspect 120, wherein the host cells comprise bacteria or fungi. Aspect 137. The method of Aspect 115, wherein the composition further comprises chitosan, a polyactive carbohydrate, or a combination thereof. Aspect 138. The method of Aspect 137, wherein chitosan is less than 1% by weight of the composition. Aspect 139. The method of Aspect 137, wherein the polyactive carbohydrate is less than 1% by weight of the composition.

EXAMPLES

(226) The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how the compounds, compositions, and methods described and claimed herein are made and evaluated, and are intended to be purely exemplary and are not intended to limit the scope of what the inventors regard as their invention. Efforts have been made to ensure accuracy with respect to numbers (e.g., amounts, temperature, etc.) but some errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, temperature is in ° C. or is at ambient temperature, and pressure is at or near atmospheric. Numerous variations and combinations of reaction conditions, e.g., component concentrations, desired solvents, solvent mixtures, temperatures, pressures, and other reaction ranges and conditions can be used to optimize the product purity and yield obtained from the desired process. Only reasonable and routine experimentation will be required to optimize such processes and conditions.

Example 1: Preparation of DNA Construct for the Production of Genelight Cultures

(227) The DNA construct was composed of the genetic components described herein and assembled in plasmid vectors (e.g., pYES2, pBAD). Sequences of genes and/or proteins with desired properties were identified in GenBank; these included a gene that expresses HSP70, a gene that expresses RuBisCO large subunit 1, a gene that expresses tonB, a gene that expresses

hydrogenase, and a gene that expresses P-type ATPase. These sequences were synthesized by CloneTex Systems, Inc. (Austin, TX). Other genetic parts were also obtained for inclusion in the DNA constructs including, for example, promoter genes (e.g., GAL1 promoter and/or araBAD promoter), reporter genes (e.g., EGFP), and terminator sequences (e.g., CYC1 terminator and/or rrnB terminator). These genetic parts included restriction sites for ease of insertion into plasmid vectors.

(228) The cloning of the DNA construct into the biological devices was performed as follows. Sequences of individual genes were amplified by polymerase chain reaction using primers that incorporated restriction sites at their 5′ ends to facilitate construction of the full sequence to be inserted into the plasmid. Genes were then ligated using standard protocols to form an insert. The plasmid was then digested with restriction enzymes according to directions and using reagents provided by the enzymes' supplier (Promega). The complete insert, containing restriction sites on each end, was then ligated into the plasmid. Successful construction of the insert and ligation of the insert into the plasmid were confirmed by gel electrophoresis.

(229) In some experiments, PCR-amplified pieces of all gene fragments were combined by using homologous recombination technology (Gibson Assembly).

(230) From 5′ to 3′, one version of the construct includes (a) a gene that expresses HSP70, (b) a gene that expresses RuBisCO large subunit 1, (c) a gene that expresses tonB, (d) a gene that expresses rrnB terminator, (e) a gene that expresses araBAD promoter, (f) a gene that expresses hydrogenase, (g) a gene that expresses P-type ATPase, and (h) a gene that expresses EGFP (FIGS. **1** and **2**).

(231) From 5′ to 3′, a second version of the construct includes (a) a gene that expresses hydrogenase, (b) a CYC1 terminator, (c) a GAL1 promoter, (d) a gene that expresses P-type ATPase, (e) a CYC1 terminator, (f) a GAL1 promoter, (g) a gene that expresses tonB, (h) a CYC1 terminator, (i) a GAL1 promoter, (j) a gene that expresses HSP70, (k) a CYC1 terminator, (1) a GAL1 promoter, (m) a gene that expresses RuBisCO large subunit 1, (n) a CYC1 terminator, (o) a GAL1 promoter, and (p) a gene that expresses EGFP (FIGS. **3** and **4**).

(232) PCR was used to enhance DNA concentration using a Mastercycler Personal 5332 ThermoCycler (Eppendorf North America) with specific sequence primers and the standard method for amplification (Sambrook, J., E. F. Fritsch, and T. Maniatis, 1989, *Molecular Cloning: A Laboratory Manual*, 2.sup.nd ed., Vol. 1, Cold Spring Harbor Laboratory Press: Cold Spring Harbor, NY). Digestion and ligation were used to ensure assembly of DNA synthesized parts using restriction enzymes and reagents (PCR master mix of restriction enzymes: XhoI, KpnI, XbaI, EcoRI, BamHI, and HindIII, with alkaline phosphatase and quick ligation kit, all from Promega). DNA was quantified using a NanoVue spectrophotometer (GE Life Sciences) and a standard UV/Visible spectrophotometer using the ratio of absorbances at 260 nm and 280 nm. In order to verify final ligations, DNA was visualized and purified via electrophoresis using a Thermo EC-150 power supply.

(233) The DNA construct was made with gene parts fundamental for expression of sequences such as, for example, native and constitutive promoters, reporter genes, and transcriptional terminators or stops. Backbone plasmids and synthetic inserts can be mixed together for ligation purposes at different ratios ranging from 1:1, 1:2, 1:3, 1:4, and up to 1:5. In one aspect, the ratio of backbone plasmid to synthetic insert is 1:4. After the vector comprising the DNA construct has been produced, the resulting vector can be incorporated into the host cells using the method described below.

(234) A first DNA device for the production of genelight cultures and extracts was constructed by assembling a plasmid (pBAD) having the following genetic components in the following order: (a) a gene that expresses HSP70 having SEQ ID NO. 1, (b) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2, (c) a gene that expresses tonB having SEQ ID NO. 3, (d) an rrnB terminator, (e) an araBAD promoter, (f) a gene that expresses hydrogenase having SEQ ID NO. 4,

(g) a gene that expresses P-type ATPase having SEQ ID NO. 5, and (h) a gene that expresses EGFP having SEQ ID NO. 6. The DNA construct having SEQ ID NO. 7 was transformed into cells, as described below, to produce the biological devices. Plasmids containing the first DNA device are shown in FIGS. **1** and **2**.

(235) A second DNA device for the production of genelight cultures and extracts were constructed by assembling a plasmid (pYES2) having the following genetic components in the following order: (a) a gene that expresses hydrogenase having SEQ ID NO. 4, (b) a CYC1 terminator, (c) a GAL1 promoter, (d) a gene that expresses P-type ATPase having SEQ ID NO. 5, (e) a CYC1 terminator, (f) a GAL1 promoter, (g) a gene that expresses tonB having SEQ ID NO. 3, (h) a CYC1 terminator, (i) a GAL1 promoter, (j) a gene that expresses HSP70 having SEQ ID NO. 1, (k) a CYC1 terminator, (1) a GAL1 promoter, (m) a gene that expresses RuBisCO large subunit 1 having SEQ ID NO. 2, (n) a CYC1 terminator, (o) a GAL1 promoter, and (p) a gene that expresses EGFP. The DNA construct having SEQ ID NO. 8 was transformed into cells, as described below, to produce the biological devices. Plasmids containing the second DNA device are shown in FIGS. **3** and **4**.

Example 2: Selection of Microorganisms

(236) In some experiments, the genelight cultures and extracts were produced using transfected yeasts (*Saccharomyces cerevisiae*, ATCC® 200892™). Bacterial devices were constructed with one *Escherichia coli* (catalogue no. C29871 from New England BioLabs.

Example 3: Development of Competent Yeast Cells

(237) Yeast cells were made competent by subjecting them to an electrochemical process adapted from Gietz and Schiestl (*Nature Protocols*, 2007, 2:35-37). Briefly, a single yeast colony was inoculated into 100 mL YPD (yeast extract peptone dextrose) growth media. Yeast was grown overnight on a shaker at 30° C. to OD.sub.600=1.0. (Acceptable results were obtained with OD.sub.600 values ranging from 0.6 to 1.8.) Cells were centrifuged at 2000 rpm in a tabletop centrifuge and resuspended in 10 mL TEL buffer (10 mM Tris-HCl, 1 mM EDTA, 0.1 M LiAc, pH=7.5) and shaken vigorously overnight at room temperature. Cells were again centrifuged and resuspended in 1 mL TEL buffer. Cells prepared in this manner could be stored in the refrigerator for up to one month.

Example 4: Transformation of Microbial Cells to Produce Genelight Devices

(238) Competent cells were stored in the freezer until needed. Cells were thawed on ice and 100 μL of competent cells in TEL buffer were placed in a sterile 1.5 mL microcentrifuge tube. To this was added 5 μL of a 10 mg/mL solution of salmon sperm DNA (carrier DNA). Transforming DNA was added in various amounts. From 1 to 5 μg was sufficient for plasmids from commercial sources, but more DNA was required when transforming yeast with artificial DNA constructs. 10 μL of the DNA device were added to the microcentrifuge tube containing the competent yeast cells and the contents of the tube were mixed. The DNA-yeast suspension was incubated for 30 min at room temperature.

(239) A PLATE solution (consisting of 40% PEG-3350 in 1×TEL buffer) was prepared. 0.7 mL of PLATE solution was added to the DNA-yeast suspension and the contents were mixed thoroughly and incubated for 1 h at room temperature. The mixture was placed in an electromagnetic chamber for 30 minutes. Cells were then heated at 42° C. for 5-10 minutes and 250 μL aliquots were plated on yeast malt agar to which selective growth compounds had been added. Plates were incubated overnight at 30° C.

(240) DNA expression and effectiveness of transformation were determined by fluorescence of the transformed cells expressed in fluorescence units (FSUs) using a 20/20 Luminometer (Promega) according to a protocol provided by the manufacturer. Plasmid DNA extraction, purification, PCR, and gel electrophoresis were also used to confirm transformation. Different transformed devices were obtained. Different types of fluorescent reporter proteins were used (e.g., yellow, red, green, and cyan) for all transformed cells and/or constructs. However, the green fluorescent protein EGFP was preferred. When no fluorescent reporter protein was assembled, no fluorescence was observed.

(241) *S. cerevisiae* cells were subjected to transformation with the construct of SEQ ID. NO 8. Transformed yeast cells were incubated for 30 min at 28-30° C. Colonies of transformed yeast cells were selected, their DNA isolated and subjected to PCR amplification. Two control treatments were also carried out: (1) a negative control involving competent yeast and nuclease free water instead of a plasmid and (2) a positive control involving competent yeast with unmodified pYES2 plasmid. Transformed yeast were selected on a synthetic complete (SC) dropout plate deficient in uracil. A well-isolated clone was selected from the SC plate and preserved in YPD medium containing 15% glycerol for storage at −80° C. until later use.
(242) Alternatively, the construct of SEQ ID. NO 8 was transformed into competent *E. coli* using a standard heat shock protocol. Four clones were selected from a transformed plate and processed for full-length DNA sequencing. A clone with 100% DNA sequence accuracy was selected for further processing and was used to obtain a high concentration of plasmid construct at a mid-scale plasmid purification level.
Example 5: Production of Microbial Extracts
(243) *E. coli* cells transformed with SEQ ID NO 7 as described herein were grown in nutrient broth (25 mL of device inoculum in 1 L of medium) with 1 μg/mL ampicillin and 100 μM isopropyl β-D-1-thiogalactopyranoside (IPTG) at 37° C. for 96 hours. The cultures were then sonicated 7 times for a total of 2 min and 30 s and then autoclaved for 15 min.
(244) *S. cerevisiae* cells transformed with SEQ ID NO 8 as described herein were grown in yeast malt medium (25 mL of device inoculum in 1 L of medium) with 1 mg/mL of glucosamine, 2% raffinose, and induction with 1% galactose at 30° C. for 72 hours. The cultures were then sonicated 7 times for a total of 2 min and 30 s and then autoclaved for 15 min.
(245) Electrochemical Properties of Microbial Extracts
(246) Electrochemical properties of the extracts produced as described above were measured using a MASTECH 19-Range Digital Multimeter (MAS830B), an Eclipse Tools LED Light Intensity Meter (MT-4617 LED), and a Hach HQ11D portable pH/ORP/mV meter for water (HQ11D53000000). For experiments requiring the construction of a circuit, 12 cm copper and zinc electrodes were used, and 5 tubes with 20 mL extract in each tube were used to construct the circuit. LEDs used were white in color. Properties for *S. cerevisiae* extracts are summarized in Table 8 below:
(247) TABLE-US-00008 TABLE 8 Selected Properties of *S. cerevisiae* Extracts Optical Density of Culture 2.86 Optical Density of Extract 4.86 Voltage 5.3 V Amperage 2.0 × 10.sup.−4 A Lux 22.6 lux pH 6.2 Conductivity 8.76 mS/cm Redox 71.3 mV
Example 6: Construction of a Microbial Battery
(248) A microbial electrical circuit was constructed as follows (FIGS. **5** and **6**A-**6**B). A series of tubes was arranged in either a linear or a parallel manner. Tubes with volume capacities of 10, 15, 20, and 50 mL were evaluated, but the 50 mL capacity tubes were preferred.
(249) 25 mL aliquots of extract produced from the *E. coli* transformed with SEQ ID NO 7 were placed in each tube. Five tubes were used for each experiment, but a higher number of tubes was workable if higher voltage was needed for the circuit.
(250) Alternatively, a slurry of lyophilized extract and liquid extract was used. In a typical experiment, 1.5 g of lyophilized material and 150 μL of liquid extract were combined. Other mixtures were evaluated (1 g lyophilized extract and 50 μL of liquid extract, 1 g lyophilized extract and 200 μL of liquid extract, 2 g lyophilized extract and 200 μL of liquid extract) but were not preferred.
(251) Following distribution of extract in the tubes, zinc and copper electrodes were introduced into each tube. Electrodes were connected to one another with appropriate wires and the circuit was closed using various colors of LED lights (e.g., white, blue, green, yellow, and red) at different luminescences or intensities. See FIG. **5** for an example microbial circuit compared to a traditional serial circuit. See also FIGS. **6**A-**6**B wherein the microbial circuit is used to power an LED.

(252) Effectiveness of different microbial extract circuits was assessed using commercial instruments. Voltage (V) and current (A) as well as intensity and luminescence (lux) were measured for the circuit and for light produced by the LEDs using a multimeter (MASTECH MAS830DB), a photometer (PROSKIT MT4617LD), and direct observation, as appropriate. Oxidation and reduction were measured with an electrode sensor (HATCH HQ11d).

Example 7: Protective Effect of Genelight Extracts on *Bacillus subtilis*

(253) The UV-protective effects of the extracts disclosed herein were tested in *Bacillus subtilis* (ATCC® 82) cultures. *S. cerevisiae* cells transformed with SEQ ID NO 8 were fermented in yeast malt medium with 2% raffinose, 1 mg/mL glucosamine, and galactose for induction at 30° C. for 72 hours. The cultures were then sonicated 7 times for a total of 2 min, 30 sec. The supernatants were filtered using 8 μm, 5 μm, 2 and 1.2 μm filters and then used as-is in further experiments. Extracts produced from *E. coli* transformed with SEQ ID NO 7 were prepared as described previously.

(254) Extinction Coefficient Determination

(255) Extinction coefficient for the extracts disclosed herein was determined at 280 nm. Absorbance was measured using a microplate reader, with the molar extinction coefficient determined by the following equation:

(256) .Math. = $\frac{A \times I}{C \times V}$

where ε is the extinction coefficient of the extract, A is absorbance, l is the area of the well in the microplate reader, c is the concentration of extract, and V is the sample volume. The extinction coefficient was based on a sample concentration of 7.7 mg/ml of genelight extract produced from *E. coli* and 9.0 mg/ml of genelight extract produced from yeast transformed with the SEQ ID NOS 7 and 8, respectively. Experimentally-determined extinction coefficients can be found in Table 9 below:

(257) TABLE-US-00009 TABLE 9 Extinction Coefficient (ε280) for Genelight Extracts Device ε A *E. coli*-based 104 2.4 *S. cerevisiae*-based 122 3.3

Effects of Genelight Extracts on *Bacillus subtilis* Cultures

(258) *B. subtilis* cultures were grown at 30° C. for 1-2 days. Aliquots were removed from culture and diluted to different concentrations to determine optimum concentration for further experiments. Experiments were spectrophotometric and optical densities (OD) of 0.5, 1.0, 1.4, and 2.0 were tested, with 1.4 found to be preferable. The above dilutions were mixed with different concentrations of genelight extracts prepared as described above to obtain different proportions (for example, a 5:2 ratio would indicate 5 parts extract versus 2 parts *B. subtilis* 1.4 OD culture). These solutions were placed in Petri dishes with a total volume of 7 mL; the experiments were conducted in triplicate.

(259) Petri dishes were placed in a UV incubator and 1 mL samples were removed at different times (30 min, 1 hour, 2 hours) and thoroughly mixed. 500 μL aliquots of these samples were placed on nutrient agar using a standard streaking method. Three agar plate replicates were used each time; remaining portions of the aliquots were reserved for measurement of ATP. The agar plates were incubated at 30° C. for 1-4 days, following which, bacterial colonies of *B. subtilis* were viewed and counted.

(260) FIGS. **7**A and **7**B shows *B. subtilis* colonies after 30 minutes (FIG. **7**A) or 2 hours (FIG. **7**B). Results from the *E. coli* based extracts are shown in the dishes marked 1, results from the *S. cerevisiae* experiments are shown in the dishes marked 2, and untreated controls with water added to make up the 7 mL volume are shown in the dishes marked 3. *B. subtilis* cultures treated with genelight extracts prior to UV irradiation exhibited higher counts of living cells and higher cell density than untreated controls.

(261) Effects of Genelight Extracts on Human Fibroblast Cells

(262) Extracts for fibroblast experiments were prepared as follows. *S. cerevisiae* cells transformed with SEQ ID NO 8 were fermented in yeast malt medium with 2% raffinose and induced with 1%

galactose at 30° C. for 72 hours. Samples were sonicated 7 times for a total of 2 min and 30 sec and the supernatant was filtered through 8 μm, 5 μm, 3 μm, 2 μm, and 1.2 μm membranes.
(263) Human skin fibroblasts (ATCC® CRL-2522) were maintained in culture media for propagation and renewal following ATCC® recommendations. Propagation medium was based on Eagle's Minimal Essential Medium including 0.025% trypsin and 0.03% EDTA. Fetal bovine serum was added to the medium to a final concentration of 10%. Medium was also renewed according to ATCC® instructions.
(264) Fibroblast cells were grown at 37° C. under 5% CO.sub.2.
(265) Different proportions of extracts were applied to fibroblast cultures. The mixtures were then exposed to UV-B radiation at 302 nm for different periods of time and incubated at 37° C. and 5% CO.sub.2. Each experiment was performed in triplicate. Aliquots of fibroblast cells were harvested and subjected to microscopic analysis, with dead, living, and apoptotic cells counted following staining with trypan blue. Cells were counted at 20× magnification using several microscopic field views. Results are presented in Table 10 below for samples composed of 5 parts genelight extract per 4 parts fibroblast culture:
(266) TABLE-US-00010 TABLE 10 UV-Protective Effect of Genelight Cultures on Human Skin Fibroblasts Control (Fibroblasts + Treatment (Fibroblasts + Type of Cell Water) Genelight Extract) (Percentages) 30 min 1 hour 30 min 1 hour Live 0 0 33 20 Dead 60 80 0 53 Apoptotic 40 20 67 27 Total Survival 40 20 100 47 Protection — 60 27
where “live” cells appear as elongated cells without blue pigmentation, “dead” cells are spherical with intense blue pigmentation, and “apoptotic” cells are slightly curved and either lack pigmentation or are faintly pigmented. Total survival percentage is the sum of living and apoptotic cells, while protection percentage is the difference between total survival percentage of treated cultures and total survival percentage for the controls. Thus, more fibroblasts survived for a longer time when treated with the genelight extracts disclosed herein.
Total Genomic DNA from Fibroblast Cells
(267) A QIAmp DNA Mini Kit (item 51304 from Qiagen) was used to determine total DNA in the fibroblast cells following UV exposure using a protocol provided by the manufacturer. DNA concentration was determined using the 260/280 nm ratio measured in a Lambda 25 UV/Vis Spectrophotometer (PerkinElmer). DNA was separated using 1.2% agarose gel electrophoresis with a voltage of 100 V. Gel results can be seen in FIG. **12** and are summarized in Table 11 below:
(268) TABLE-US-00011 TABLE 11 Total Genomic DNA from Fibroblast Cells Treatment Group DNA Concentration (ng/μL) Cells Prior to Treatment 37.8 Cells Treated with *E. coli* Extracts 31.8 Cells Treated with *S. cerevisiae* Extracts 10.2 Untreated Cells (Water Control) 14.2
(269) FIG. **12** shows quantitation of genomic DNA from human fibroblasts with and without treatment using the devices and extracts disclosed herein. Lane 1 shows DNA expression prior to any irradiation or treatment. Lane 2 shows DNA expression in human fibroblasts treated with extracts from *E. coli* devices as disclosed herein and then exposed to UV irradiation. Lane 3 shows DNA expression in human fibroblasts treated with extracts from *S. cerevisiae* devices as disclosed herein and then exposed to UV irradiation. Lane 4 shows DNA expression in human fibroblasts treated with water only and then exposed to UV irradiation. Lane 5 is a 1 kb DNA ladder.
Example 8: Effect of Genelight Extracts on Plant Hormone Production
(270) The effect of the microbial devices and extracts disclosed herein on root and plant development was assessed as follows. Extracts were prepared by culturing *E. coli* transformed with SEQ ID NO 7 in LB broth at 37° C. and 150 rpm with 1 μg/mL ampicillin. Induction was initiated with 2% raffinose and 100 μM IPTG. The culture was then sterilized in an autoclave for 15 min and sonicated 3 times for a total of 2 min and 30 sec.
(271) Golf course soil was fertilized with Hou-Actinite prior to placing sod pieces of approximately 2,025 cm.sup.2 in size. Experimental treatments were applied after the sod was placed on the soil and roots were not yet visible. Sod treated with the genelight cultures and extracts disclosed herein

showed earlier rooting (within the first 48 hours) compared to a control (fertilized soil alone, with rooting after 48 hours). Treatment resulted in an improved appearance of the grass after two weeks.
(272) Two applications of genelight extracts per week were made to the sod samples, using either 75 mL of extract per square meter of sod or 150 mL per square meter of sod. Three measurements were taken for each treatment and sampling time using a ruler to measure root length, and the mean of these measurements was calculated.
(273) Root tension per area was measured using a tension meter and is expressed in Newtons (kg.Math.m/s.sup.2). The tension meter was placed on different parts of the sod (i.e., center and edges) and a mean from these three tension measurements was calculated. Tension forces were also determined for each plant and these results corroborate the larger-scale results. A higher value indicates more rooting per unit area, which means better anchoring of the plant to the soil. Sod treated with the extracts disclosed herein exhibited higher tension at about two times the value for the control of fertilizer alone. Root length and tension results for bacterial devices are summarized in Table 12 below:
(274) TABLE-US-00012 TABLE 12 Root Length and Tension for Bacterial Devices Root Force Root Force for Dose (mL/ for Sod Single Plant Root Length (in) square (Tension/ (Tension/ Days Treatment meter) Pressure) Pressure) 2 7 10 *E. coli* 150 21.33 4.33 1.5 2 2.5 extract *E. coli* 75 19.33 4.33 2 2.5 2.2 extract
(275) Yeast extracts were also tested. Root tension was measured for a 45×45 cm area chosen from a 1-square meter area of sod as well as for individual plants. Three replicates were tested for each measurement. Controls were also assessed wherein the area of sod was not treated but was only watered as normal. Root lengths were also measured as three replicates. Root length and tension results for yeast-based devices are summarized in Table 13 below:
(276) TABLE-US-00013 TABLE 13 Root Length and Tension for Yeast Devices Root Force for Root Force for Root Length (in) Sod Single Plant Week Treatment (Tension/Area) (Tension/Area) 1 2 5 Control (Water) 14 5.3 0.2 0.6 1.5 Yeast Extract 22 6.66 1 1.8 2.5
Example 9: Effect of Genelight Extracts on Uptake of Mineral Ions
(277) The optimum concentration of the disclosed compositions and extracts to enhance uptake of mineral nutrients in Bermuda grass was assessed as follows. Extracts prepared from *E. coli* biological devices as described herein were used undiluted (1×) or were diluted 10-fold (10×), 50-fold (50×), 100-fold (100×), and 1000-fold (1000×). Extracts were used alone or in combination with chitosan and were compared to controls of polyactive carbohydrate as defined herein and of fertilizer (Hou-actinite) alone. Mineral nutrient levels for phosphorus, iron, magnesium, calcium, and potassium were evaluated.
(278) For some experiments, metal concentration was measured using a Perkin-Elmer Nexlon 350X ICP-MS operating in a dual detector mode. Bermuda grass samples were processed as follows prior to performing ICP-MS analysis. Samples were ground to powder in dry ice and transferred in a metal-free tube. 2 mL (10× dilution) using 5% hydrogen peroxide and 1% nitric acid solution were added to the sample. The sample mixture was centrifuged at 1000 rpm for 2 min in order to obtain a homogeneous dispersion. An internal standard (indium, 10 ng/mL) was also added to the solution. Blank subtraction was applied after internal standard correction, with a 3-point calibration curve typically used to quantify compounds. The accuracy of the ICP-MS analytical protocol was periodically evaluated via the analysis of certified reference materials (serum and urine toxicology controls) from UTAK Laboratories, Inc. Results in mg of metal per kg of sample are presented in Tables 14A-14B below:
(279) TABLE-US-00014 TABLE 14A Mineral Ion Analysis in Bermuda Grass for *E. coli* Devices Treatment Li Na Mg Al P K Ca Ti V Cr Mn Fe Chitosan 0.02% 0.713 970 710 104 893 4508 2550 3.95 0.803 52.3 84.2 1281 Polyactive Carbohydrate 0.586 213 195 41 <LOQ 1060 890 2.43 0.317 12.2 13.6 253 1X Genelight Extract + Chitosan 0.617 461 477 163 1006 4064 3085 6.11 0.691 6.15 51.1 520 1X Genelight Extract + Chitosan + Fertilizer 0.555 336 234 66 <LOQ 1173 1515 3.33

0.534 7.39 36.7 371 Polyactive Carbohydrate + Chitosan + Fertilizer 0.571 104 92 50 <LOQ 728 588 2.47 0.354 19.1 15.9 336 No Treatment 0.638 457 357 72 <LOQ 1355 1741 3.37 0.482 10.1 28.1 394 1X Genelight Extract 0.587 645 422 52 1051 3064 1566 3.28 0.491 24.3 52.8 521 10X Genelight Extract 0.691 394 343 167 673 2529 2474 5.08 1.02 9.62 74.1 835 50X Genelight Extract 0.730 714 417 120 1411 6202 3327 4.30 1.34 89.6 66.1 2073 100X Genelight Extract 0.670 420 399 111 1028 4617 3783 4.08 0.913 23.2 53.7 778 1000X Genelight Extract 0.548 156 193 57 <LOQ 1654 1560 3.51 0.377 6.49 53.2 262

(280) TABLE-US-00015 TABLE 14B Mineral Ion Analysis in Bermuda Grass for *E. coli* Devices Treatment Co Cu Ge As Rb Sr Zr Mo Cs Ba La W Tl Chitosan 0.02% 0.489 9.18 0.148 0.307 2.40 20.9 <LOQ 2.46 0.063 24.6 0.959 0.124 0.029 Polyactive Carbohydrate 0.196 4.08 0.346 0.318 0.82 5.50 <LOQ 0.923 <LOQ 8.34 3.21 0.036 0.004 1X Genelight Extract + 0.307 5.16 0.171 0.461 2.86 19.9 0.498 1.10 0.099 16.1 1.37 0.069 0.015 Chitosan 1X Genelight Extract + 0.332 4.30 0.114 0.300 0.86 8.62 <LOQ <LOQ <LOQ 10.7 0.728 0.145 0.004 Chitosan + Fertilizer Polyactive Carbohydrate + 0.198 3.69 0.127 <LOQ 0.65 4.15 <LOQ 0.956 <LOQ 9.53 0.648 <LOQ 0.004 Chitosan + Fertilizer No Treatment 0.243 4.62 <LOQ 0.297 0.88 11.5 <LOQ 1.35 <LOQ 15.9 0.873 0.034 0.005 1X Genelight Extract 0.311 5.96 0.135 <LOQ 1.73 9.77 <LOQ 1.46 <LOQ 14.6 1.02 0.089 0.005 10X Genelight Extract 0.777 5.49 0.135 0.335 1.49 13.3 0.445 1.15 0.072 13.6 0.926 0.188 0.005 50X Genelight Extract 0.677 6.96 0.275 0.497 2.79 20.9 0.459 3.44 <LOQ 20.5 2.29 0.131 0.006 100X Genelight Extract 0.392 6.28 <LOQ 0.367 2.35 17.5 0.414 1.56 <LOQ 13.6 0.744 0.088 0.006 1000X Genelight Extract 0.218 4.18 0.127 <LOQ 1.00 6.95 <LOQ 1.04 0.062 10.6 1.01 0.052 0.005

(281) These results demonstrate the efficacy of the 50× and 100× dilutions for enhancing uptake of mineral nutrients compared to a fertilizer control. 1× and 10× dilutions were also more effective than the control. The 1000× dilution showed higher potassium uptake and equal phosphorus uptake compared to a fertilizer control. Thus, treatment with the disclosed extracts over a wide range of dilutions equates to greater mineral uptake and healthier, better developed plants.

Example 10: Effect of Genelight Extracts on Chlorophyll Production

(282) Chlorophyll a is the pigment in plants that acts directly in the reactions of photosynthesis requiring light, while chlorophyll b is an accessory pigment that acts indirectly in photosynthesis by transferring energy from the light it absorbs to chlorophyll a. The ratio of chlorophyll a to chlorophyll b in a typical chloroplast is about 3:1. Bermuda grass was treated with the extracts disclosed herein undiluted (B1) and with various dilutions (10× or B10; 50× or B50, 100× or B100, and 1000× or B1000) as well as with a control of fertilizer alone (Hou-actinite).

(283) Samples were prepared as follows: 25 mL polypropylene tubes were labeled. Acetone to be used as the extraction solvent, previously prepared to include a vitamin E internal standard, was removed from the freezer and brought to room temperature. Grass samples were ground to powder and placed in the polypropylene tubes. 5 mL of 100% acetone containing the internal standard was added to each tube and the tubes were immediately covered with Parafilm and transferred to a −25° C. freezer for at least 30 min.

(284) The samples were removed from the freezer and Parafilm was removed. A 47 mm GF/F sample filter was added to each tube and the tubes were re-covered. The samples were again transferred to a −25° C. freezer for 1 hour.

(285) Samples were removed from the freezer and filters were macerated using an ultrasonic probe with a duty cycle of 90% and an output level of 5. Tubes were partially submerged in an ice and water bath during sonication to minimize heat accumulation. The condition of the sonicator tip was periodically monitored and polished using fine grade sandpaper to yield a smooth surface, and was rinsed with acetone and wiped with a Kimwipe between samples to prevent cross-contamination. This procedure required approximately 10 s and could be timed by counting the pulses emitted by the probe. Cavitation was prevented by pausing during sonication to push the filter slurry back down to the bottom of the tube if necessary.

(286) After processing, samples were placed in a −25° C. freezer for 3-6 hours, then removed and filtered using a 0.45 μm PTFE syringe cartridge filter attached to a disposable plastic syringe. The pigment extract was collected in glass scintillation vials with foil-lined, labeled caps and the vials were placed in a dark environment.
(287) After filtration, sample extracts were vortexed. Approximately 500 μL of each extract was transferred to an amber HPLC vial and capped with an HPLC cap having a silicone/PTFE liner. The vials were placed in a pre-chilled temperature-controlled autosampler and HPLC analysis was started. Unused extracts were stored in a freezer until the HPLC analyses were successfully completed.
(288) HPLC analysis was conducted using a Supelcosil LC318 C18 column (25 cm×4.6 mm with 5 μm particle size for computer modeling work and 10 cm×4.6 mm with 5 μm particle size for pigment isolations). Flow rate was 1 mL/min. Mobile phase A was 70:30 (v/v) methanol and 28 mM aqueous TEAA, pH 6.5, while mobile phase B was methanol. A linear gradient of 5% (0 minutes) to 100% mobile phase B (20 minutes) was used for elution.
(289) Relative amounts of chlorophyll a and chlorophyll b are given in Table 15 below:
(290) TABLE-US-00016 TABLE 15 Primary Pigments per μg of Grass Sample after Treatment with *E. coli* Extracts Sample Chlorophyll a Chlorophyll b B1 4835.148 1050.309 B10 3054.033 817.857 B50 843.357 249.964 B100 5918.972 1343.685 B1000 3509.713 978.343 Control 2827.165 680.853
(291) The disclosed extracts and dilutions induce greater levels of both chlorophyll a and chlorophyll b compared to a fertilizer control, with the 100× (B100) dilution having the strongest effect.
Example 11: Effect of Genelight Extracts on Plant Hormone Production
(292) Bermuda grass was grown for two weeks; during this time period, either extracts from the *E. coli* devices disclosed herein or a fertilizer control (Hou-actinite) were applied three times.
(293) Samples were analyzed for plant hormones, chlorophyll, and minerals as described above. Hormone concentration and root length are based on the mean of three replicates, while mineral concentrations reported herein are based on single samples. Summarized results are presented in Table 16 below, while Tables 17A-17B contain a more detailed breakdown of additional plant hormones:
(294) TABLE-US-00017 TABLE 16 Effect of Genelight Extracts on Plant Hormone Production Hormone Quantity (ng/sample) Mean Root Other Jasmonic Salicylic Length Chlorophyll b Pigments Mineral Analysis (mg/kg) Treatment Auxins Acid Acid (in) (μg/sample) (μg/sample) Mg K P Ca Genelight 243.33 25.94 3444.94 2.9 4291.216 2479.246 558 4064 1248 2165 Extract (1X) Control 115.56 2.21 859.43 1.5 3548.283 866.046 383 4340 1218 1590 (fertilizer)
(295) TABLE-US-00018 TABLE 17A Plant Hormone Production in Bermuda Grass for *E. coli* Devices Methyl IAA- Treatment ABA.sup.a SA.sup.b cZ.sup.c cZR.sup.c tZr.sup.c GA8.sup.d GA12.sup.d GA19.sup.d GA24.sup.d GA53.sup.d IAA.sup.e IAA.sup.e Asp.sup.e Chitosan 0.02% 75.694 427.058 0.323 0.288 ND 39.451 35.310 5.115 9.665 29.506 330.577 0.749 ND Polyactive Carbohydrate 85.055 406.050 0.301 0.178 ND 39.766 40.928 6.896 4.572 35.746 344.079 1.075 ND 1X Genelight Extract + 1.817 33.112 0.050 0.352 0.598 0.987 ND 1.706 ND 5.668 8.385 0.283 1.018 Chitosan 1X Genelight Extract + 61.213 184.349 0.162 0.100 ND 28.684 33.537 6.282 ND 29.909 96.527 0.749 ND Chitosan + Fertilizer Polyactive Carbohydrate + 47.455 202.524 0.182 0.202 ND 21.862 29.511 5.157 ND 30.244 76.893 0.452 ND Chitosan + Fertilizer No Treatment 69.007 121.804 0.217 0.160 ND 24.916 38.246 4.338 5.815 29.860 32.888 0.291 ND 1X Genelight Extract 60.067 125.501 0.126 0.111 ND 18.436 18.518 3.173 2.699 19.036 118.216 0.248 ND 10X Genelight Extract 120.354 644.789 0.366 0.374 ND 57.394 30.603 6.039 18.698 45.071 632.977 0.496 ND 50X Genelight Extract 12.628 20.616 0.054 0.271 0.113 10.381 6.566 1.773 1.968 7.333 15.234 0.230 4.476 100X Genelight Extract 172.878 1440.829 0.411 0.328 ND 48.494 20.796 5.523 7.323 13.656 410.832 0.816 ND 1000X Genelight Extract 91.438 318.869 0.260 0.304 ND

34.344 24.863 4.980 6.895 11.291 289.535 0.401 ND
(296) TABLE-US-00019 TABLE 17B Plant Hormone Production in Bermuda Grass for *E. coli* Devices IAA- JA- Treatment Trp.sup.e JA.sup.f ILE.sup.f OPDA.sup.g Strigoi.sup.h Chitosan 0.02% ND 17.804 ND 45.422 ND Polyactive ND 15.020 ND 16.942 ND Carbohydrate 1X Genelight ND 1.645 0.094 3.447 ND Extract + Chitosan 1X Genelight ND 6.944 ND 5.142 ND Extract + Chitosan + Fertilizer Polyactive ND 8.982 ND 15.776 ND Carbohydrate + Chitosan + Fertilizer No Treatment ND 5.926 ND 5.437 275.530 1X Genelight Extract ND 4.403 ND 2.743 ND 10X Genelight Extract ND 33.824 ND 96.835 ND 50X Genelight Extract ND 12.865 0.893 73.048 ND 100X Genelight Extract ND 27.209 ND 83.050 ND 1000X Genelight Extract 0.509 8.859 ND 24.891 ND .sup.aABA = abscisic acid .sup.bSA = salicylic acid .sup.ccZ = cis-zeatin; cZr = cis-zeatin riboside; tZr = trans-zeatin riboside .sup.dGA8 = gibberellin A(8); GA12 = gibberellin A(12); GA19 = gibberellin A(19); GA24 = gibberellin A(24); GA53 = gibberellin A(53) .sup.eIAA = indole-3-acetic acid; MethylIAA = methyl-IAA; IAA-Asp = IAA conjugated with aspartic acid; IAA-Trp = IAA conjugated with tryptophan .sup.fJA = jasmonic acid; JA-Ile = jasmonic acid conjugated with isoleucine .sup.gOPDA = 12-oxo-phytodienoic acid .sup.hStrigol = (+)-strigol
(297) These results indicate that higher induction of hormones related to root growth and stimulation of disease resistance in plants, as well as longer root lengths, occur with treatment using the disclosed extracts compared to a fertilizer control. Higher levels of chlorophyll b and magnesium are also found in grass treated with the disclosed extracts, indicating the disclosed extracts are better able to stimulate photosynthesis compared to a control.
(298) Higher induction of other pigments and higher uptake of minerals including phosphorus, potassium, and calcium also occurred in grass treated with the disclosed compounds and extracts, compared to a fertilizer control.
Example 12. Characterization of Extract
(299) Samples for Raman Spectroscopy:
(300) 1. Biolight *E.Coli* Device Extract, prepared through fermentation of *E.Coli* device in LB Broth at 37° C. 2. Culture was allowed to ferment for 48 hours then centrifuged, pellets were retained and media removed. 3. Pellets were then redissolved in sterile deionized water and sonicated to lyze cells. 4. The lysate was centrifuged and filtered through a 0.20 um filter and filtrate was retained and labeled as Biol *E.Coli* Device Extract. 5. The extract was purified and concentrated through Millipore Tangential Flow Filtration with a Pall filtration cartridge with a 5K molecular weight cut off. 6. The retentate was collected and 10 mL were lyophilized in a Harvest Right Scientific Freeze Drier at −40° C. 7. Approximately 20 mg of lyophilized sample was analyzed with a Renishaw Invia Confocal Raman Spectrometer. 8. The sample was placed on a glass slide directly in the confocal laser path. 9. The laser selected was 785 nm excitation laser, with 10 sec exposure and 4 acquisitions at 25° C. 10. Spectra was processed and analyzed using Wiley KnowitAll informatics systems 2020.
(301) As seen on FIG. **13**, genelight *E. coli* extract is a mixture of proteins. Strong markers for proteins are the Amide I, II and III peaks at 1660, 1525, and 1297 cm′, which are present in our Raman spectra. In addition, the presence of the monomeric protein thioredoxin is detected, since this protein has very specific peaks at 2300 & 2500 to 2600 cm′ region.
(302) The aromatic amino acids Phenylalanine and Tyrosine were detected. The respective peaks at 1000 cm.sup.−1, 1050 cm.sup.−1 and 620 cm.sup.−1 for the Phenylalanine and the 1170 cm.sup.−1, the fermi doublet at 830, 850 cm.sup.−1 and the 650 cm.sup.−1 peaks for Tyrosine are all present in the sample's spectra.
(303) In addition Biolight 1 *E. Coli* Device Extract seems to have a mixture of unodered secondary protein structures and alpha helix structures noted by the strong peaks at 1257 cm.sup.−1 (unordered) and 1297 cm.sup.−1 (alpha helix).
(304) Based on the ELISA assay determination, the Biolight 1 *E. Coli* Device Extract includes

Rubisco and HSP70.

(305) Throughout this application, various publications are referenced. The disclosures of these publications in their entireties are hereby incorporated by reference into this application in order to more fully describe the compounds, compositions, and methods described herein.

(306) Various modifications and variations can be made to the compounds, compositions, and methods described herein. Other aspects of the compounds, compositions, and methods described herein will be apparent from consideration of the specification and practice of the compounds, compositions, and methods disclosed herein. It is intended that the specification and examples be considered as exemplary.

Claims

1. A method for improving a physiological property of a plant, the method comprising: applying to the plant a composition comprising an extract prepared from host cells cultured in a culture medium, said host cells transformed with a DNA construct comprising the following genetic components: (1) a gene that expresses a heat shock protein having at least 90% sequence identity to SEQ ID NO: 1, (2) a gene that expresses RuBisCO large subunit 1 having at least 90% sequence identity to SEQ ID NO: 2, (3) a gene that expresses tonB having at least 90% sequence identity to SEQ ID NO: 3, (4) a gene that expresses hydrogenase having at least 90% sequence identity to SEQ ID NO: 4, and (5) a gene that expresses p-type ATPase having at least 90% sequence identity to SEQ ID NO: 5, wherein the property is improved compared to the property of the same plant that has not been applied the composition, and wherein the physiological property comprises increased root tension, root length, hormone production, drought tolerance, disease resistance, photosynthesis, or any combination thereof.
2. The method of claim 1, wherein the property comprises increased root tension, root length, hormone production, drought tolerance, disease resistance, photosynthesis, or any combination thereof.
3. The method of claim 1, wherein the plant is grass, trees, bushes, shrubs, flower, vines, coffee, soybean, or cotton.
4. The method of claim 3, wherein the grass is growing on a golf course, a lawn, or an athletic playing field.
5. The method of claim 1, wherein the heat shock protein is HSP70.
6. The method of claim 1, wherein the construct further comprises a gene that expresses a reporter protein.
7. The method of claim 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having at least 90% sequence identity to SEQ ID NO: 1, (2) a gene that expresses RuBisCO large subunit 1 having at least 90% sequence identity to SEQ ID NO: 2, (3) a gene that expresses tonB having at least 90% sequence identity to SEQ ID NO: 3, (4) a gene that expresses hydrogenase having at least 90% sequence identity to SEQ ID NO: 4, and (5) a gene that expresses p-type ATPase having at least 90% sequence identity to SEQ ID NO: 5.
8. The method of claim 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having at least 90% sequence identity to SEQ ID NO: 1, (2) a gene that expresses RuBisCO large subunit 1 having at least 90% sequence identity to SEQ ID NO: 2, (3) a gene that expresses tonB having at least 90% sequence identity to SEQ ID NO: 3, (4) an rrnB terminator, (5) an araBAD promoter, (6) a gene that expresses hydrogenase having at least 90% sequence identity to SEQ ID NO: 4, and (7) a gene that expresses p-type ATPase having at least 90% sequence identity to SEQ ID NO: 5.
9. The method of claim 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having at least 90%

sequence identity to SEQ ID NO: 4, (2) a gene that expresses p-type ATPase having at least 90% sequence identity to SEQ ID NO: 5, (3) a gene that expresses tonB having at least 90% sequence identity to SEQ ID NO: 3, (4) a gene that expresses a heat shock protein having at least 90% sequence identity to SEQ ID NO: 1, and (5) a gene that expresses RuBisCO large subunit 1 having at least 90% sequence identity to SEQ ID NO: 2.

10. The method of claim 1, wherein the construct further comprises a gene that expresses a dehydrogenase.

11. The method of claim 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses a heat shock protein having at least 90% sequence identity to SEQ ID NO: 1, (2) a gene that expresses dehydrogenase having at least 90% sequence identity to SEQ ID NO: 9, (3) a gene that expresses RuBisCO large subunit 1 having at least 90% sequence identity to SEQ ID NO: 2, (4) a gene that expresses tonB having at least 90% sequence identity to SEQ ID NO: 3, (5) a gene that expresses hydrogenase having at least 90% sequence identity to SEQ ID NO: 4, and (6) a gene that expresses p-type ATPase having at least 90% sequence identity to SEQ ID NO: 5.

12. The method of claim 1, wherein the DNA construct comprises from 5′ to 3′ the following genetic components in the following order: (1) a gene that expresses hydrogenase having at least 90% sequence identity to SEQ ID NO: 4, (2) a gene that expresses p-type ATPase having at least 90% sequence identity to SEQ ID NO: 5, (3) a gene that expresses tonB having at least 90% sequence identity to SEQ ID NO: 3, (4) a gene that expresses a heat shock protein having at least 90% sequence identity to SEQ ID NO: 1, (5) a gene that expresses dehydrogenase having at least 90% sequence identity to SEQ ID NO: 9, and (6) a gene that expresses RuBisCO large subunit 1 having at least 90% sequence identity to SEQ ID NO: 2.

13. The method of claim 1, wherein the DNA construct has at least 90% sequence identity to SEQ ID NOs: 7, 8, 10 or 11.

14. The method of claim 1, wherein the DNA construct is incorporated in a vector.

15. The method of claim 14, wherein the vector is a plasmid selected from the group consisting of pWLNEO, pSV2CAT, pOG44, pXTI, pSG, pSVK3, pBSK, pBR322, pYES, pYES2, pBSKII, pUC, or pBAD.

16. The method of claim 1, wherein the host cells comprise bacteria or fungi.

17. The method of claim 1, wherein the composition further comprises chitosan, a polyactive carbohydrate, or a combination thereof.

18. The method of claim 17, wherein chitosan is less than 1% by weight of the composition.

19. The method of claim 17, wherein the polyactive carbohydrate is less than 1% by weight of the composition.